



Chapter 18

Control of Gene Expression

Lecture Presentations by
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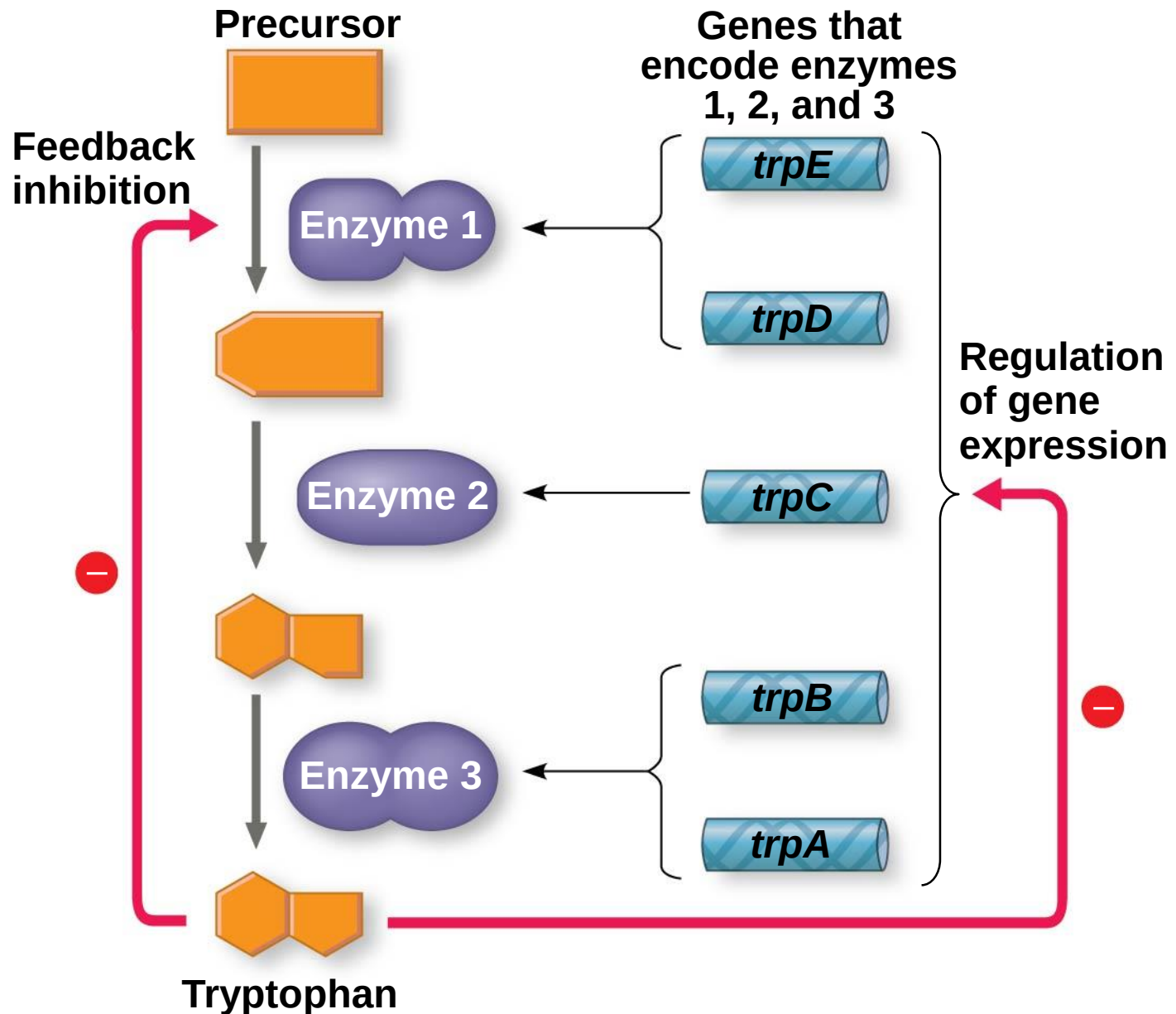
Beauty in the Eye of the Beholder

- Prokaryotes and eukaryotes precisely regulate gene expression in response to environmental conditions
- In multicellular eukaryotes, gene expression regulates development and is responsible for differences in cell types
- RNA molecules play many roles in regulating gene expression in eukaryotes

Concept 18.1: Bacteria often respond to environmental change by regulating transcription

- Natural selection has favored bacteria that produce only the gene products needed by that cell
- A cell can regulate the production of enzymes by feedback inhibition or by gene regulation
- One mechanism for control of gene expression in bacteria is the operon model

Figure 18.2



(a) Regulation of enzyme activity

(b) Regulation of enzyme production

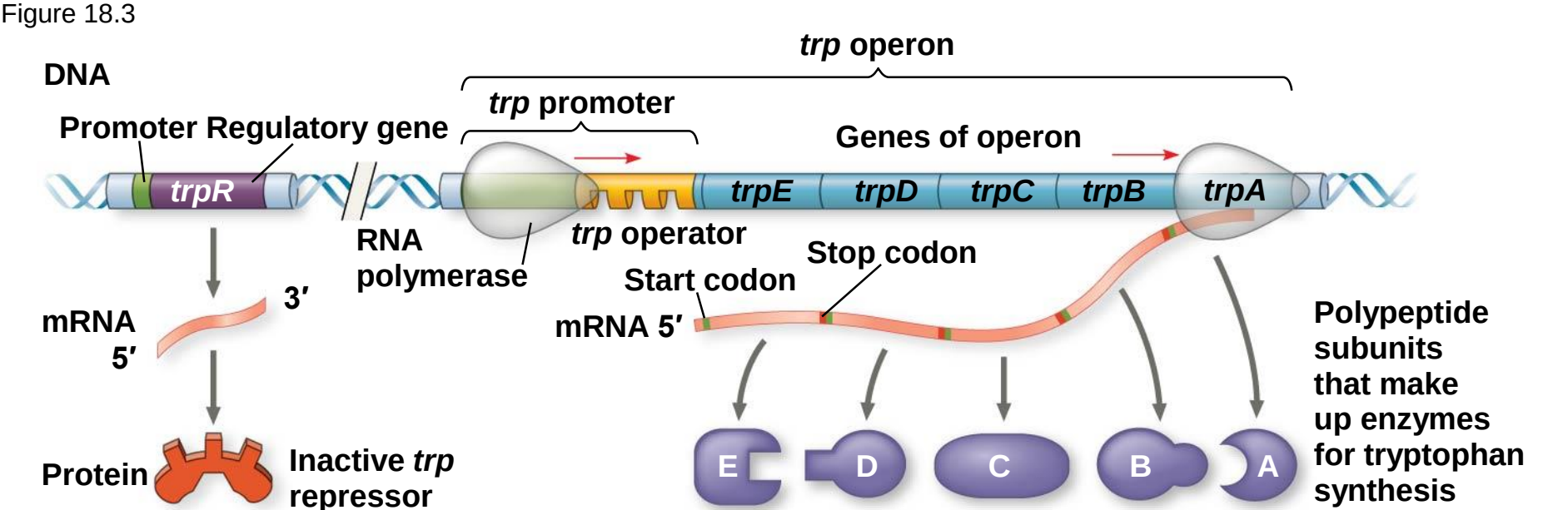
Operons: The Basic Concept

- A cluster of functionally related genes can be coordinately controlled by a single “on-off switch”
- The switch is a segment of DNA called an **operator**, usually positioned within the promoter
- An **operon** is the entire stretch of DNA that includes the operator, the promoter, and the genes that they control

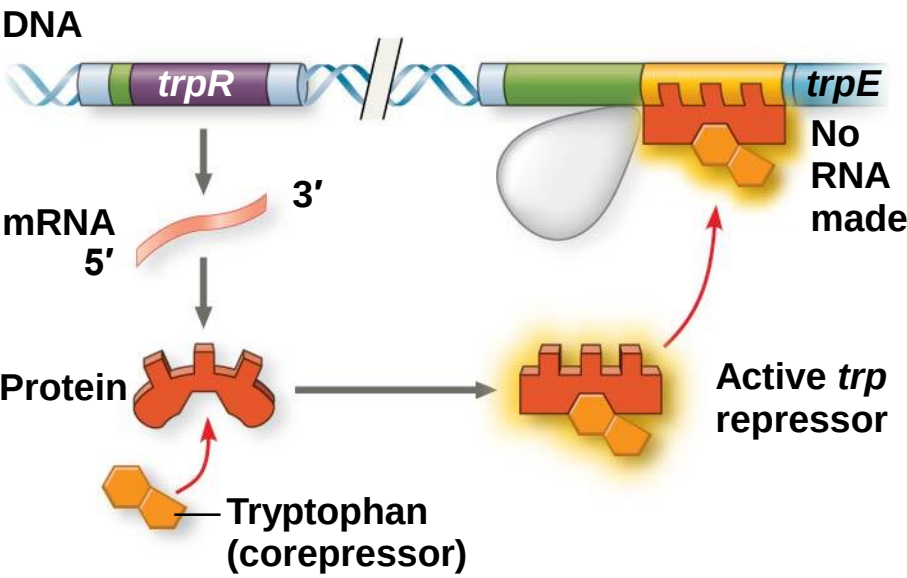
- The operon can be switched off by a protein **repressor**
- The repressor prevents gene transcription by binding to the operator and blocking RNA polymerase
- The repressor is the product of a separate **regulatory gene**, located some distance from the operon itself

- The repressor can be in an active or inactive form, depending on the presence of other molecules
- A **corepressor** is a molecule that cooperates with a repressor protein to switch an operon off
- For example, *E. coli* can synthesize the amino acid tryptophan when it has insufficient tryptophan

- By default, the *trp* operon is on and the genes for tryptophan synthesis are transcribed
- When tryptophan is present, it binds to the *trp* repressor protein, which turns the operon off
- The repressor is active only in the presence of its corepressor tryptophan; thus the *trp* operon is turned off (repressed) if tryptophan levels are high

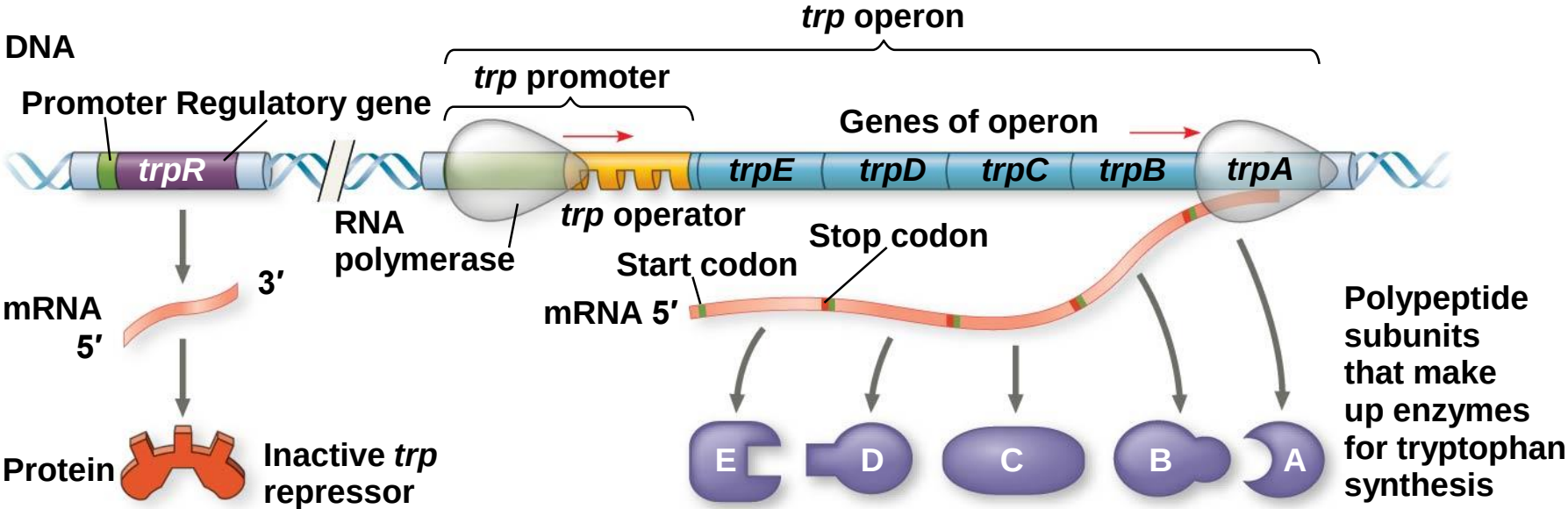


(a) Tryptophan absent, repressor inactive, operon on.



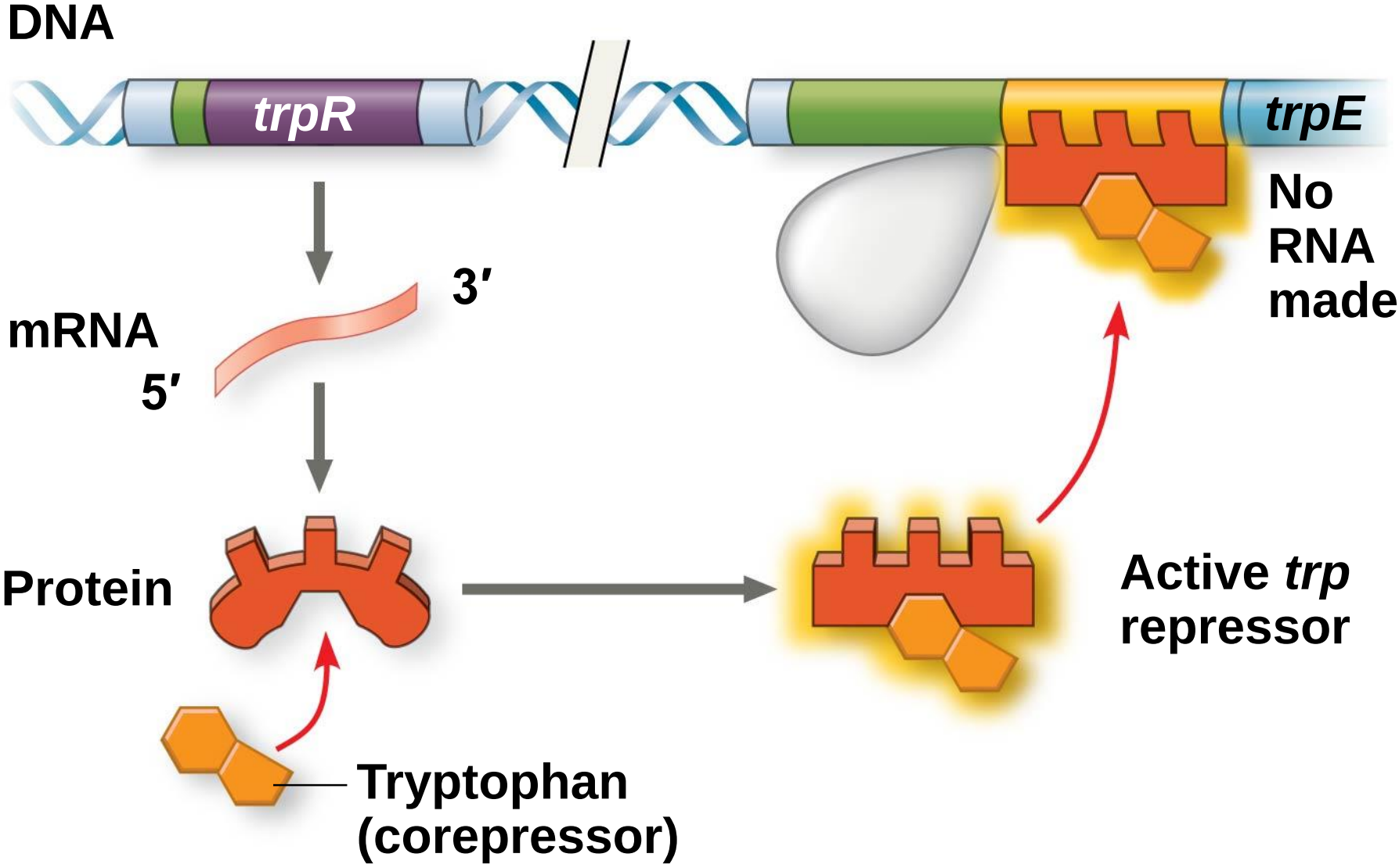
(b) Tryptophan present, repressor active, operon off.

Figure 18.3a



(a) Tryptophan absent, repressor inactive, operon on.

Figure 18.3b



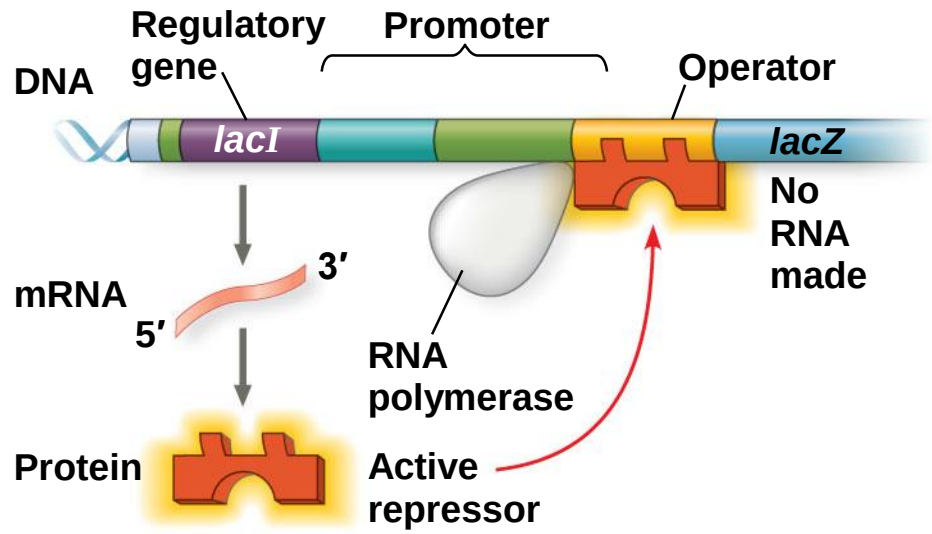
(b) Tryptophan present, repressor active, operon off.

Repressible and Inducible Operons: Two Types of Negative Gene Regulation

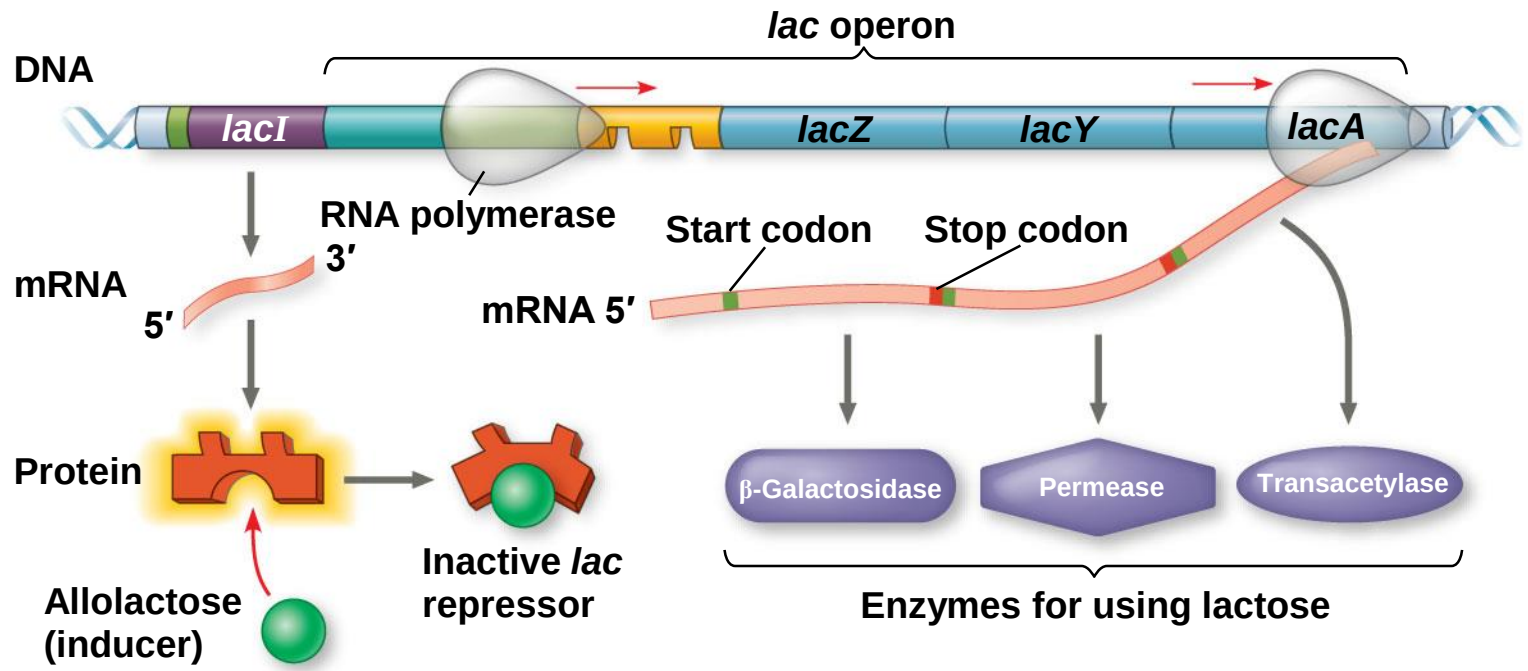
- A repressible operon is one that is usually on; binding of a repressor to the operator shuts off transcription
- The *trp* operon is a repressible operon
- An inducible operon is one that is usually off; a molecule called an inducer inactivates the repressor and turns on transcription

- The *lac* operon is an inducible operon and contains genes that code for enzymes used in the hydrolysis and metabolism of lactose
- By itself, the *lac* repressor is active and switches the *lac* operon off
- A molecule called an **inducer** inactivates the repressor to turn the *lac* operon on

Figure 18.4

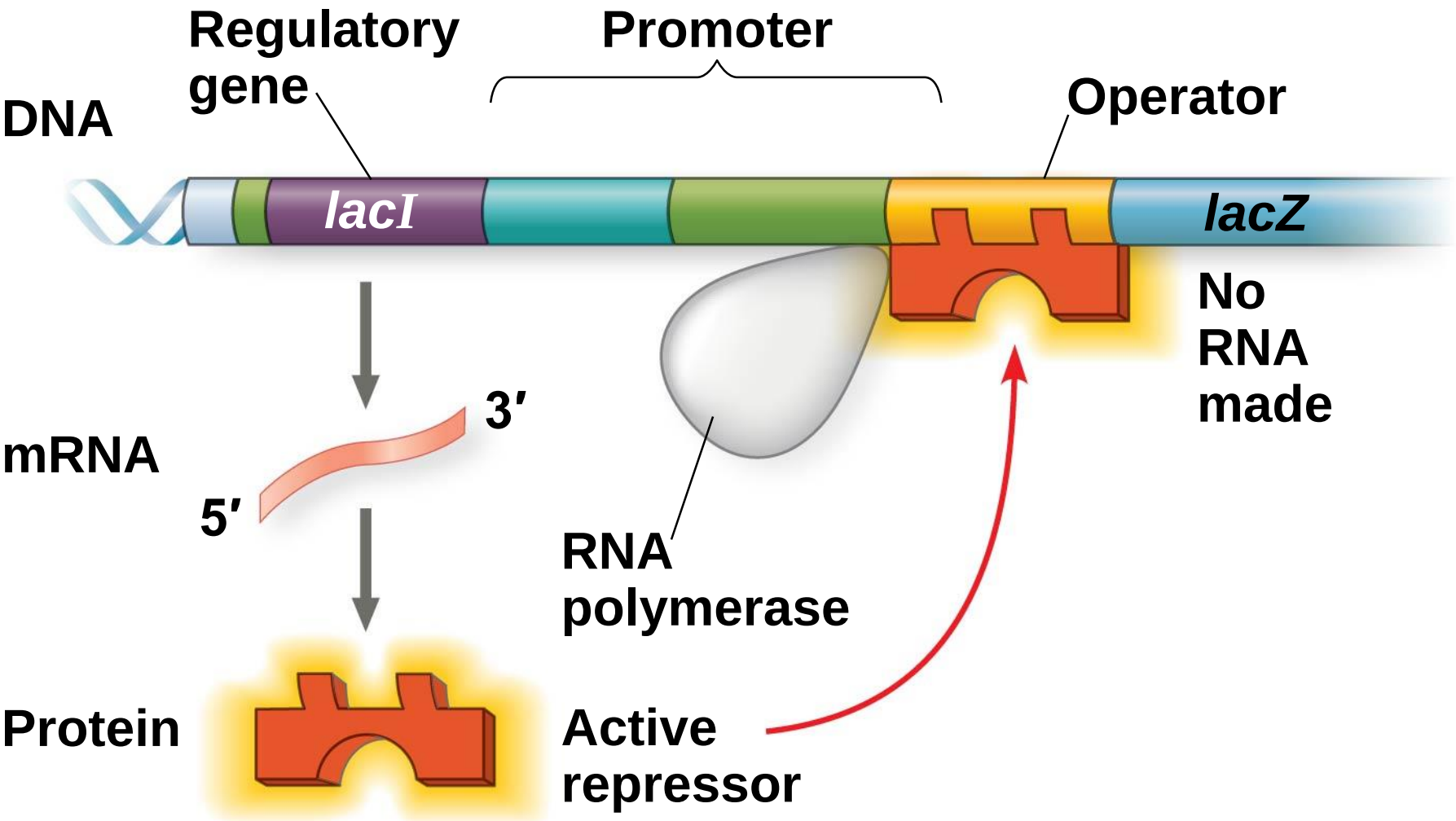


(a) Lactose absent, repressor active, operon off.



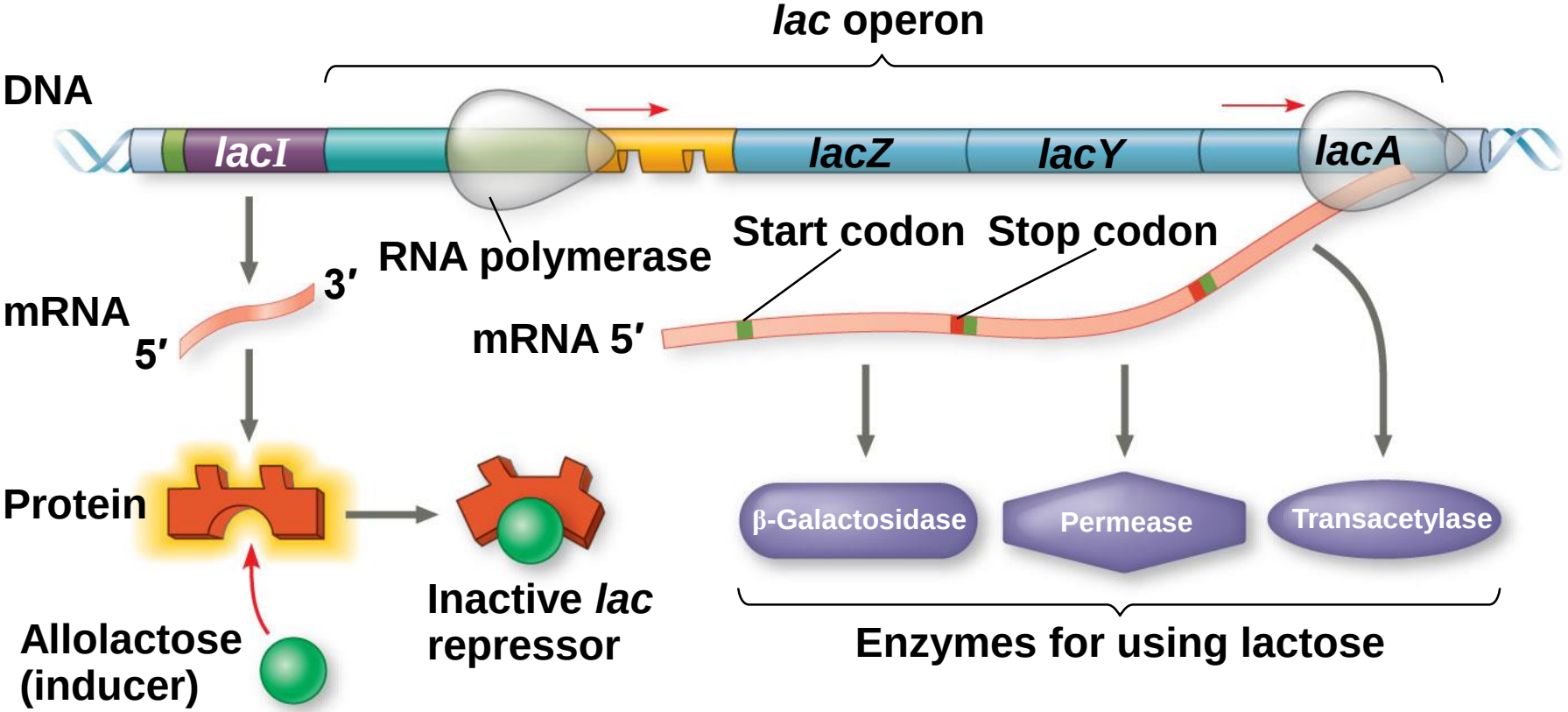
(b) Lactose present, repressor inactive, operon on.

Figure 18.4a



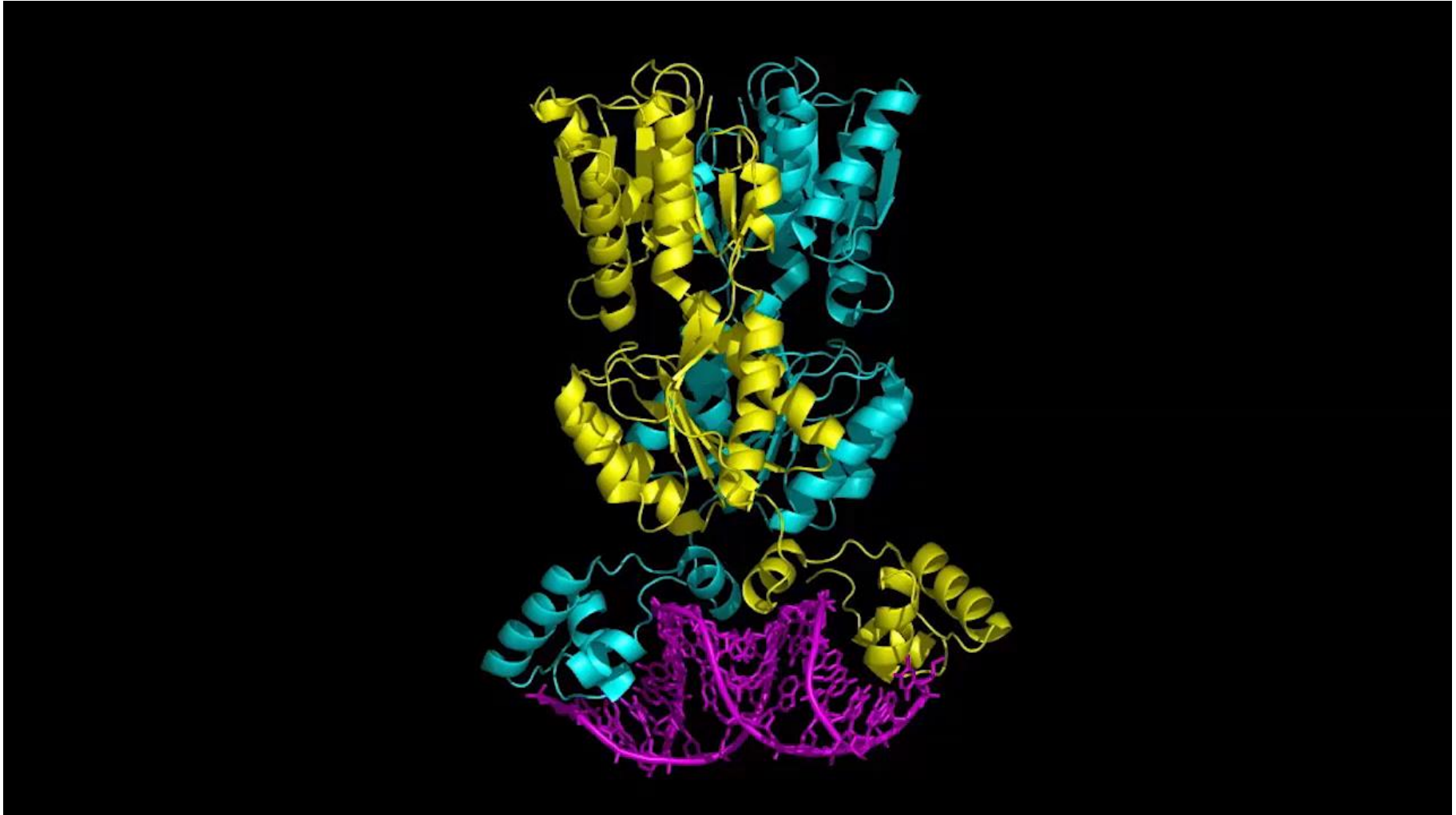
(a) Lactose absent, repressor active, operon off.

Figure 18.4b



(b) Lactose present, repressor inactive, operon on.

Video: Cartoon Rendering of the *lac* Repressor from *E. coli*



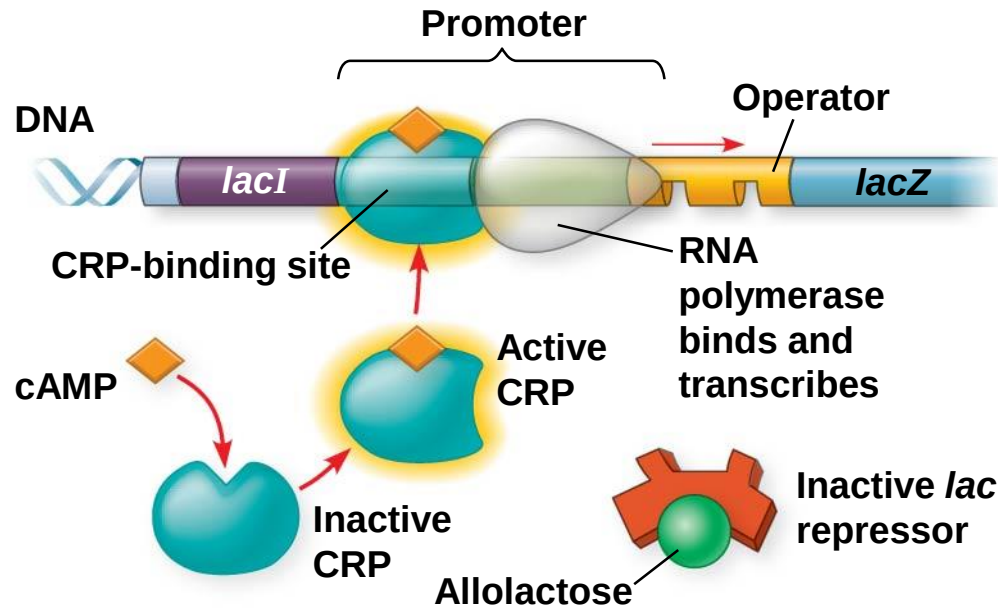
- Inducible enzymes usually function in catabolic pathways; their synthesis is induced by a chemical signal
- Repressible enzymes usually function in anabolic pathways; their synthesis is repressed by high levels of the end product
- Regulation of both the *trp* and *lac* operons involves negative control of genes because operons are switched off by the active form of the repressor

Positive Gene Regulation

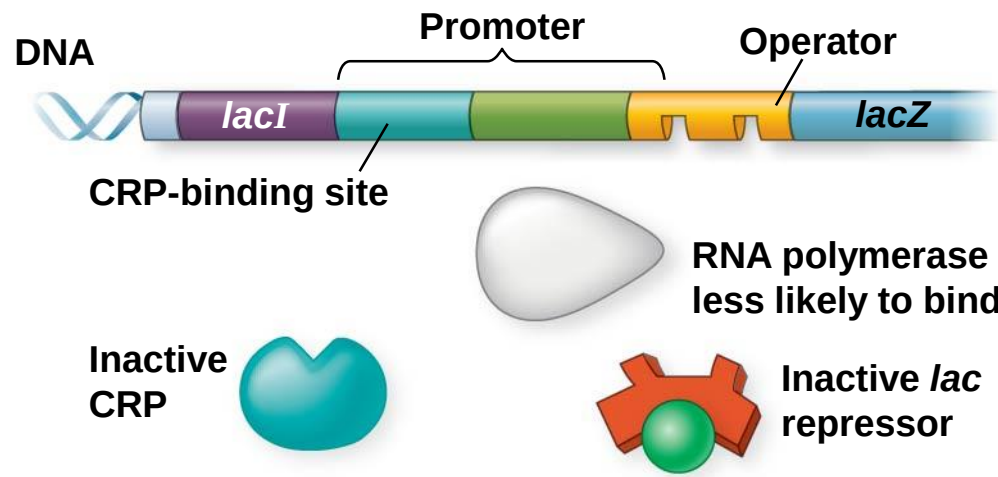
- Some operons are also subject to positive control through a stimulatory protein, such as cyclic AMP receptor protein (CRP), an **activator** of transcription
- When glucose (a preferred food source of *E. coli*) is scarce, CRP is activated by binding with **cyclic AMP (cAMP)**
- Activated CRP attaches to the promoter of the *lac* operon and increases the affinity of RNA polymerase, thus accelerating transcription

- When glucose levels increase, CRP detaches from the *lac* operon, and transcription returns to a normal rate
- CRP helps regulate other operons that encode enzymes used in catabolic pathways
- The ability to catalyze compounds like lactose enables cells deprived of glucose to survive
- The compounds present in any given cell determine which genes are switched on

Figure 18.5

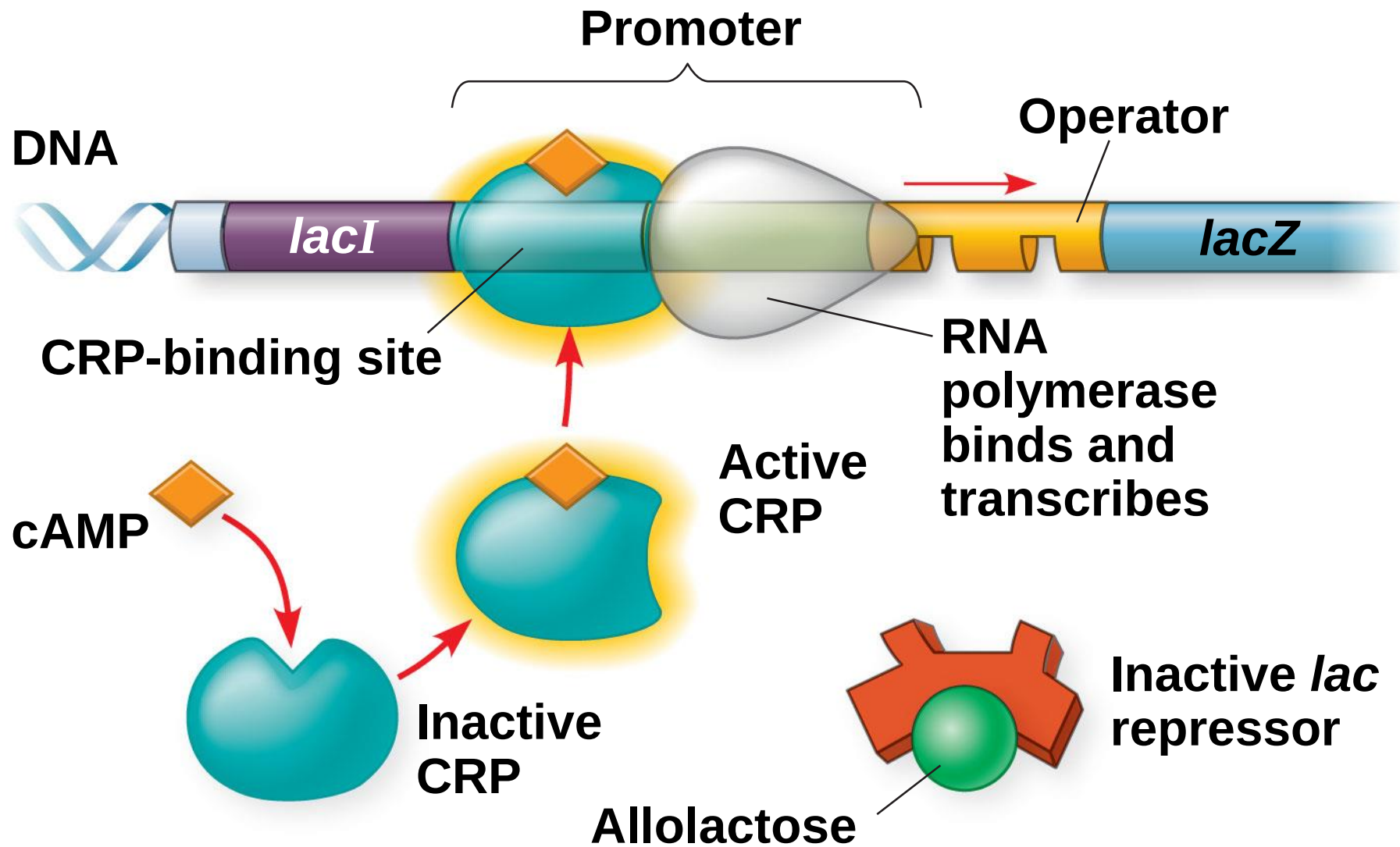


(a) Lactose present, glucose scarce (cAMP level high): abundant *lac* mRNA synthesized.



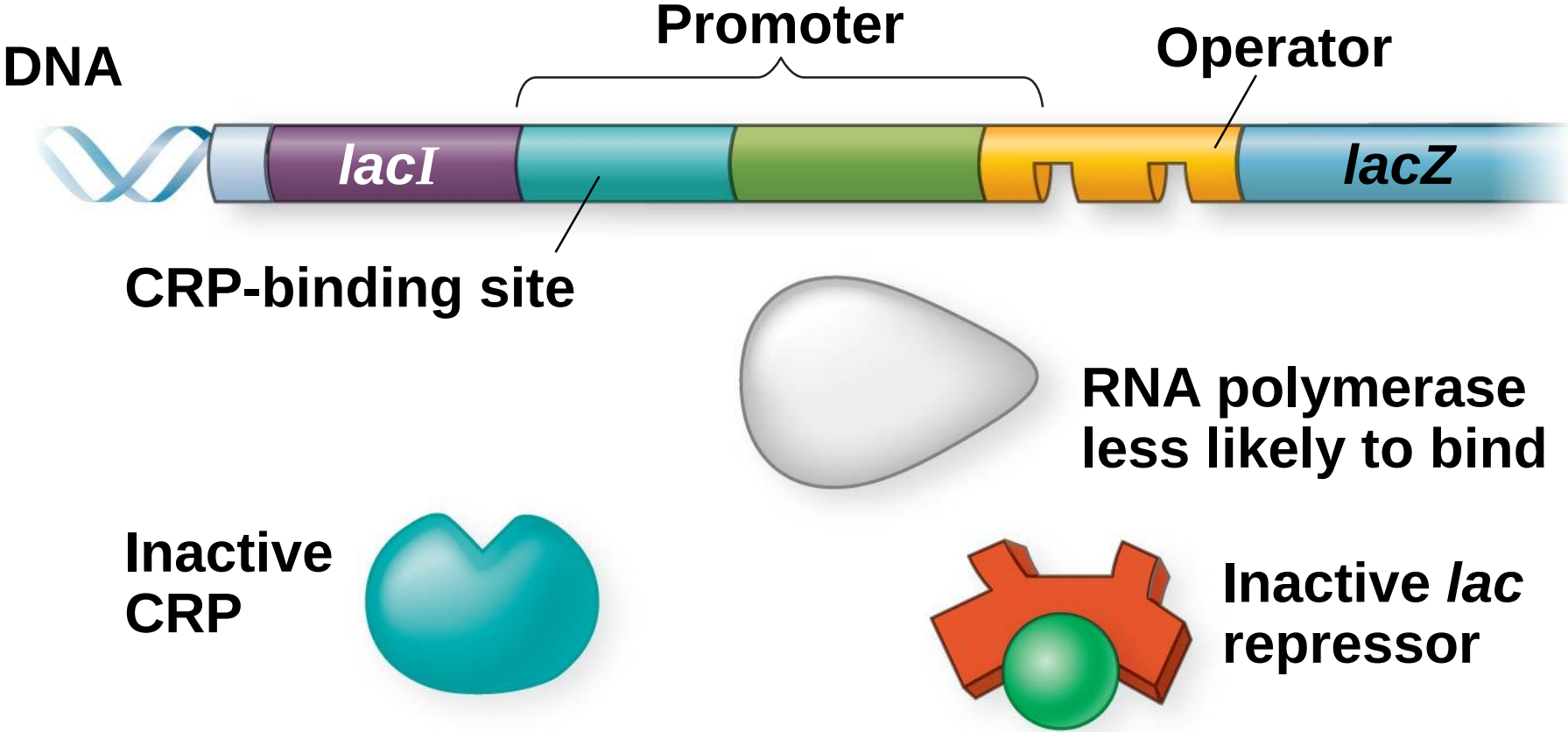
(b) Lactose present, glucose present (cAMP level low): little *lac* mRNA synthesized.

Figure 18.5a



(a) Lactose present, glucose scarce (cAMP level high): abundant *lac* mRNA synthesized.

Figure 18.5b



**(b) Lactose present, glucose present (cAMP level low):
little *lac* mRNA synthesized.**

Concept 18.2: Eukaryotic gene expression is regulated at many stages

- All organisms must regulate which genes are expressed at any given time
- Genes are turned on and off in response to signals from their external and internal environments
- In multicellular organisms, regulation of gene expression is essential for cell specialization

Differential Gene Expression

- Almost all the cells in an organism contain an identical genome
- Differences between cell types result from **differential gene expression**, the expression of different genes by cells with the same genome
- Abnormalities in gene expression can lead to diseases including cancer
- Gene expression is regulated at many stages, but is often equated with transcription

Figure 18.6

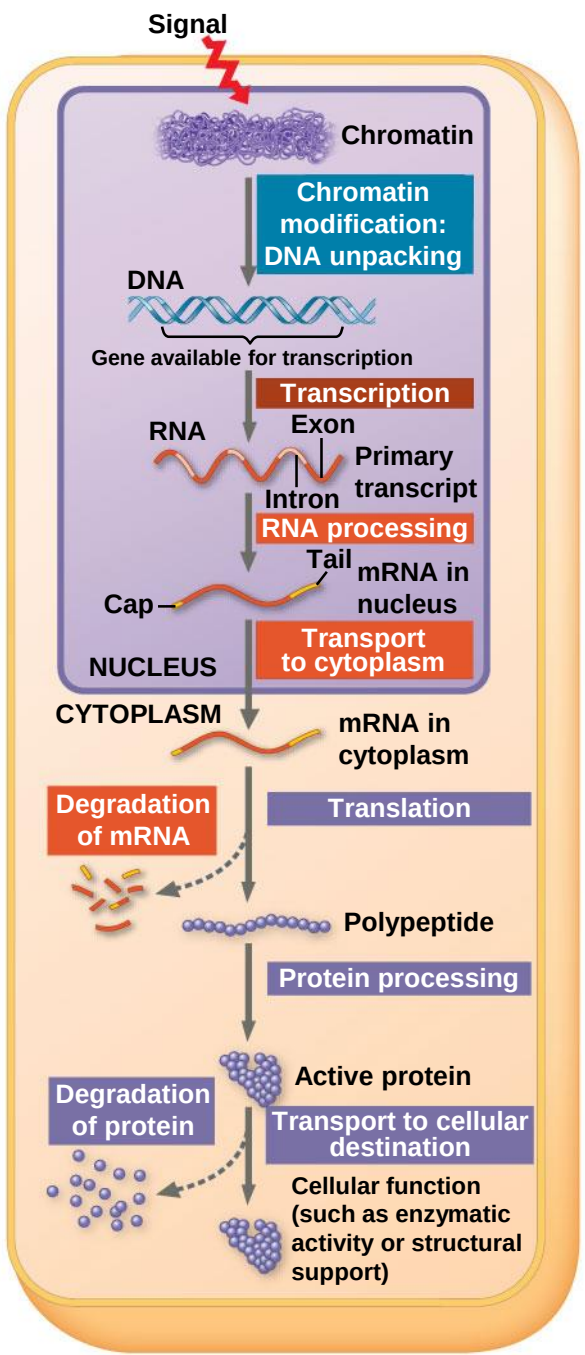


Figure 18.6a

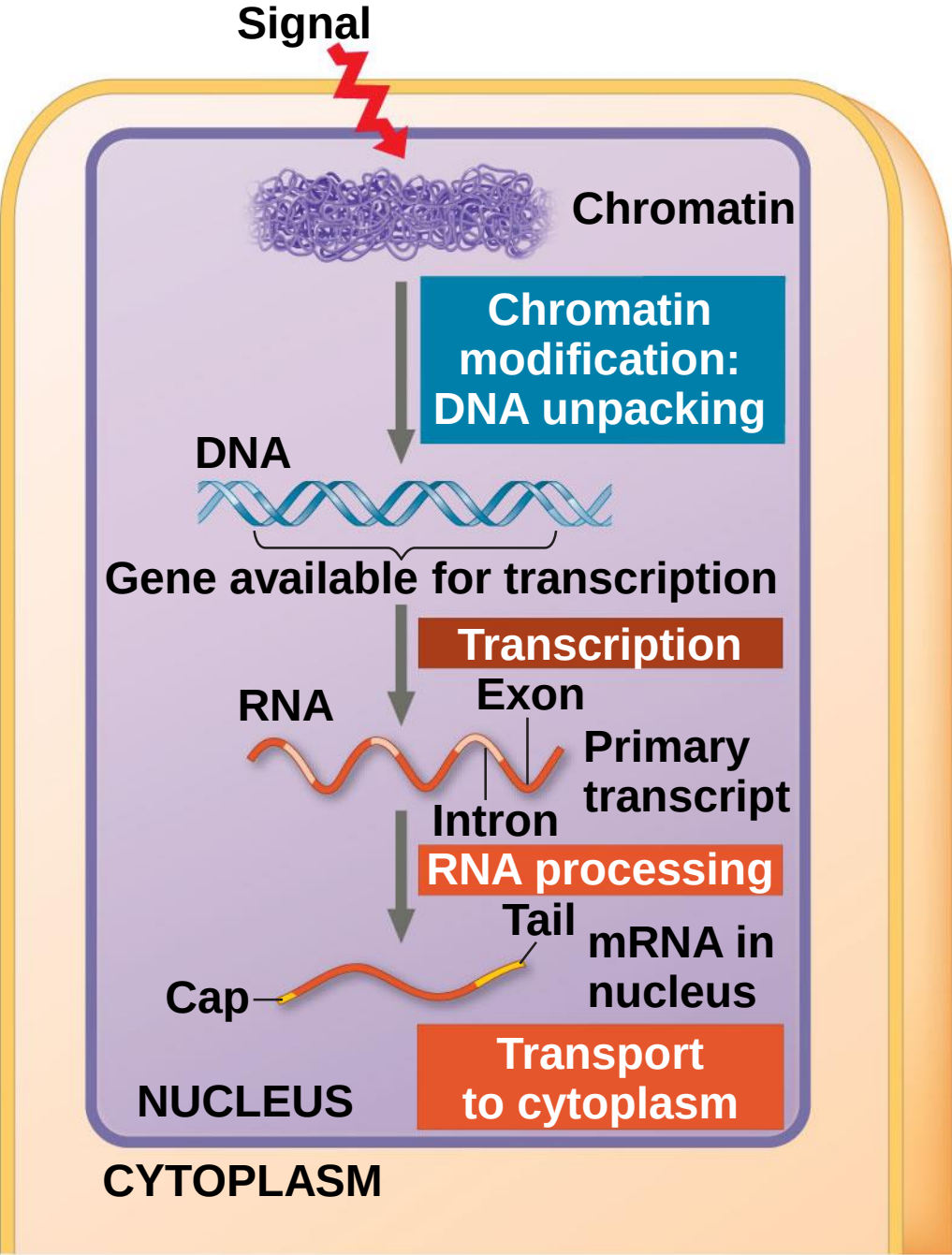
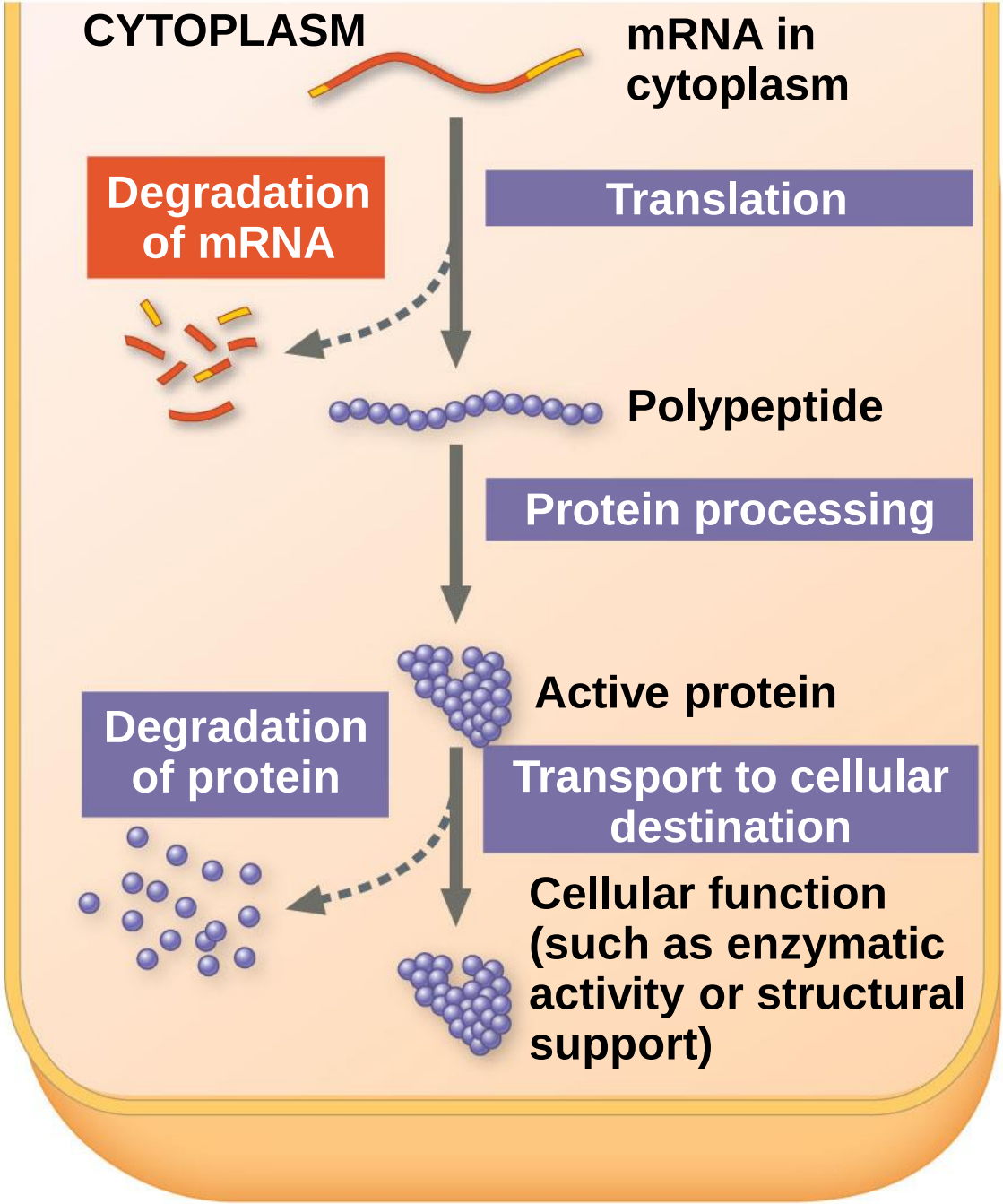
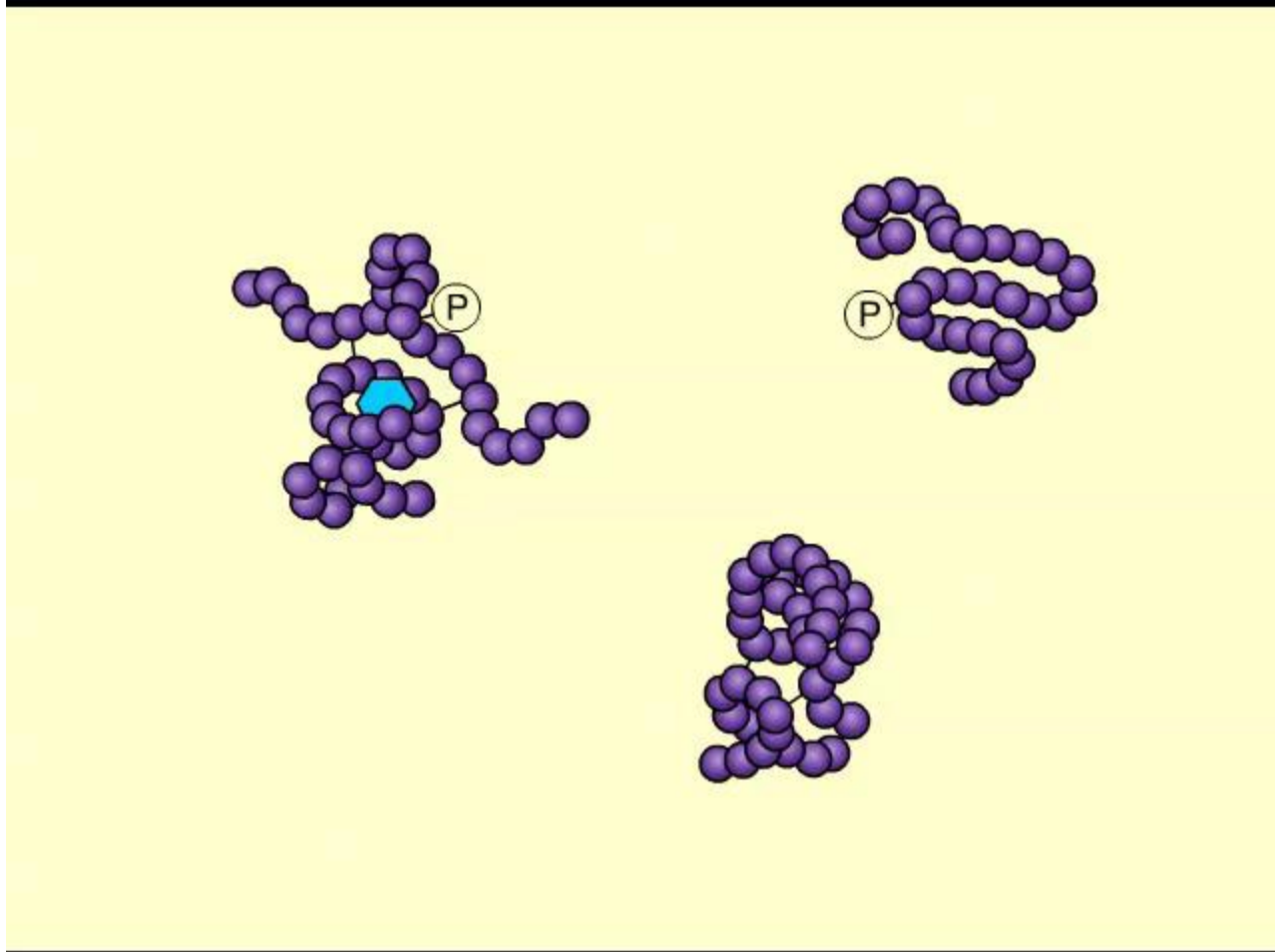


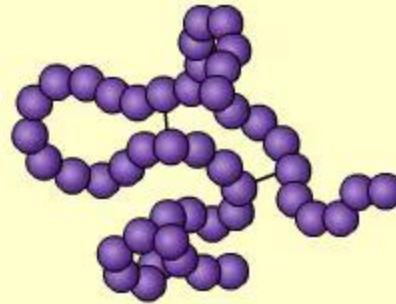
Figure 18.6b



Animation: Protein Degradation



Animation: Protein Processing



Polypeptide

Animation: Blocking Translation



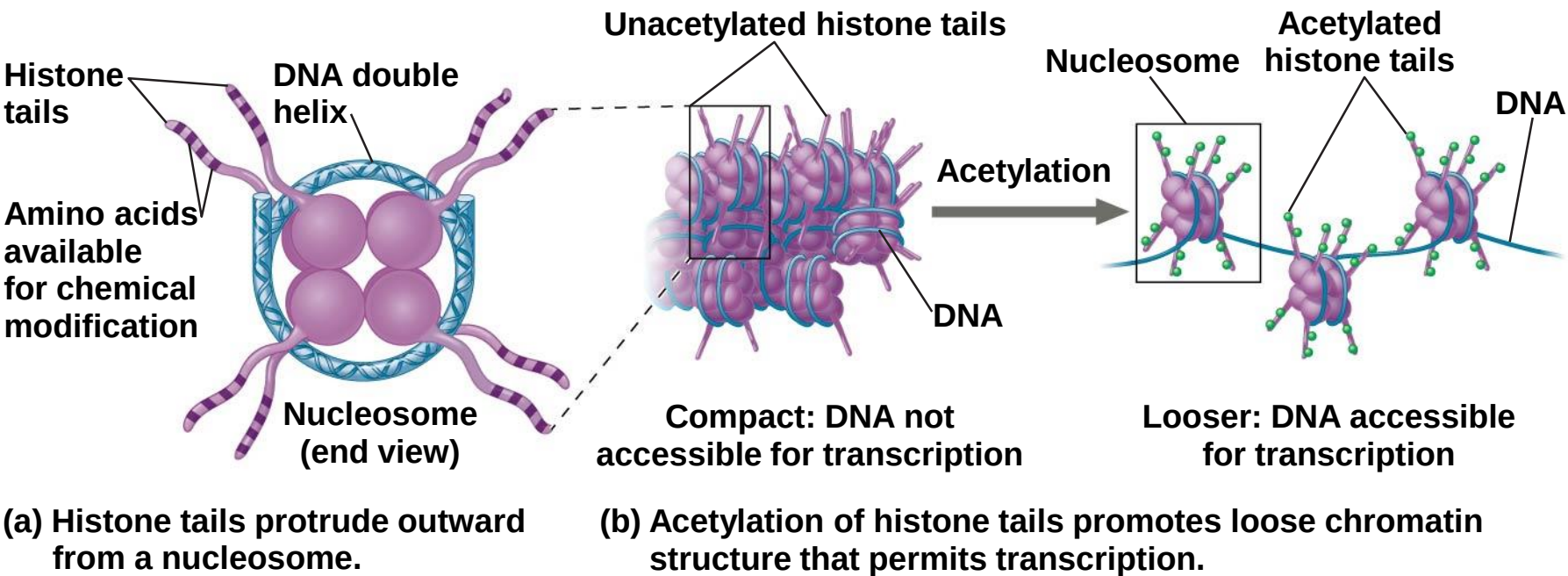
Regulation of Chromatin Structure

- The structural organization of chromatin helps regulate gene expression in several ways
- Genes within highly packed heterochromatin are usually not expressed
- Chemical modifications to histones and DNA of chromatin influence both chromatin structure and gene expression

Histone Modifications and DNA Methylation

- In **histone acetylation**, acetyl groups are attached to an amino acid in a histone tail
- This appears to open up the chromatin structure, thereby promoting the initiation of transcription
- The addition of methyl groups (methylation) can condense chromatin and reduce transcription

Figure 18.7



- **DNA methylation**, the addition of methyl groups to certain bases in DNA, is associated with reduced transcription in some species
- DNA methylation can cause long-term inactivation of genes in cellular differentiation
- In genomic imprinting, methylation regulates expression of either the maternal or paternal alleles of certain genes at the start of development

Epigenetic Inheritance

- Although the chromatin modifications just discussed do not alter DNA sequence, they may be passed to future generations of cells
- The inheritance of traits transmitted by mechanisms not directly involving the nucleotide sequence is called **epigenetic inheritance**

Regulation of Transcription Initiation

- Chromatin-modifying enzymes provide initial control of gene expression by making a region of DNA either more or less able to bind the transcription machinery

Organization of a Typical Eukaryotic Gene and Its Transcript

- Associated with most eukaryotic genes are multiple **control elements**, segments of noncoding DNA that serve as binding sites for transcription factors that help regulate transcription
- Control elements and the transcription factors they bind are critical to the precise regulation of gene expression in different cell types

Figure 18.8

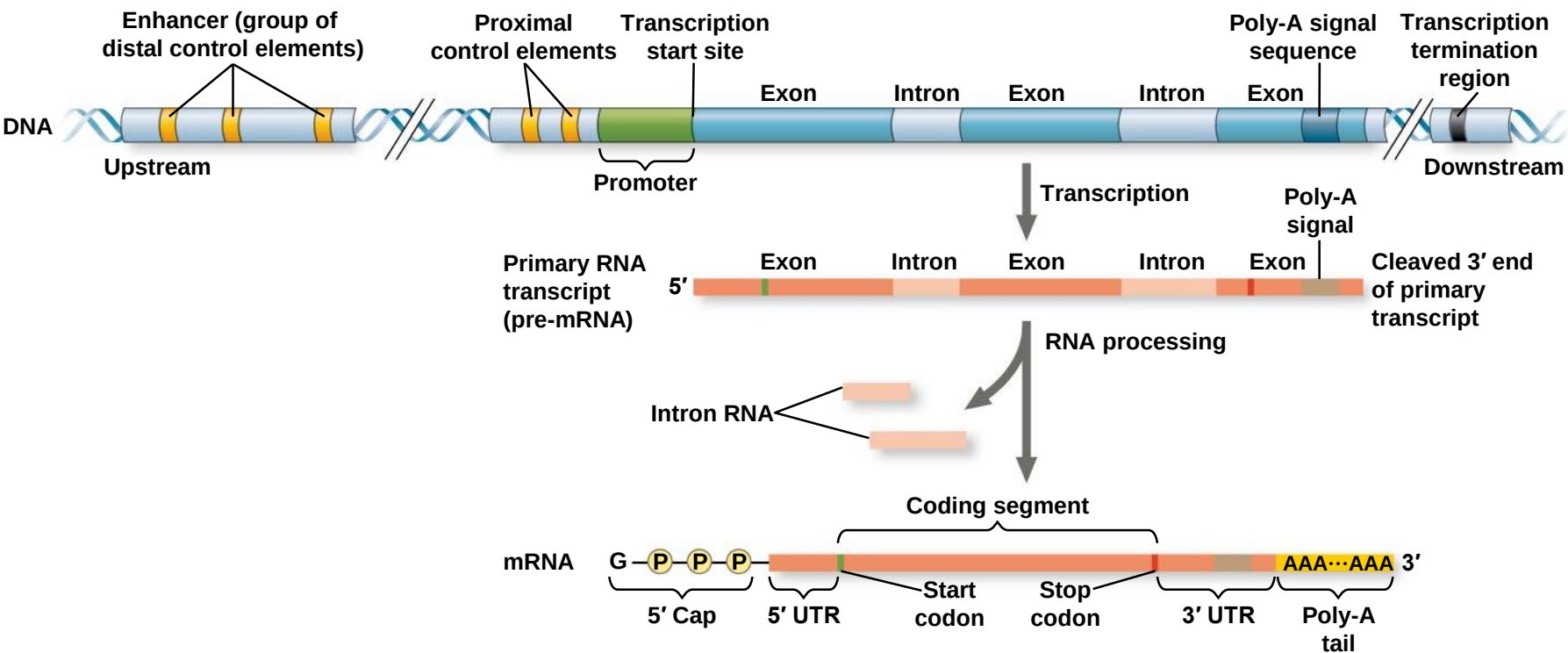


Figure 18.8a

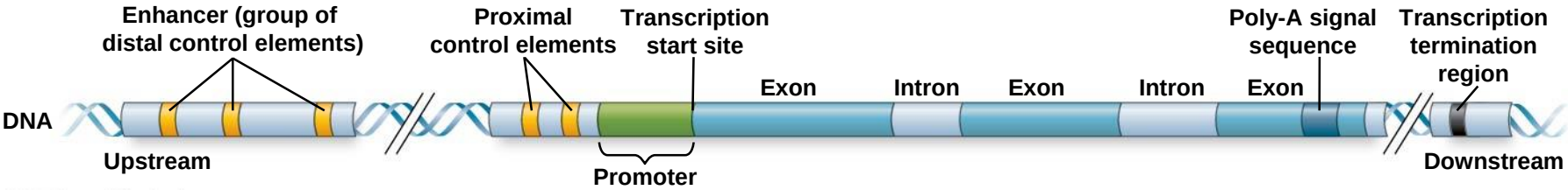


Figure 18.8b_1

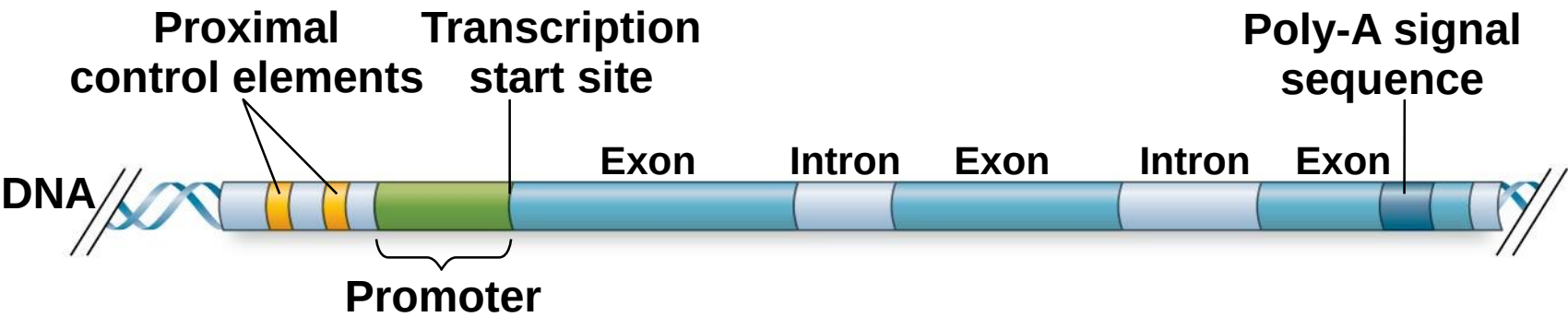


Figure 18.8b_2

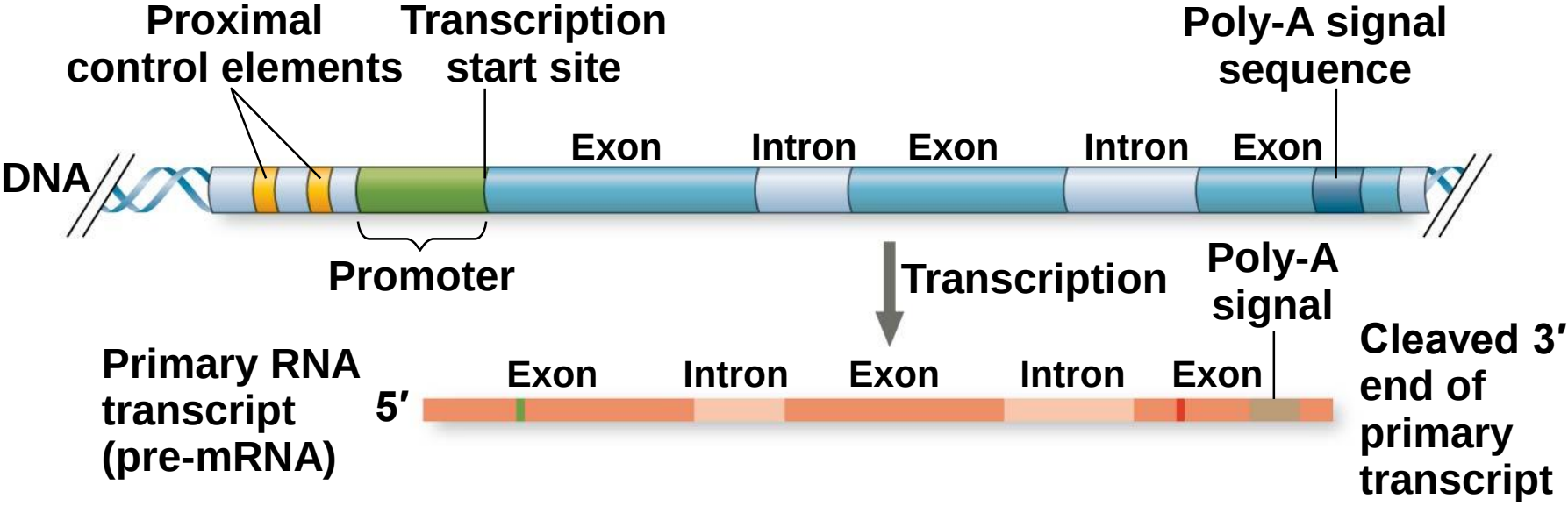
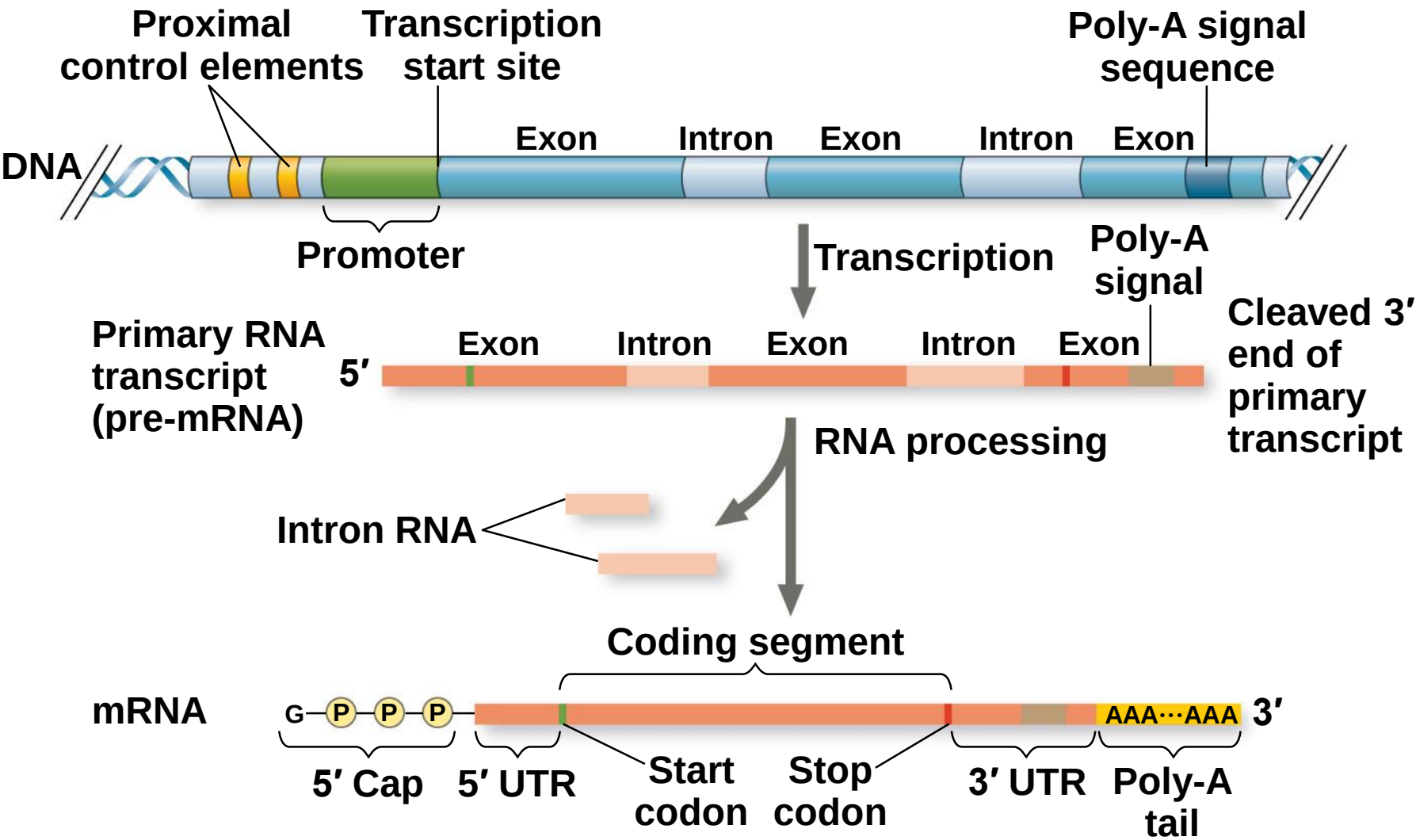


Figure 18.8b_3



Animation: mRNA Degradation



Longer-lived mRNA



Shorter-lived mRNA

The Roles of General and Specific Transcription Factors

- General transcription factors are essential for the transcription of all protein-coding genes
- In eukaryotes, high levels of transcription of particular genes depend on control elements interacting with specific transcription factors

General Transcription Factors at the Promoter

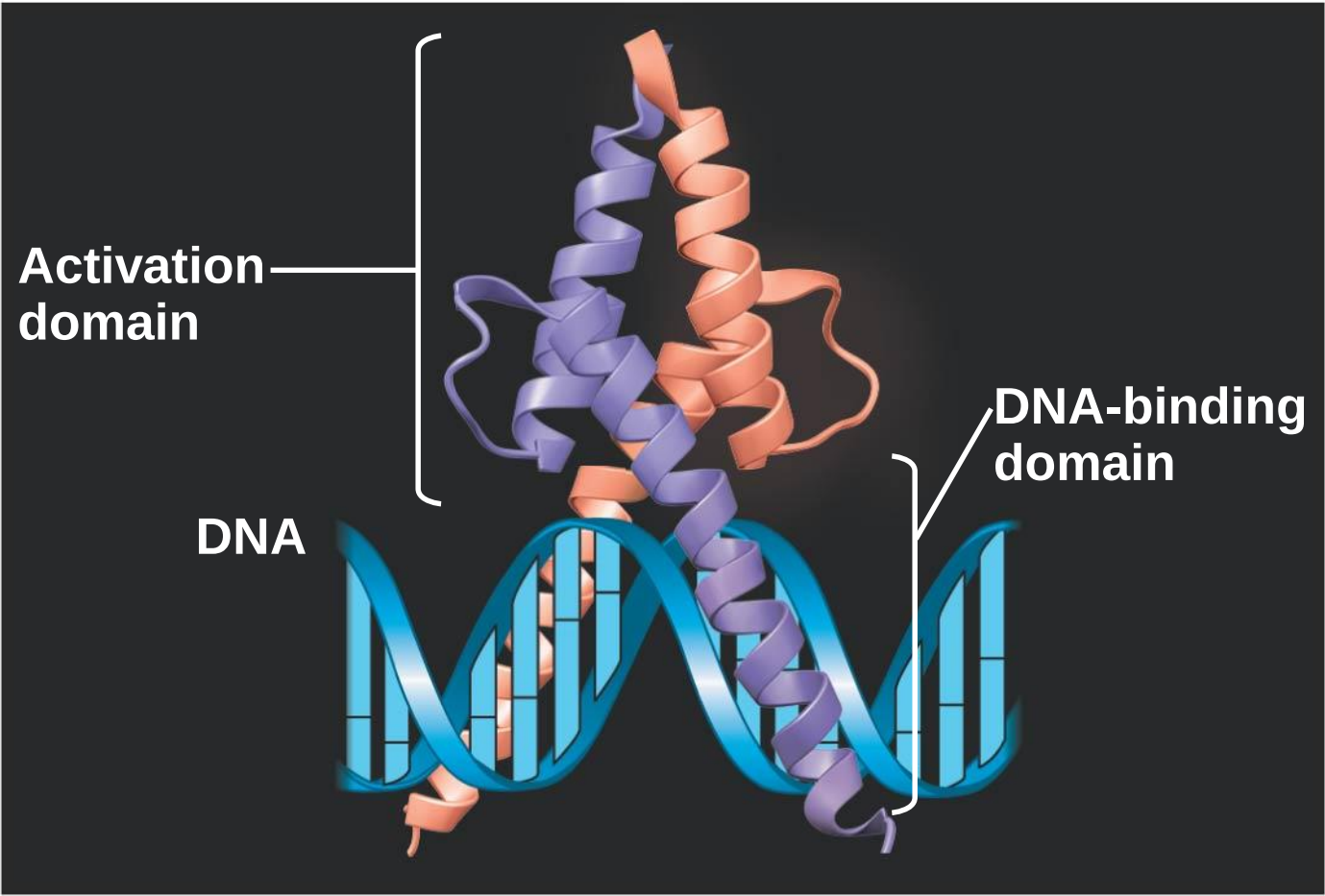
- RNA polymerase requires the assistance of transcription factors to initiate transcription
- General transcription factors are essential for the transcription of all protein-coding genes
- A few bind to the TATA box within the promoter
- Many bind to proteins, including other transcription factors and RNA polymerase II

- Only when the complete initiation complex has assembled can the RNA polymerase begin to move along the template strand of the DNA
- It produces a complementary strand of RNA
- For genes that are not expressed all the time, high levels of transcription depend on the presence of another set of factors, specific transcription factors

Enhancers and Specific Transcription Factors

- Proximal control elements are located close to the promoter
- Distal control elements, groupings of which are called **enhancers**, may be far away from a gene or even located in an intron
- Each enhancer is generally associated with only one gene and no other

Figure 18.9



- An activator is a protein that binds to an enhancer and stimulates transcription of a gene
- Activators have two domains, one that binds DNA and a second that activates transcription
- Bound activators facilitate a sequence of protein-protein interactions that result in transcription of a given gene

- The currently accepted model suggests that protein-mediated bending of the DNA brings the bound activators into contact with a group of mediator proteins
- The mediator proteins interact with general transcription factors at the promoter
- This helps assemble and position the preinitiation complex

Figure 18.10_1



Figure 18.10_2

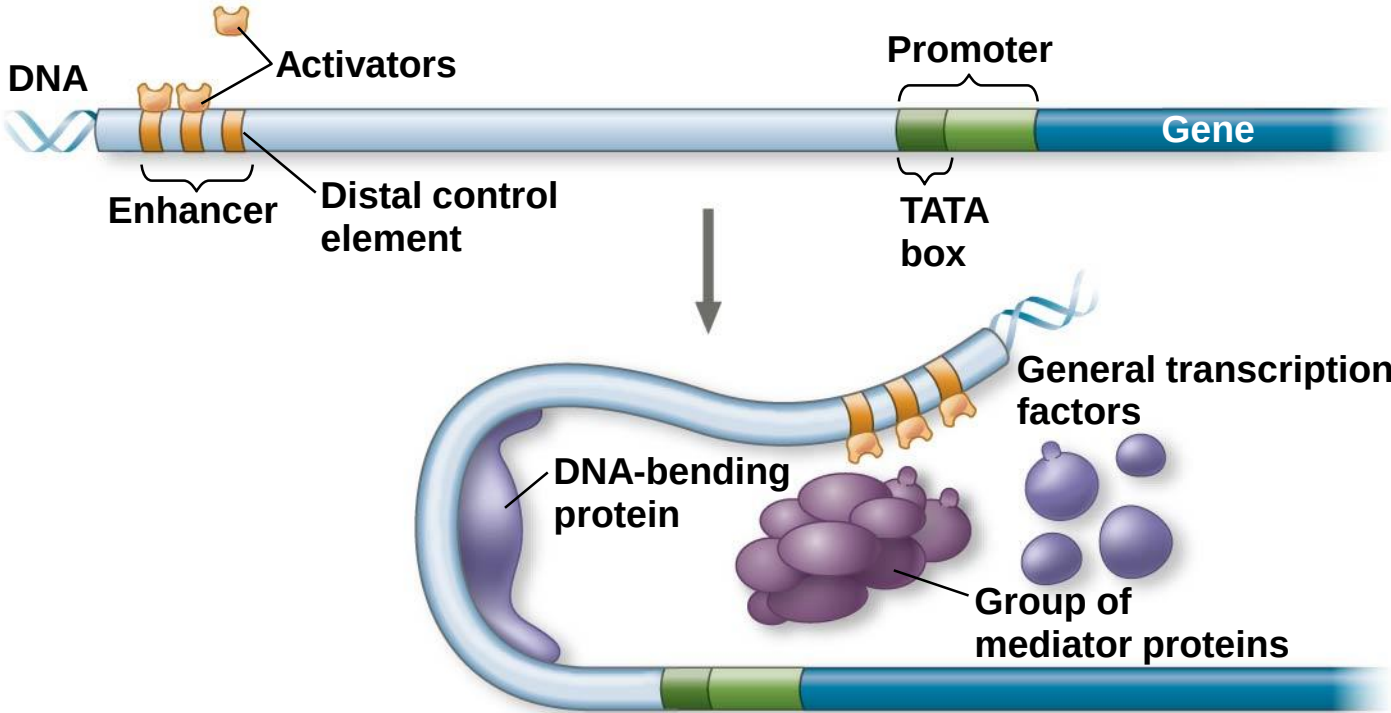
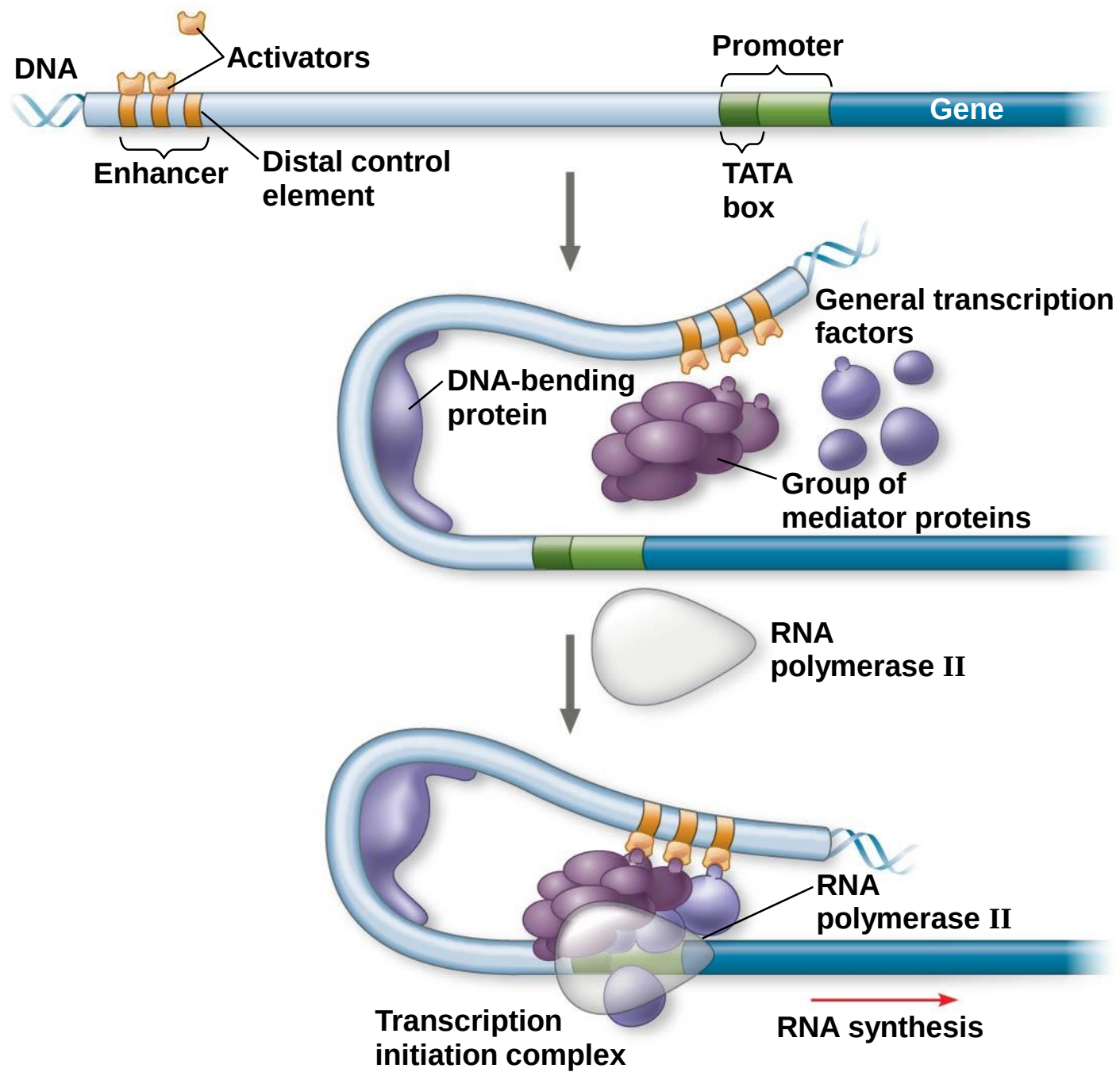
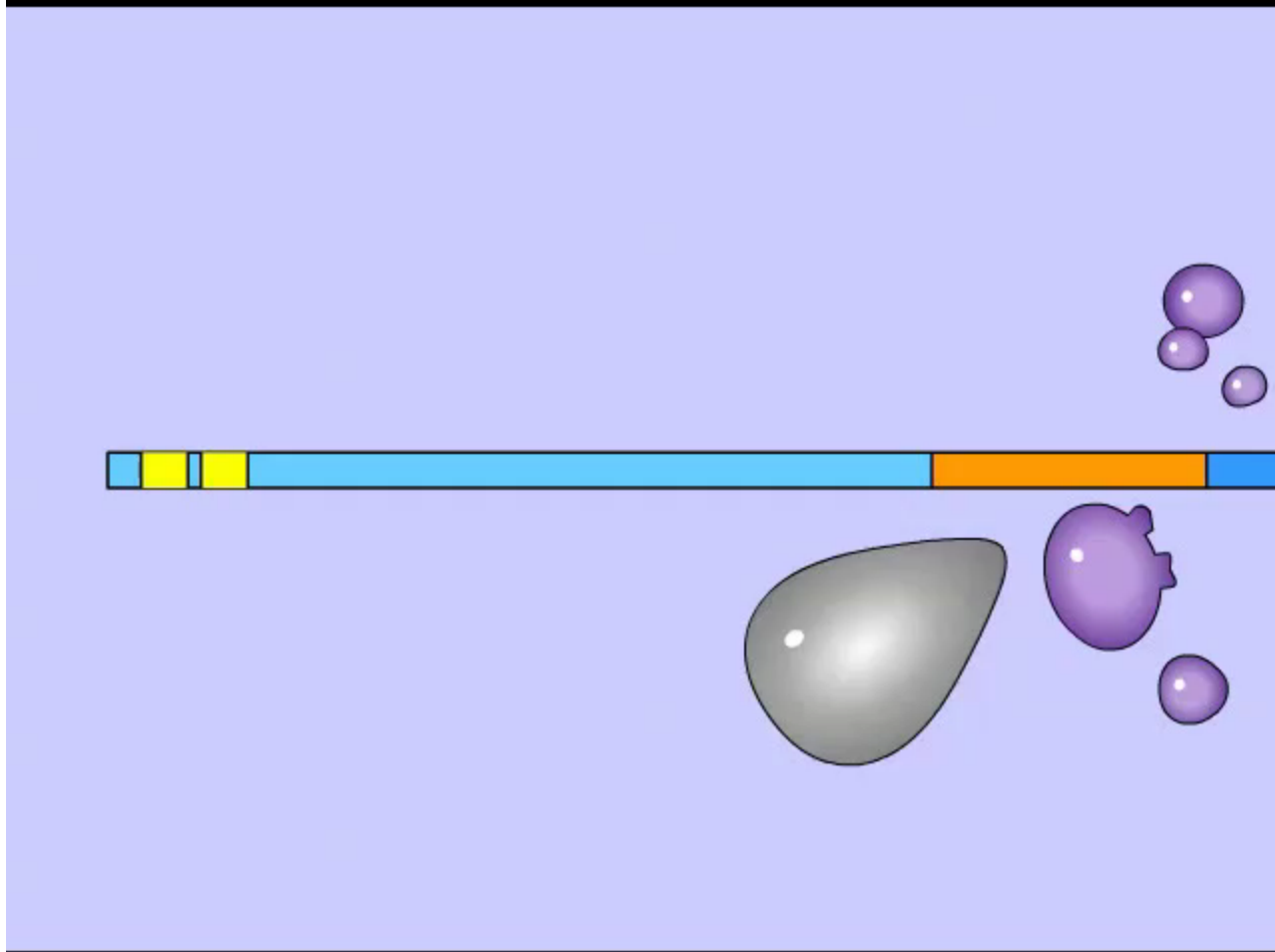


Figure 18.10_3



Animation: Initiation of Transcription



- Some transcription factors function as repressors, inhibiting expression of a particular gene in several different ways
- Some activators and repressors act indirectly by influencing chromatin structure to promote or silence transcription

Combinatorial Control of Gene Activation

- A particular combination of control elements can activate transcription only when the appropriate activator proteins are present
- With only a dozen or so control elements, a large number of potential combinations is possible

Figure 18.11

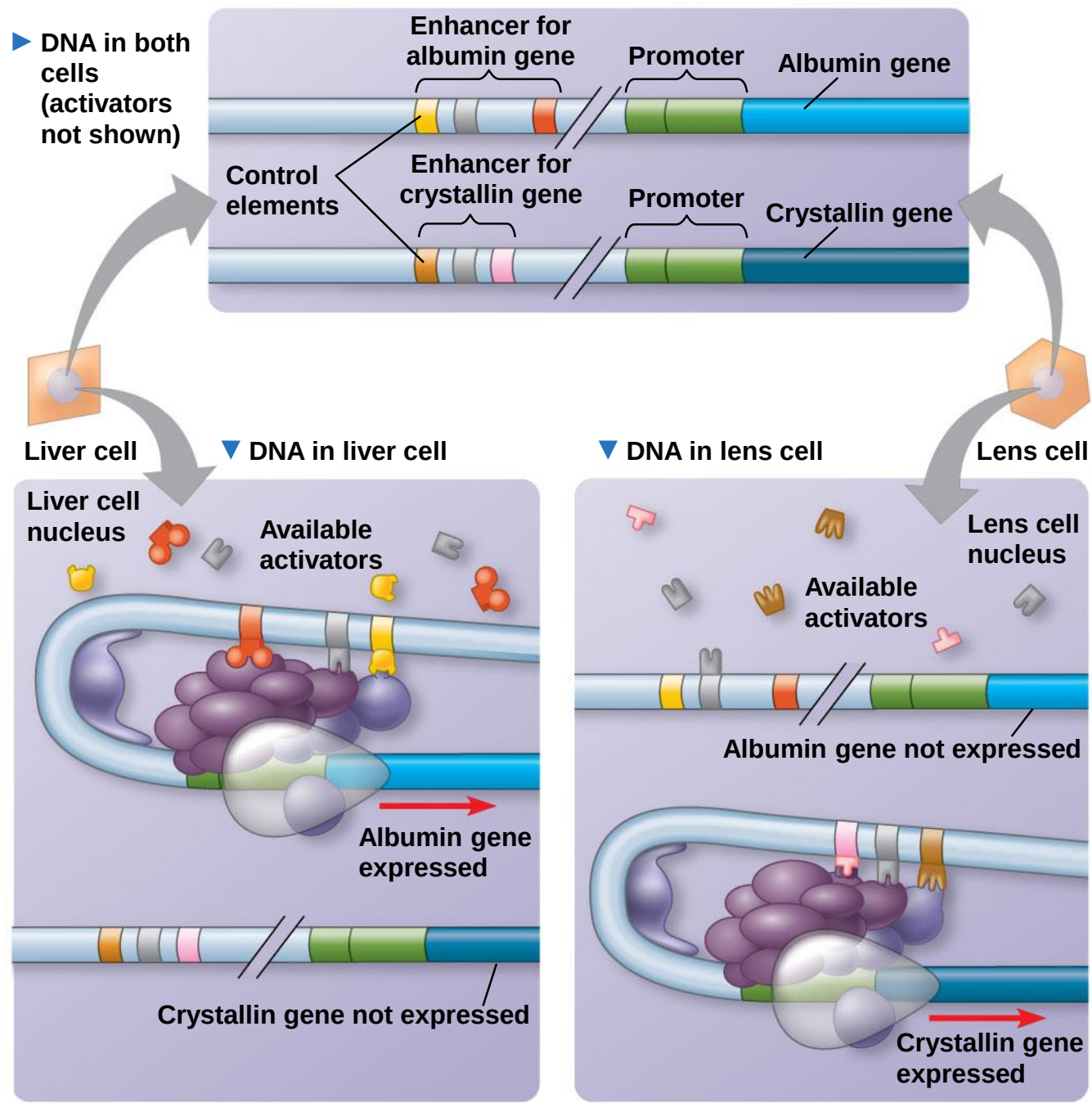


Figure 18.11a

► DNA in both cells
(activators not shown)

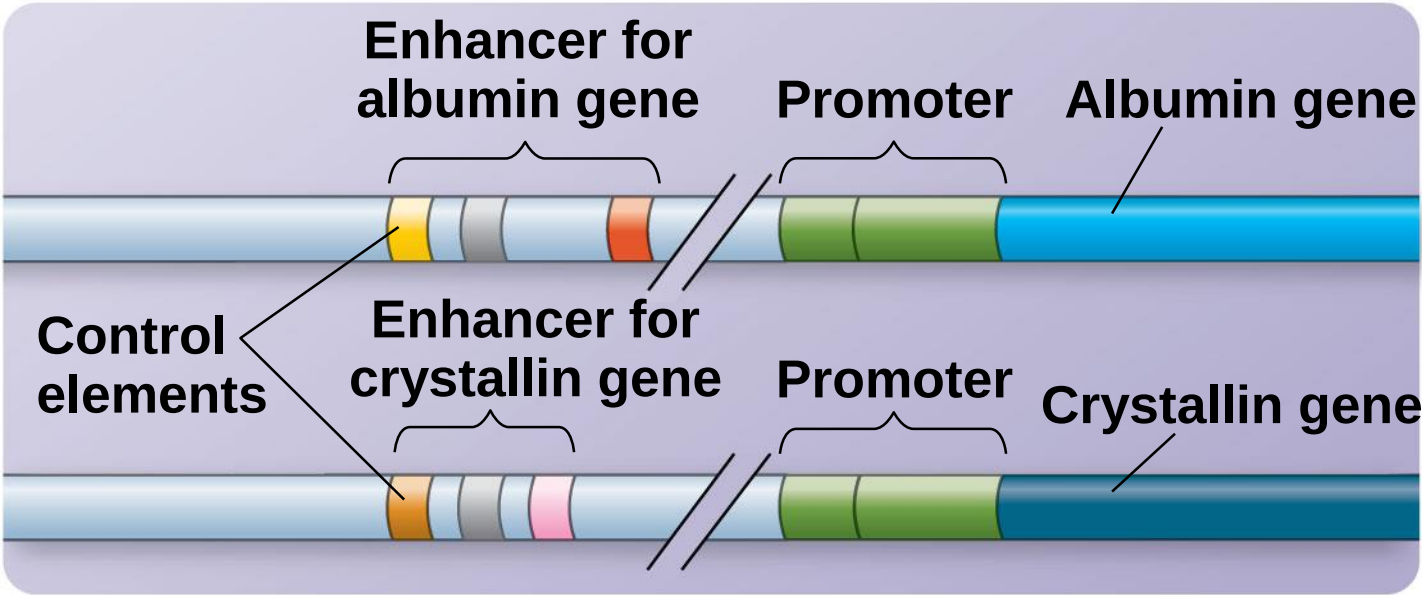


Figure 18.11b

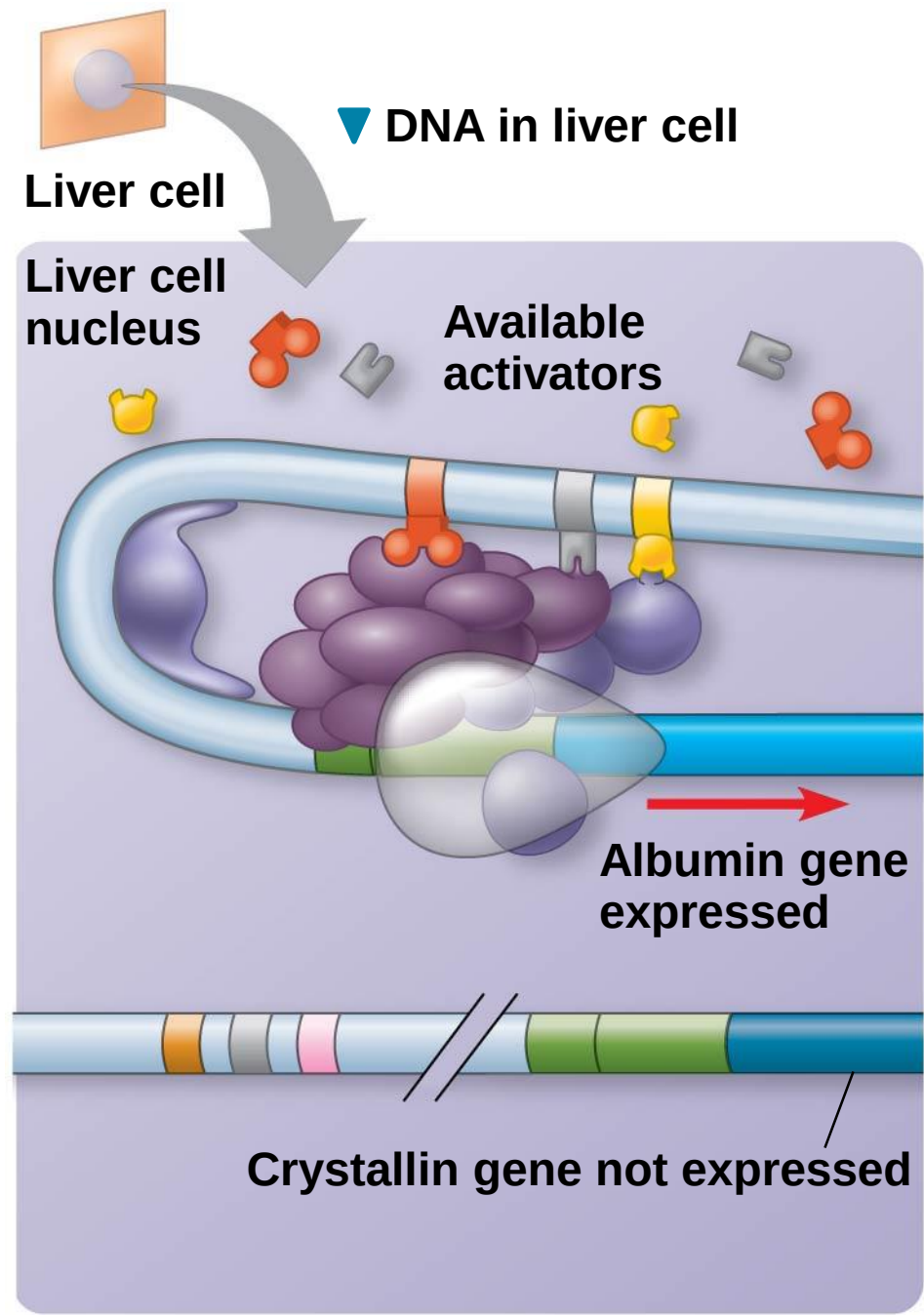
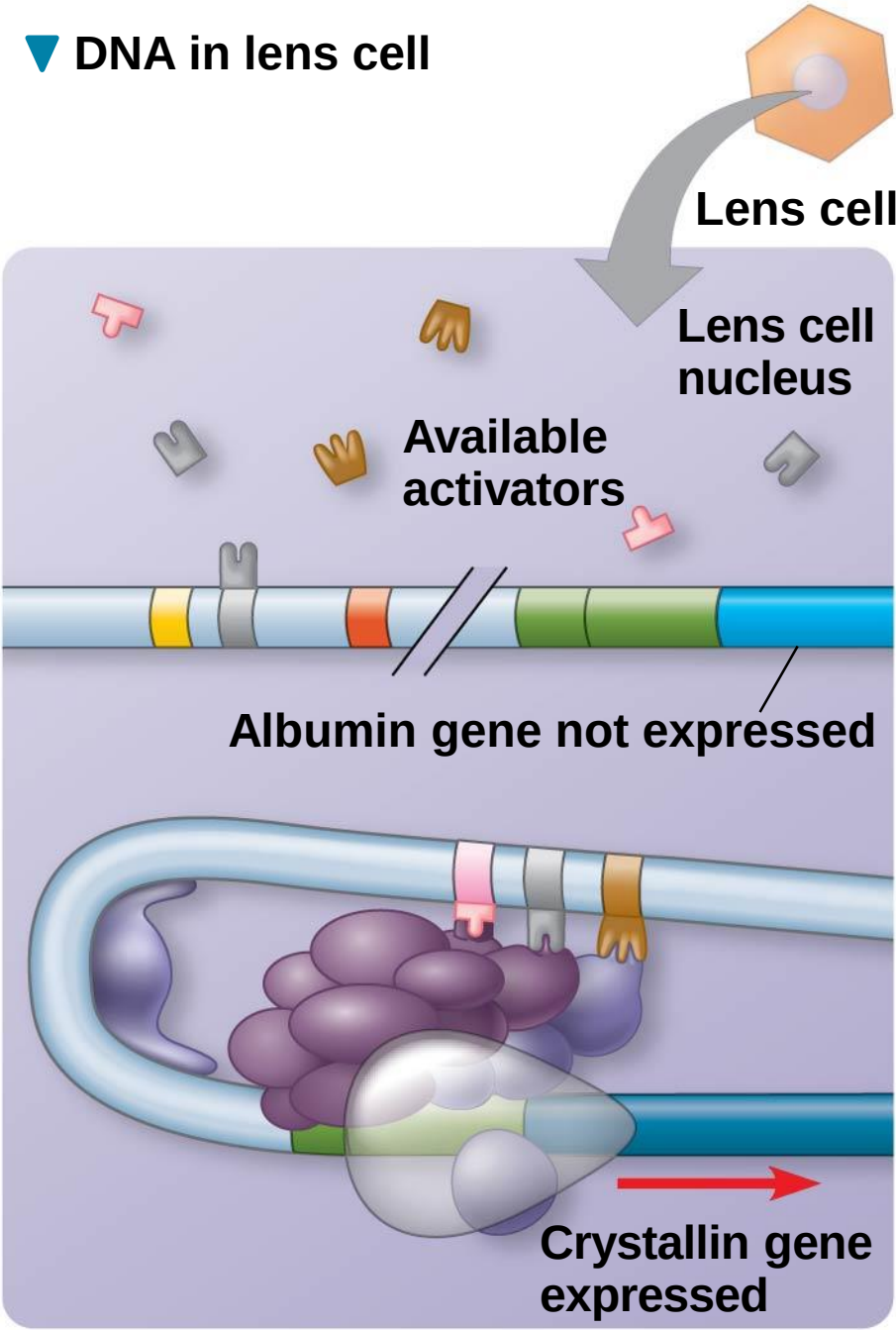


Figure 18.11c

▼ DNA in lens cell



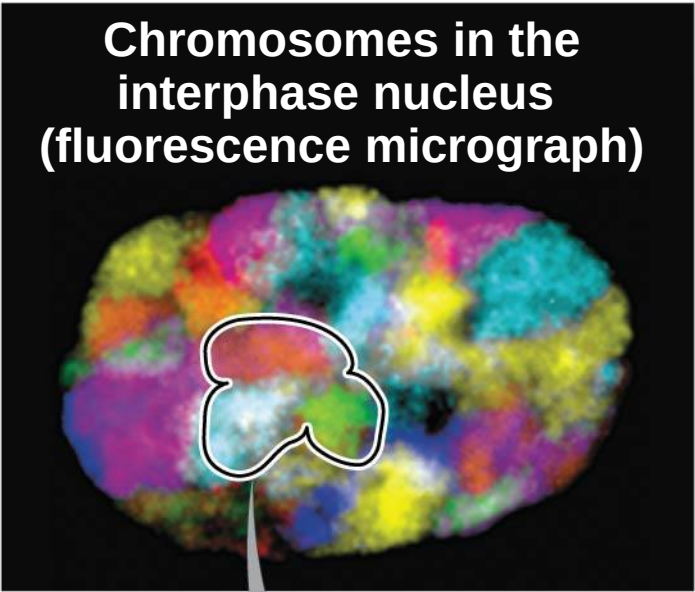
Coordinately Controlled Genes in Eukaryotes

- Co-expressed eukaryotic genes are not organized in operons (with a few exceptions)
- These genes can be scattered over different chromosomes, but each has the same combination of control elements
- Activator proteins in the nucleus recognize specific control elements and promote simultaneous transcription of the genes

Nuclear Architecture and Gene Expression

- Chromosome conformation capture techniques allow identification of regions of chromosomes that interact with each other
- Loops of chromatin from different chromosomes may congregate at particular sites, some of which are rich in transcription factors and RNA polymerases
- These transcription factories are thought to be areas specialized for a common function

Figure 18.12



5 μm

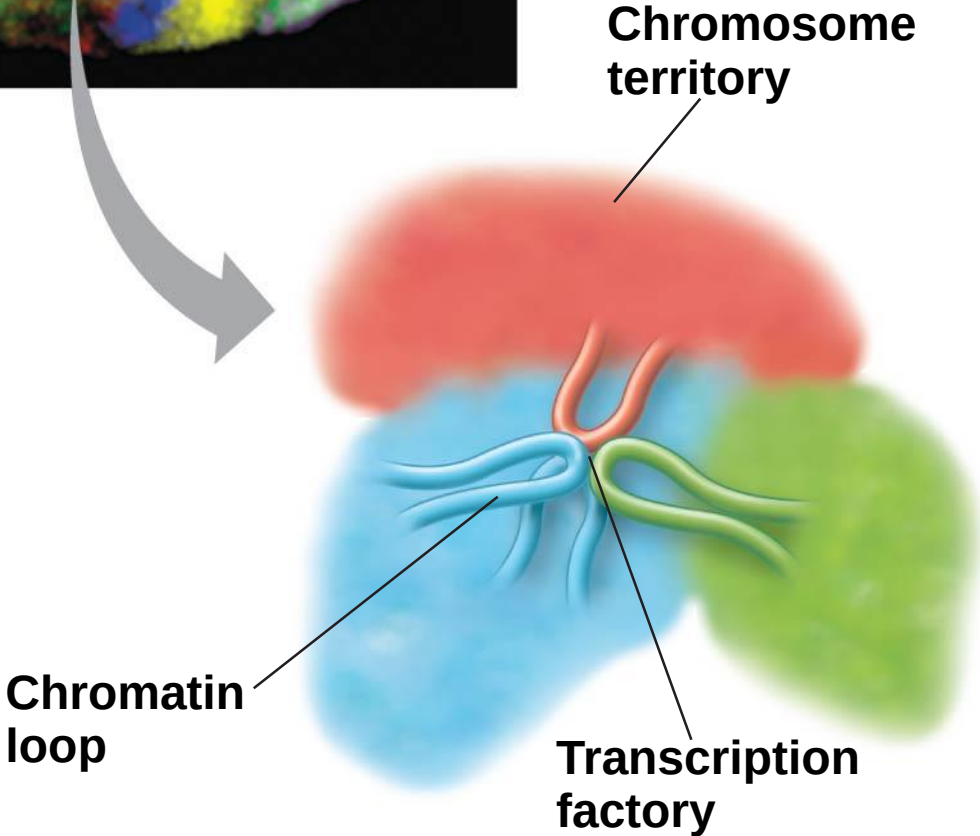
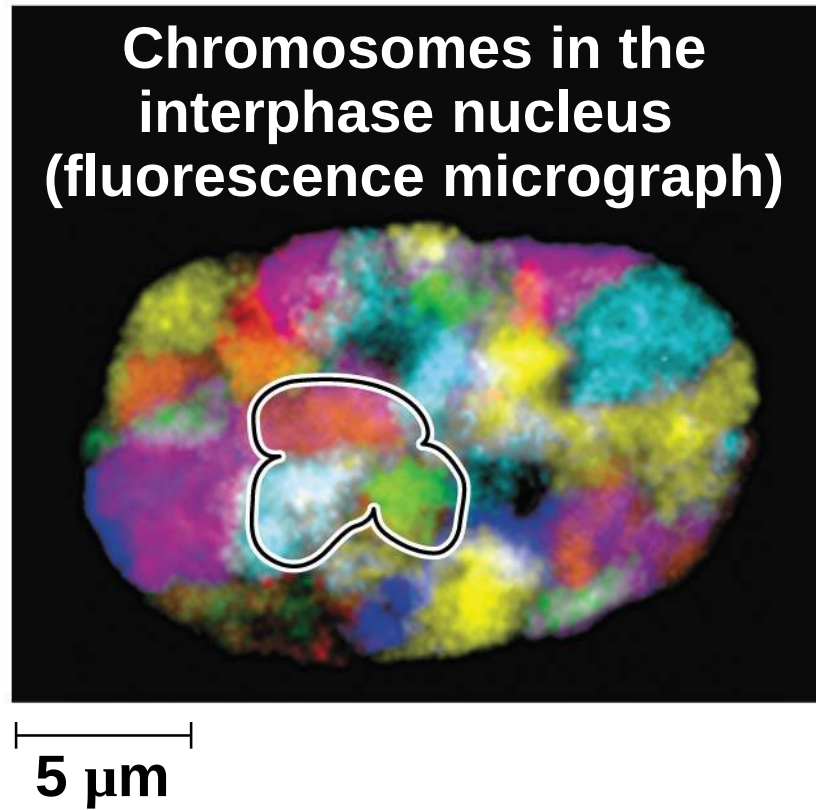


Figure 18.12a



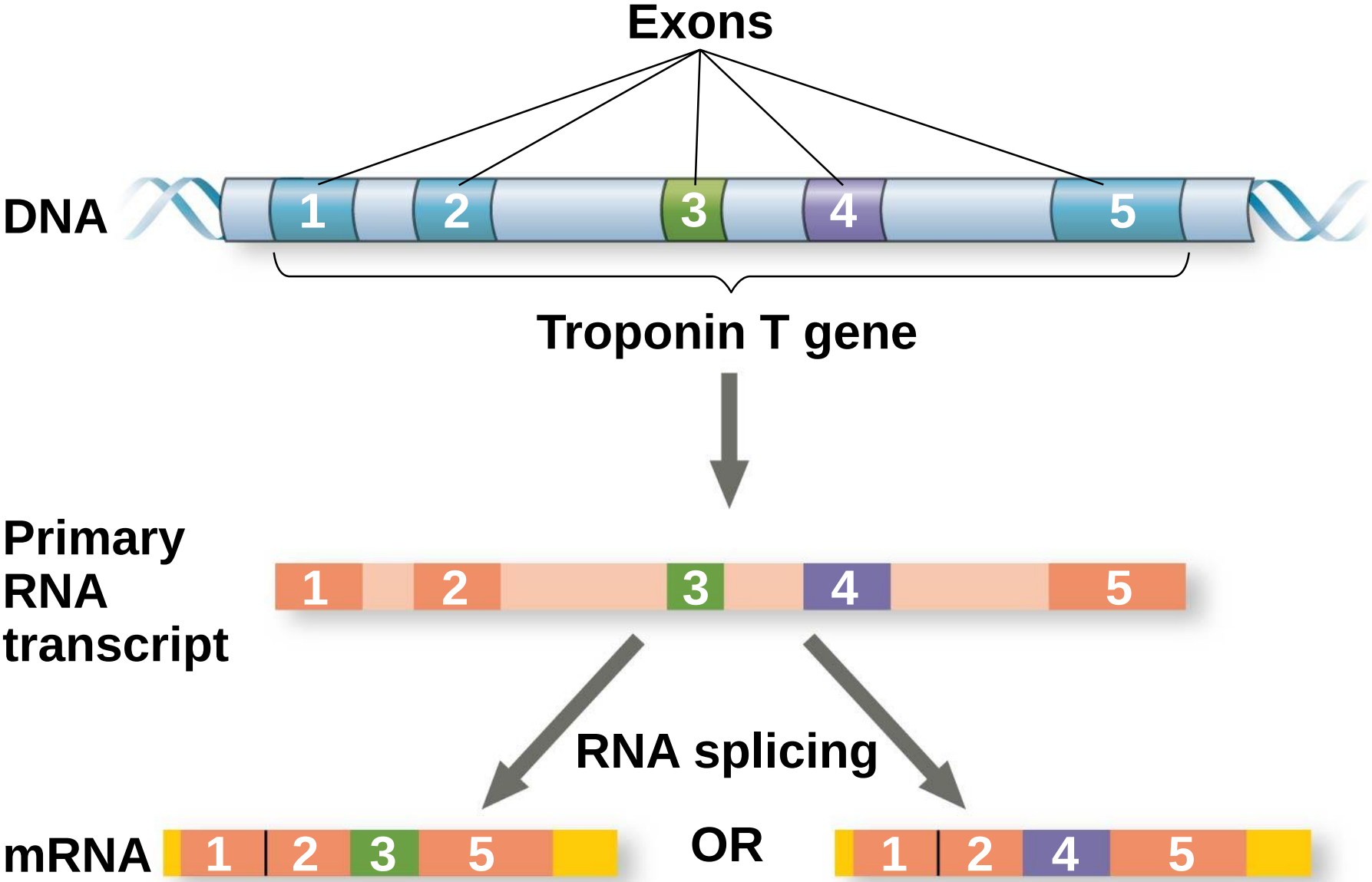
Mechanisms of Post-Transcriptional Regulation

- Transcription alone does not constitute gene expression
- Regulatory mechanisms can operate at various stages after transcription
- Such mechanisms allow a cell to fine-tune gene expression rapidly in response to environmental changes

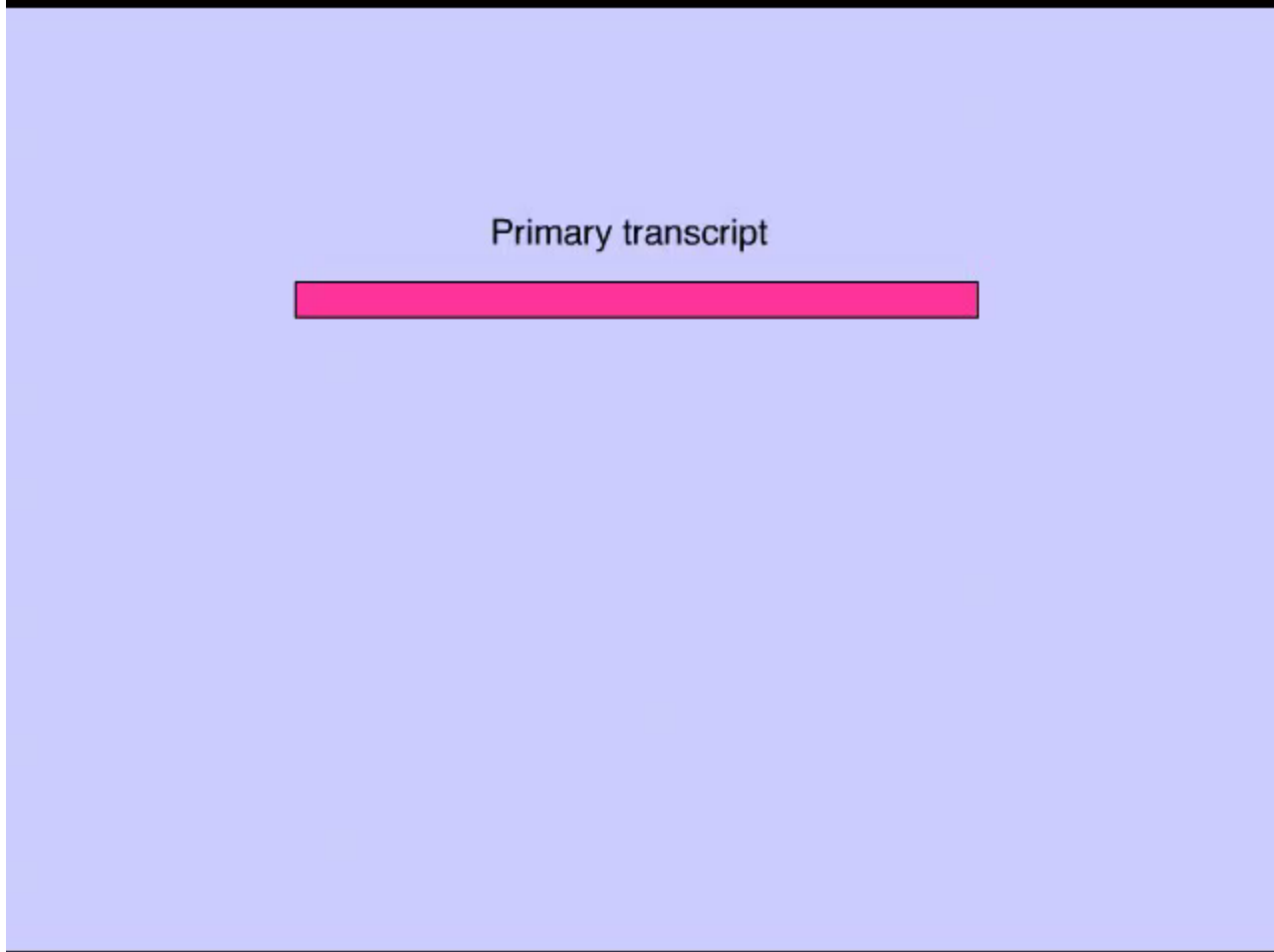
RNA Processing

- In **alternative RNA splicing**, different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as exons and which as introns
- Alternative RNA splicing can significantly expand the repertoire of a eukaryotic genome
- It is a proposed explanation for the surprisingly low number of genes in the human genome
- More than 90% of the human protein-coding genes undergo alternative splicing

Figure 18.13



Animation: RNA Processing



Initiation of Translation and mRNA Degradation

- The initiation of translation of selected mRNAs can be blocked by regulatory proteins that bind to sequences or structures of the mRNA
- Alternatively, translation of all mRNAs in a cell may be regulated simultaneously
- For example, translation initiation factors are simultaneously activated in an egg following fertilization

- The life span of mRNA molecules in the cytoplasm is important in determining the pattern of protein synthesis in a cell
- Eukaryotic mRNA is more long-lived than prokaryotic mRNA
- Nucleotide sequences that influence the life span of mRNA in eukaryotes reside in the untranslated region (UTR) at the 3' end of the molecule

Protein Processing and Degradation

- After translation, polypeptides undergo processing, including cleavage, and chemical modifications
- The length of time each protein functions is regulated by selective degradation
- Cells mark proteins for degradation by attaching ubiquitin to them
- This mark is recognized by proteasomes, which recognize and degrade the proteins

Concept 18.3: Noncoding RNAs play multiple roles in controlling gene expression

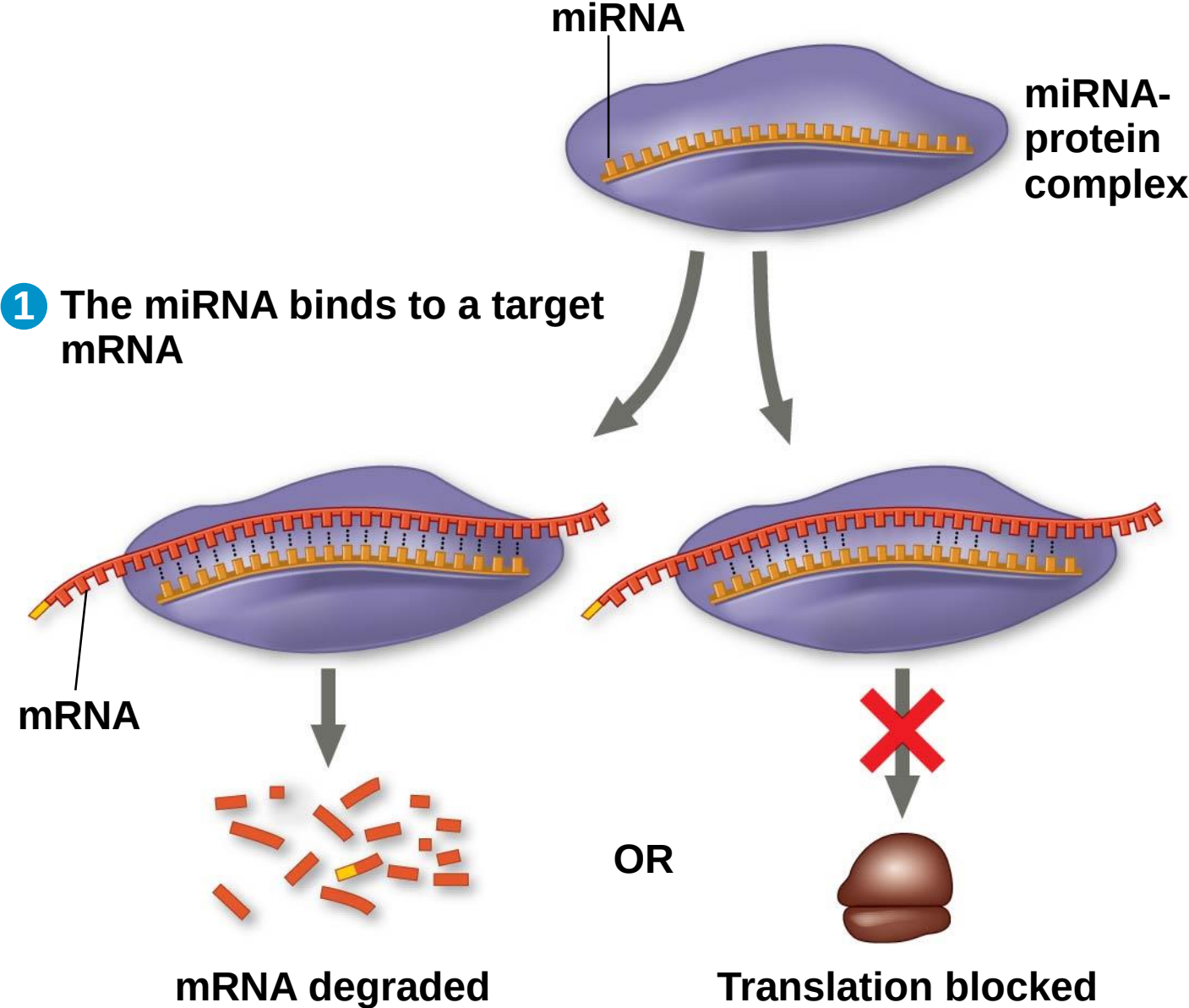
- A small fraction of DNA codes for proteins, and a very small fraction of the non-protein-coding DNA consists of genes for RNA such as rRNA and tRNA
- In the past, genes that did not encode a protein product or known functional RNA were considered “junk DNA”
- A flood of recent data has contradicted this idea

- A significant fraction of the genome may be transcribed into noncoding RNAs (ncRNAs)
- Researchers are uncovering more evidence of biological roles for these ncRNAs every day
- This represents a major shift in the the thinking of biologists

Effects on mRNAs by MicroRNAs and Small Interfering RNAs

- **MicroRNAs (miRNAs)** are small, single-stranded RNA molecules that can bind complementary sequences in mRNA
- The miRNAs and associated proteins cause degradation of the target mRNA or sometimes block its translation
- Biologists estimate that expression of at least one-half of human genes may be regulated by miRNAs

Figure 18.14



1 The miRNA binds to a target mRNA

2 If bases are complementary, mRNA is degraded (left); if the match is less complete, translation is blocked (right).

- **Small interfering RNAs (siRNAs)** are similar to miRNAs in size and function
- The blocking of gene expression by siRNAs is called **RNA interference (RNAi)**
- RNAi is used in the laboratory as a means of disabling genes to investigate their function

Chromatin Remodeling and Effects on Transcription by ncRNAs

- Some ncRNAs act to bring about remodeling of chromatin structure
- In some yeasts, siRNAs re-form heterochromatin at centromeres after chromosome replication

Figure 18.15

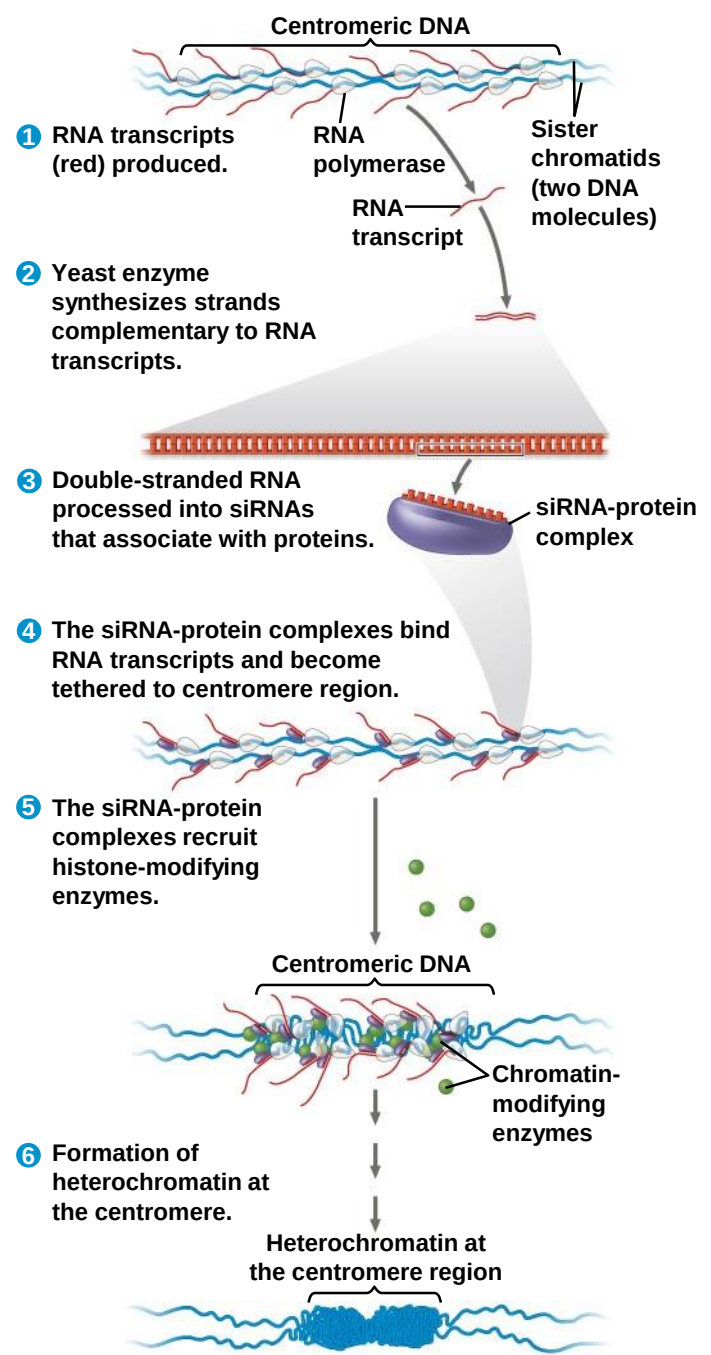
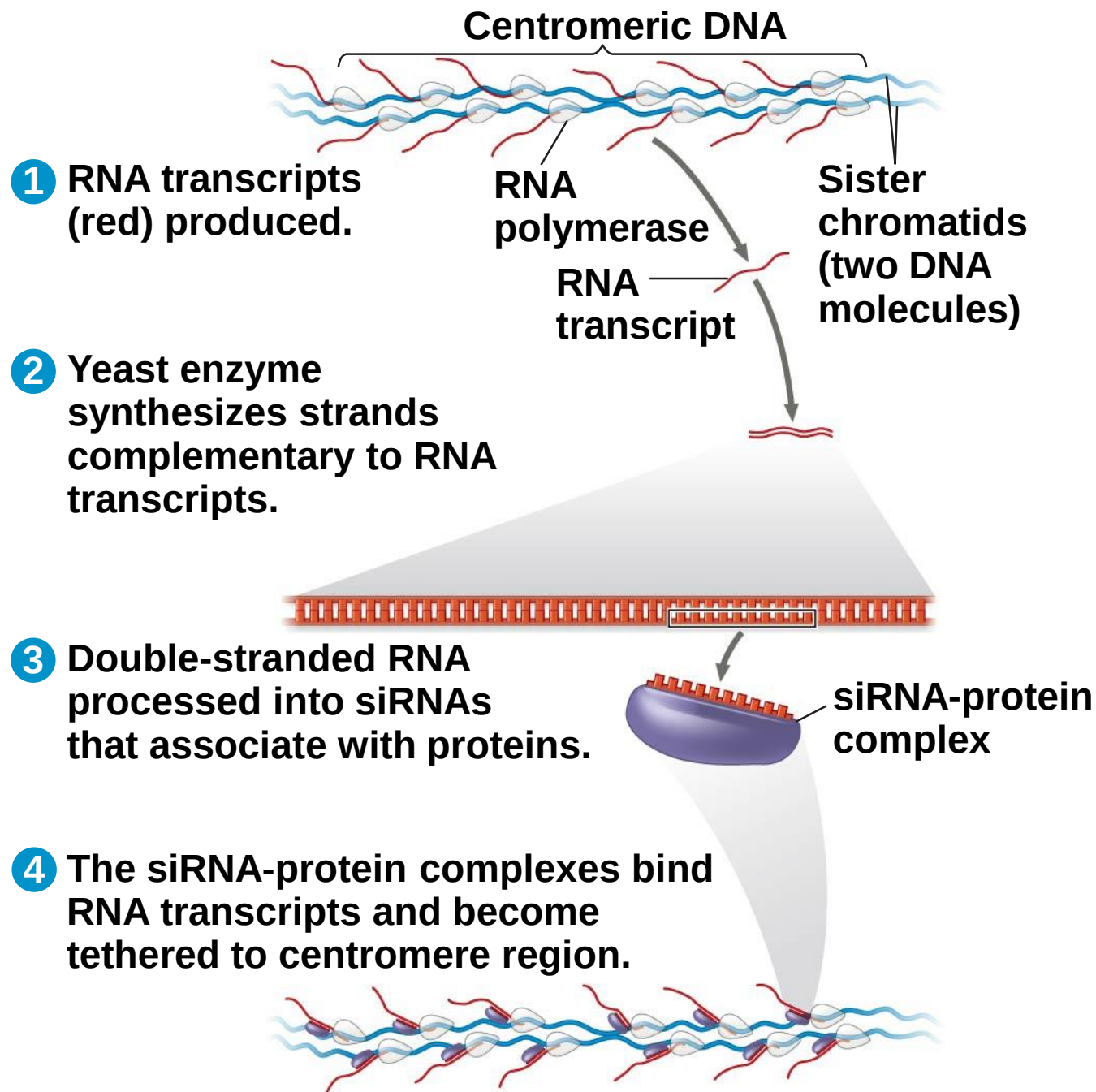
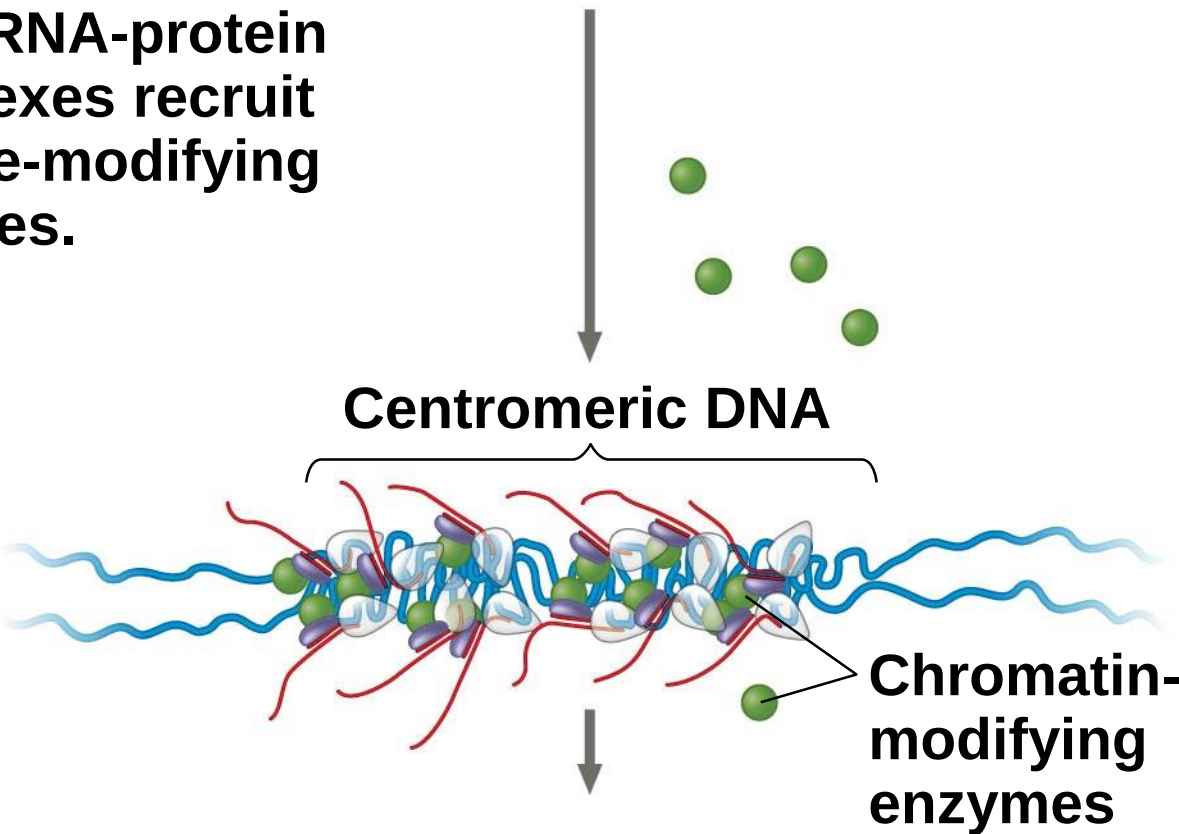


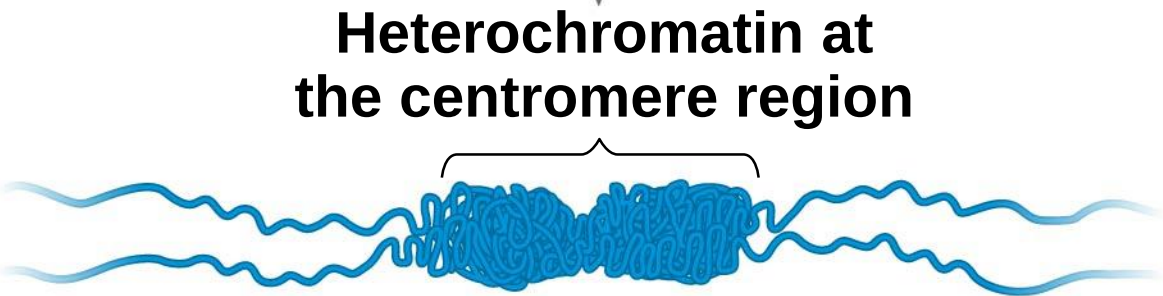
Figure 18.15a



5 The siRNA-protein complexes recruit histone-modifying enzymes.



6 Formation of heterochromatin at the centromere.



- Small ncRNAs called piwi-interacting RNAs (piRNAs) induce formation of heterochromatin, blocking the expression of parasitic DNA elements in the genome known as transposons
- piRNAs help to reestablish appropriate methylation patterns during gamete formation in many animal species

- **Long noncoding RNAs (lncRNAs)** range from 200 to hundreds of thousands of nucleotides in length
- One type of lncRNA is responsible for X chromosome inactivation
- RNA-based regulation of chromatin structure plays an important role in gene regulation

The Evolutionary Significance of Small ncRNAs

- Small ncRNAs can regulate gene expression at multiple steps and in many ways
- An increase in the number of miRNAs in a species may have allowed morphological complexity to increase over evolutionary time
- siRNAs may have evolved first, followed by miRNAs and later piRNAs

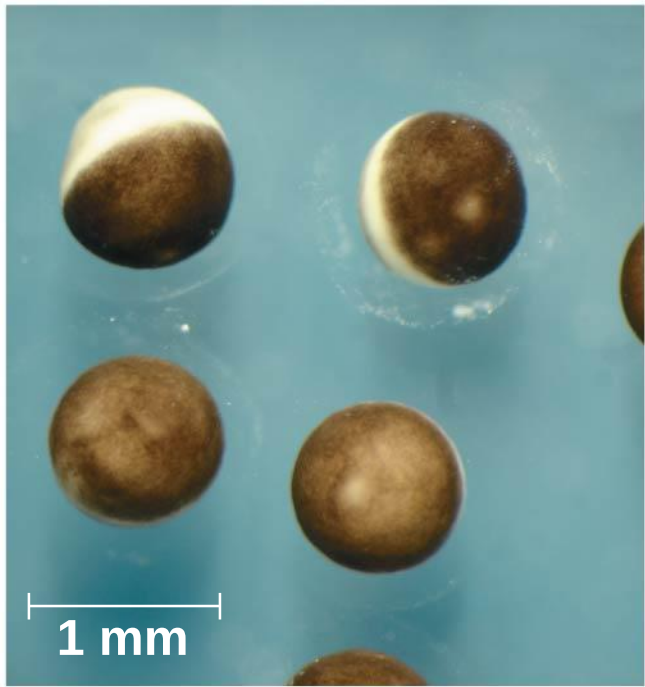
Concept 18.4: A program of differential gene expression leads to the different cell types in a multicellular organism

- During embryonic development, a fertilized egg gives rise to many different cell types
- Cells are organized successively into tissues, organs, organ systems, and the whole organism
- Gene expression orchestrates the developmental programs of animals

A Genetic Program for Embryonic Development

- The transformation from zygote to adult results from cell division, cell differentiation, and morphogenesis

Figure 18.16



(a) Fertilized eggs of a frog



(b) Newly hatched tadpole

- Cell **differentiation** is the process by which cells become specialized in structure and function
- The physical processes that give an organism its shape constitute **morphogenesis**
- Differential gene expression results from genes being regulated differently in each cell type
- Materials in the egg set up a program of gene regulation that is carried out as cells divide

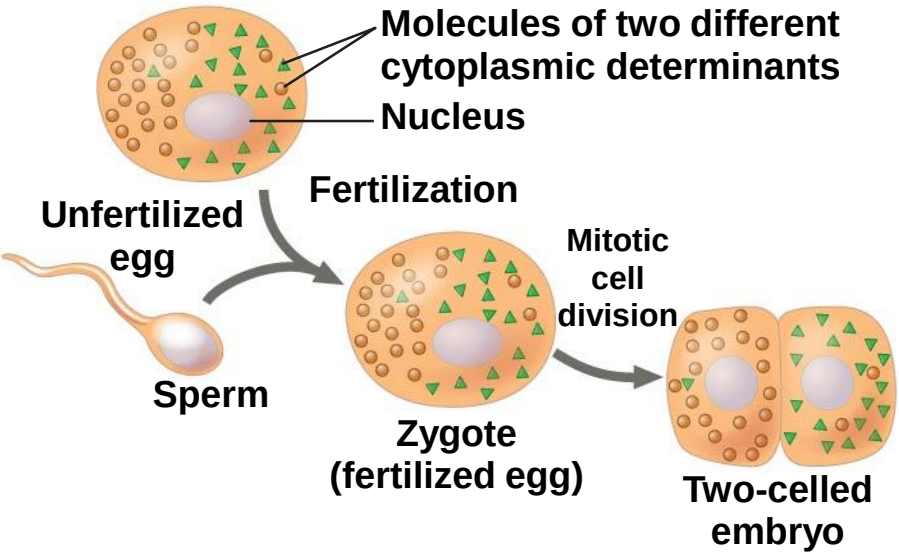
Cytoplasmic Determinants and Inductive Signals

- An egg's cytoplasm contains RNA, proteins, and other substances that are distributed unevenly in the unfertilized egg
- **Cytoplasmic determinants** are maternal substances in the egg that influence early development
- As the zygote divides by mitosis, cells contain different cytoplasmic determinants, which lead to different gene expression

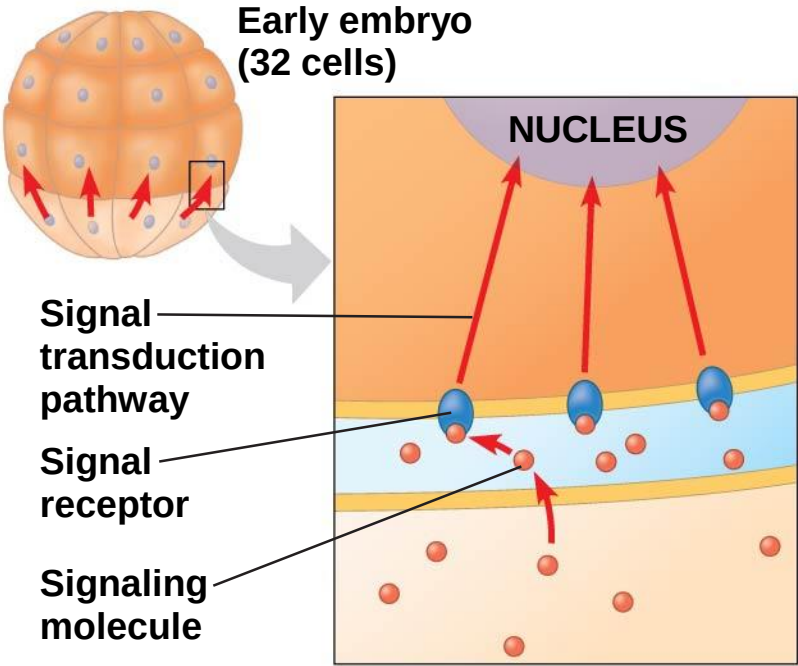
- The other major source of developmental information is the environment around the cell, especially signals from nearby embryonic cells
- In the process called **induction**, signal molecules from embryonic cells cause changes in nearby target cells
- Thus, interactions between cells induce differentiation of specialized cell types

Figure 18.17

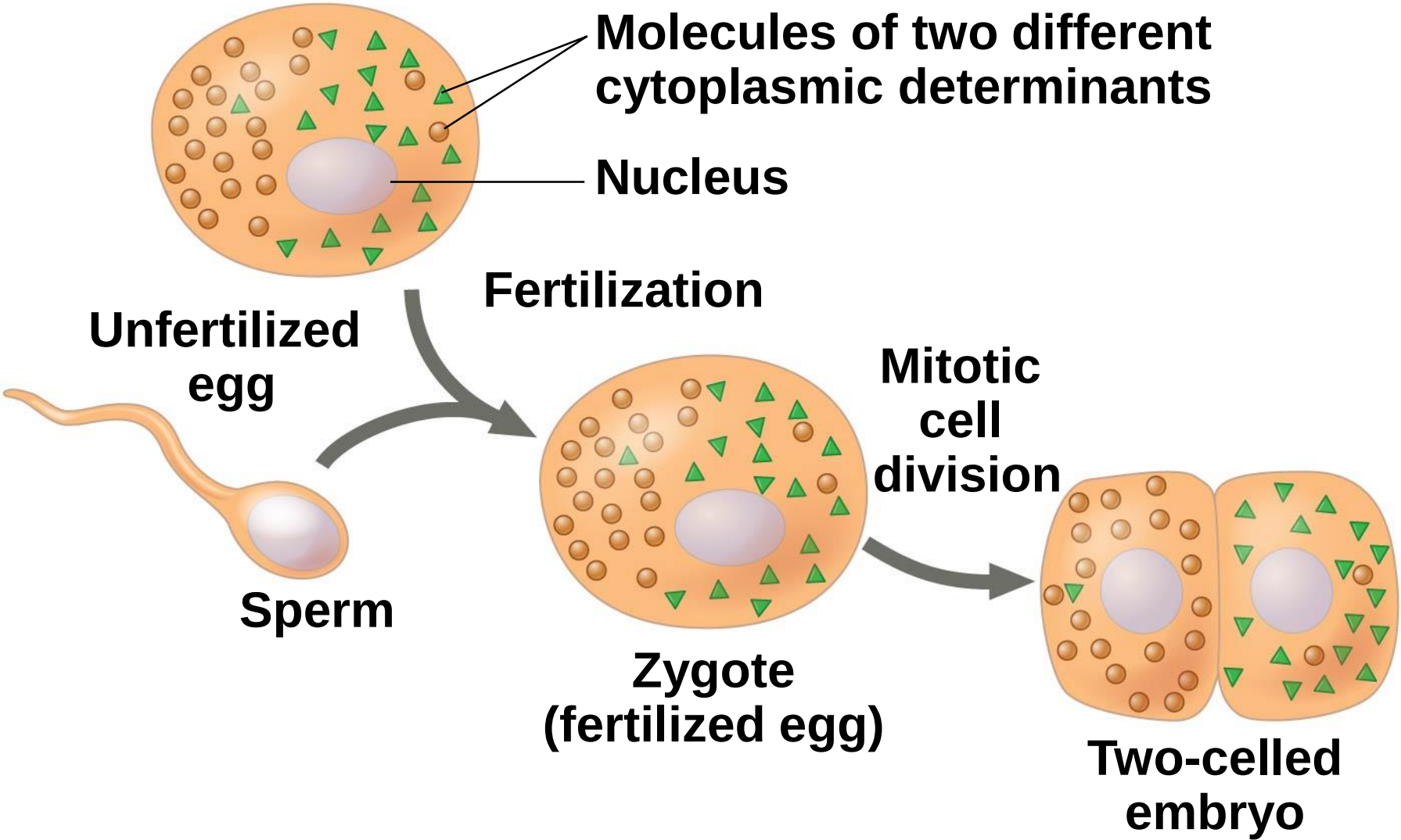
(a) Cytoplasmic determinants in the egg



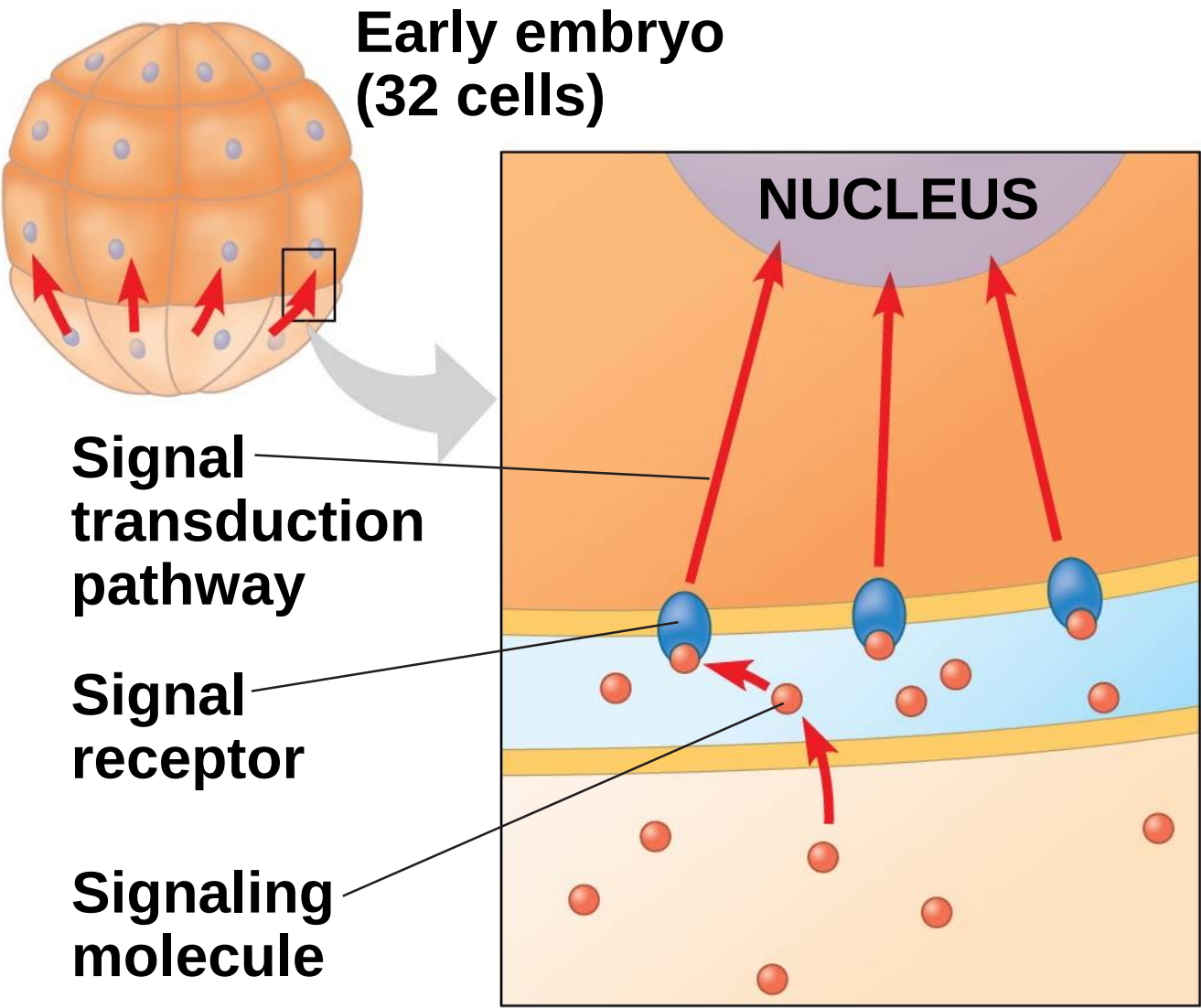
(b) Induction by nearby cells



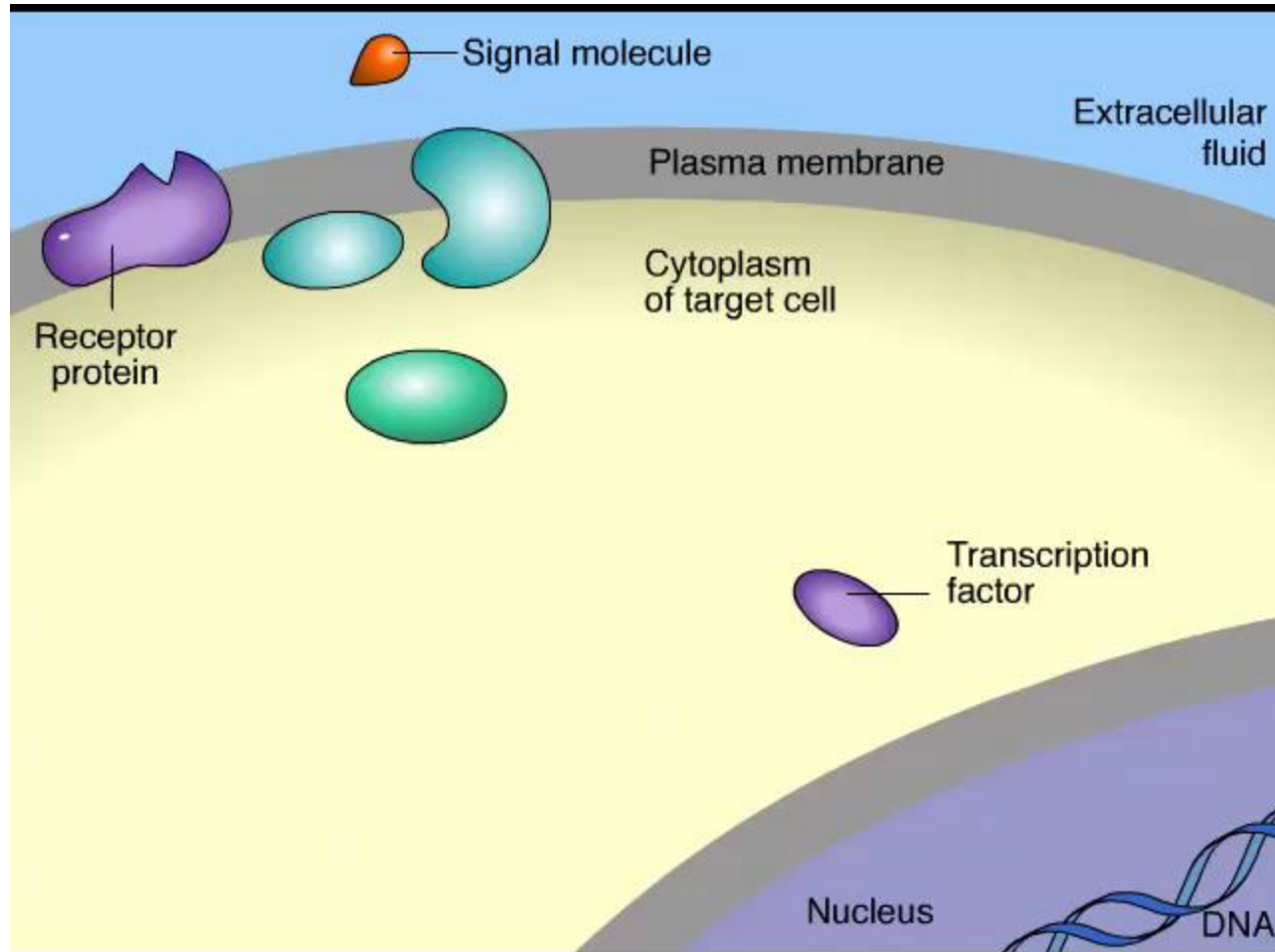
(a) Cytoplasmic determinants in the egg



(b) Induction by nearby cells



Animation: Cell Signaling



Sequential Regulation of Gene Expression During Cellular Differentiation

- **Determination** irreversibly commits a cell to becoming a particular cell type
- Determination precedes differentiation
- Cell differentiation is marked by the production of tissue-specific proteins

- Myoblasts are cells determined to form muscle cells and produce large amounts of muscle-specific proteins
- *MyoD* is a “master regulatory gene” that encodes a transcription factor that commits the cell to becoming skeletal muscle
- Some target genes for MyoD (protein) encode additional muscle-specific transcription factors

Figure 18.18_1

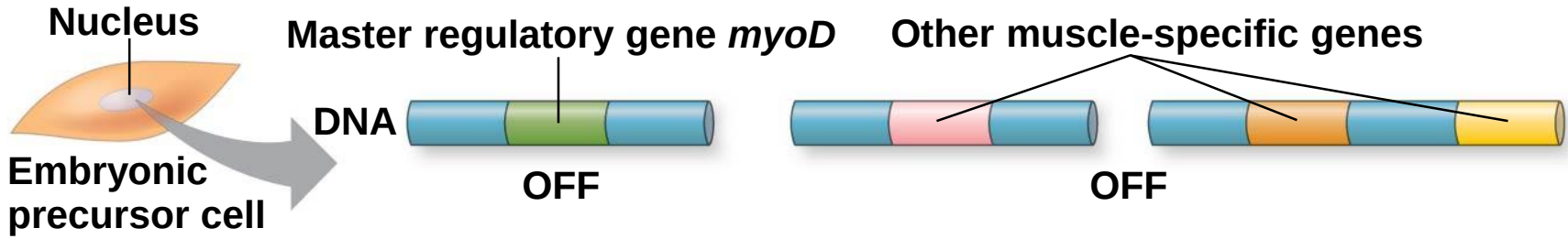


Figure 18.18_2

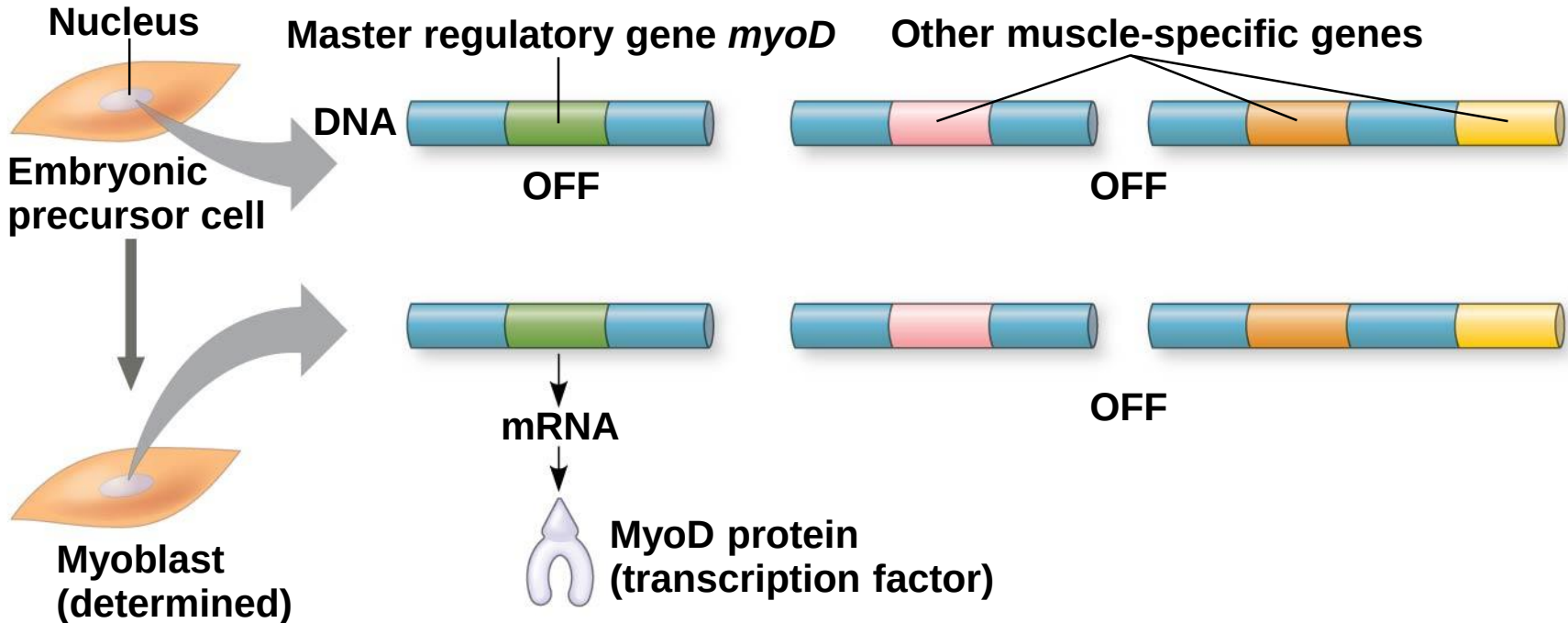
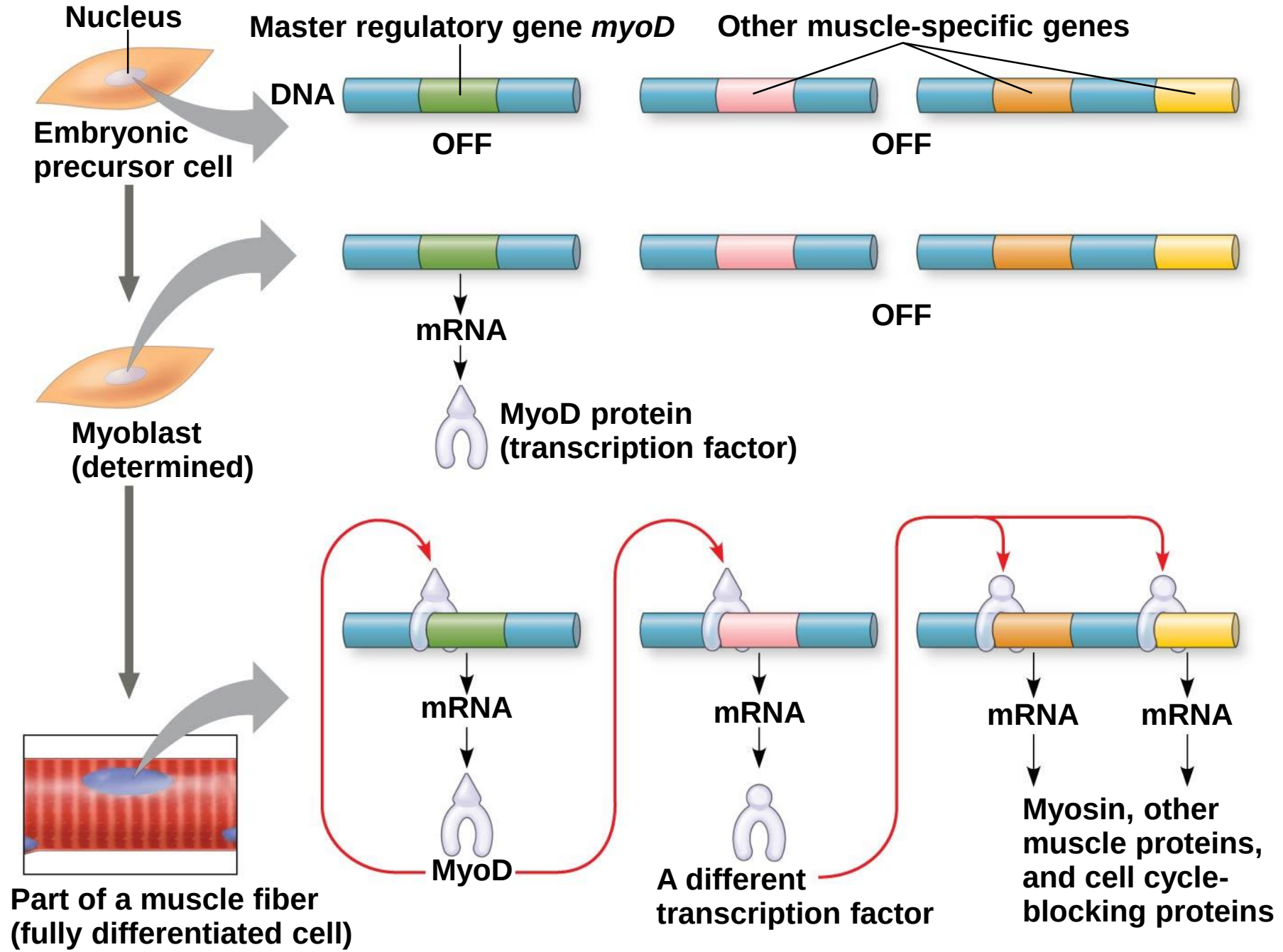


Figure 18.18_3



Pattern Formation: Setting Up the Body Plan

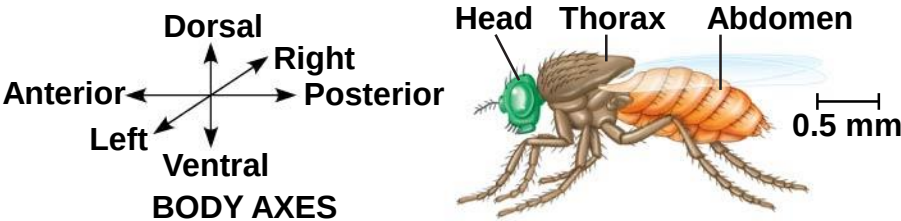
- **Pattern formation** is the development of a spatial organization of tissues and organs
- In animals, pattern formation begins with the establishment of the major axes
- **Positional information**, the molecular cues that control pattern formation, tells a cell its location relative to the body axes and to neighboring cells

- Pattern formation has been extensively studied in the fruit fly *Drosophila melanogaster*
- Combining anatomical, genetic, and biochemical approaches, researchers have discovered developmental principles common to many other species, including humans

The Life Cycle of Drosophila

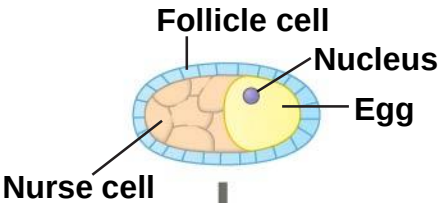
- In *Drosophila*, cytoplasmic determinants in the unfertilized egg determine the axes before fertilization
- After fertilization, the embryo develops into a segmented larva with three larval stages
- The larva then forms a pupa, which undergoes metamorphosis into the adult fly

Figure 18.19

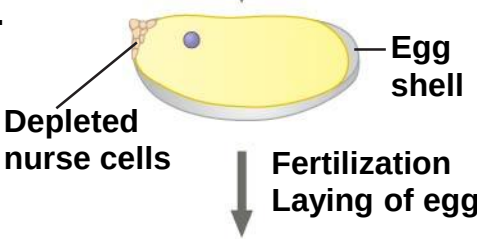


(a) Adult.

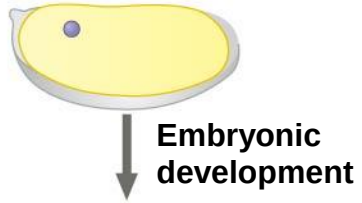
① Developing egg



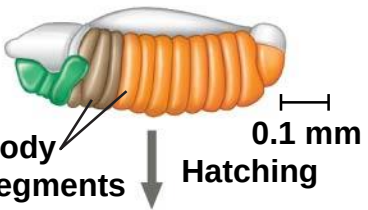
② Mature, unfertilized egg.



③ Fertilized egg.



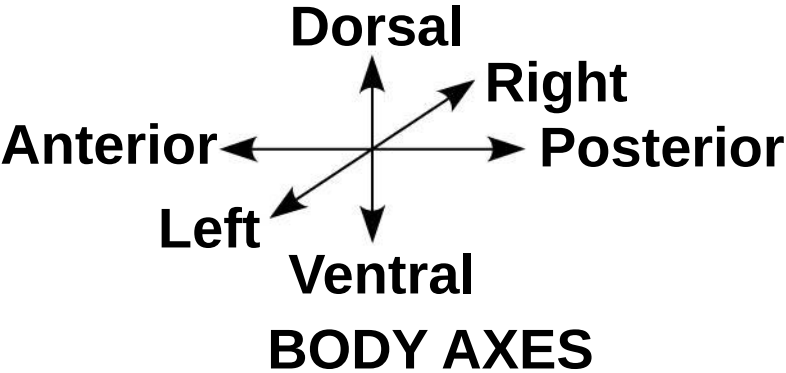
④ Segmented embryo.



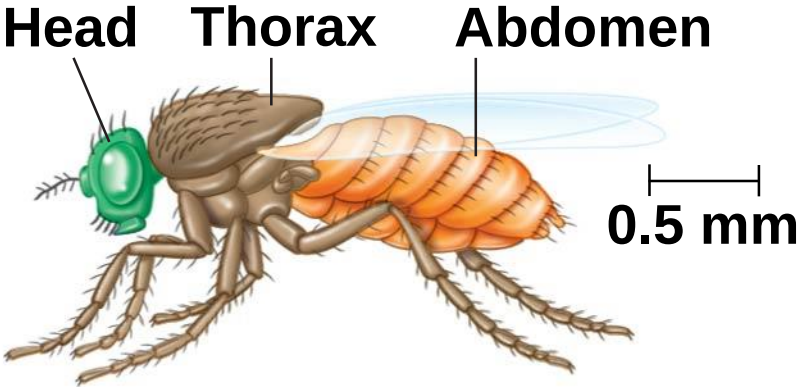
⑤ Larva.



(b) Development from egg to larva.



(a) Adult.



1 Developing egg.

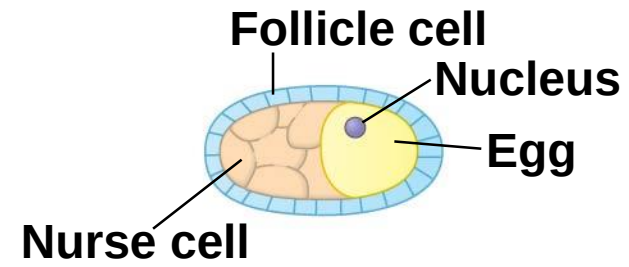
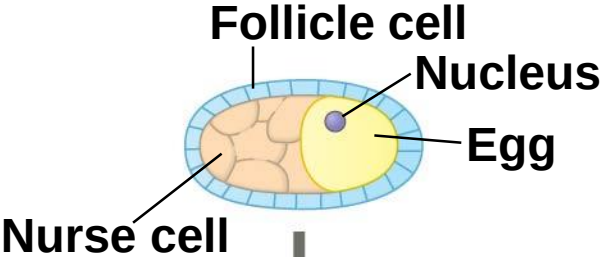


Figure 18.19b_2

1 Developing egg.



2 Mature, unfertilized egg.

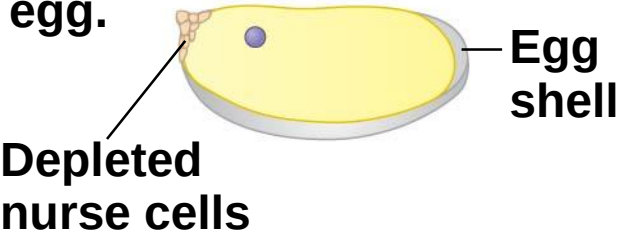
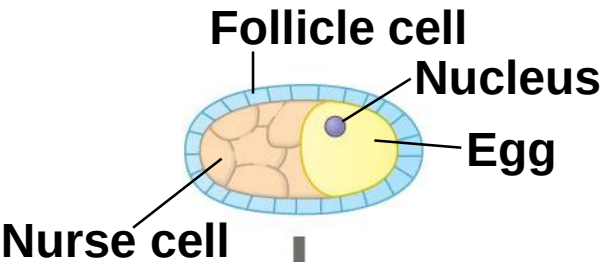
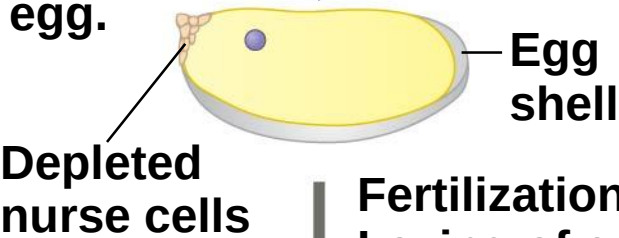


Figure 18.19b_3

1 Developing egg.



2 Mature, unfertilized egg.

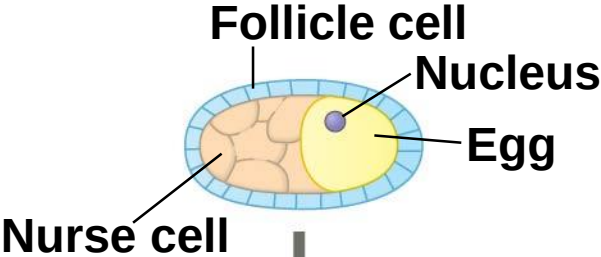


**Fertilization
Laying of egg**

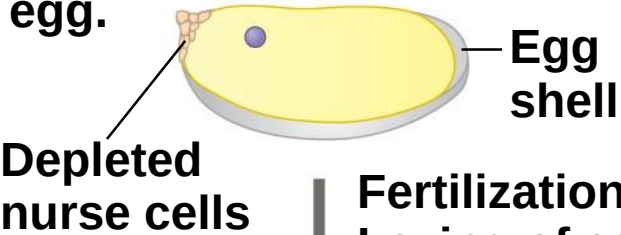
3 Fertilized egg.



1 Developing egg.



2 Mature, unfertilized egg.



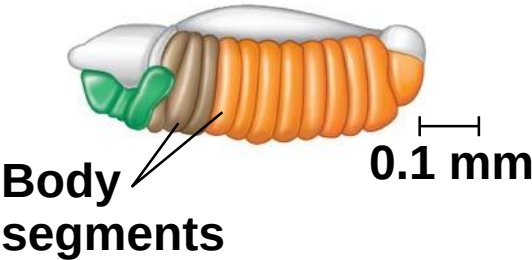
Fertilization
Laying of egg

3 Fertilized egg.

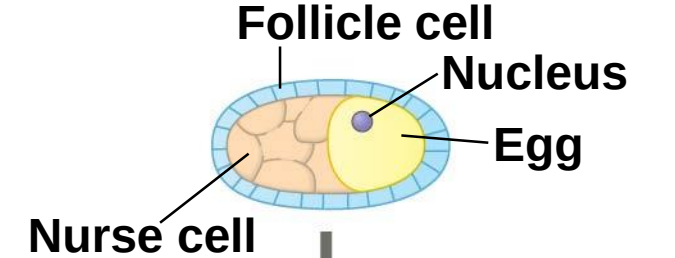


Embryonic
development

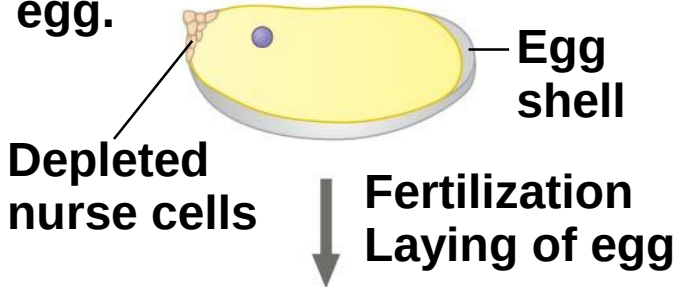
4 Segmented embryo.



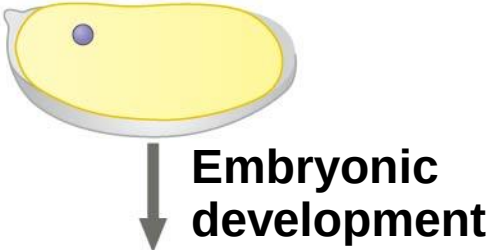
1 Developing egg.



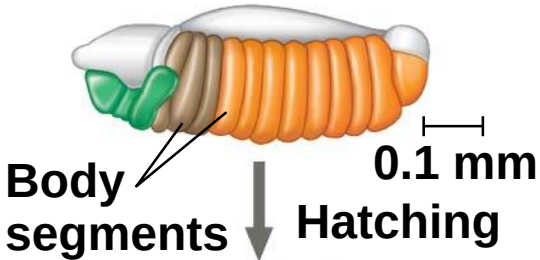
2 Mature, unfertilized egg.



3 Fertilized egg.



4 Segmented embryo.



5 Larva.

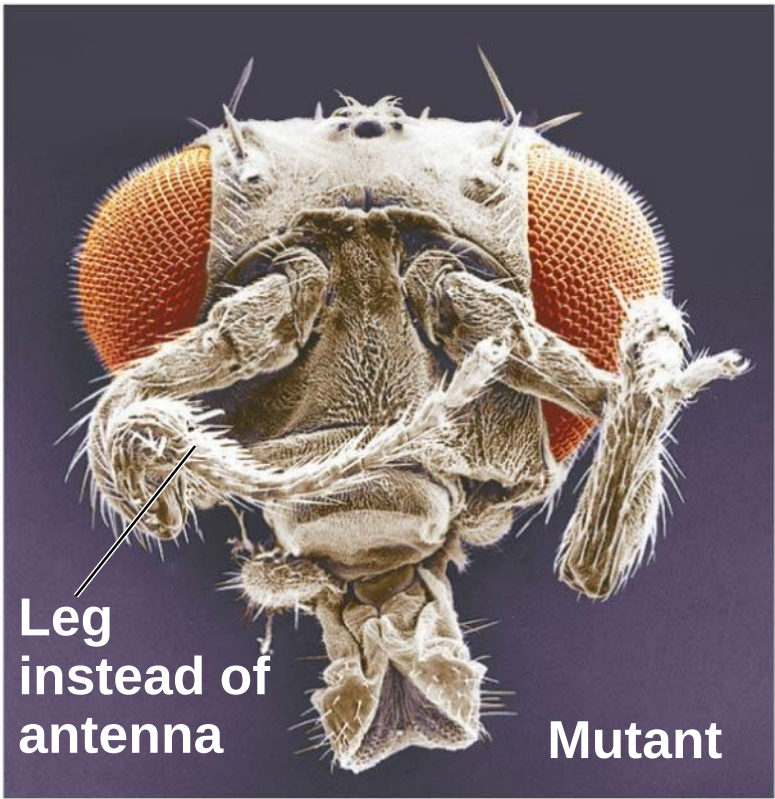
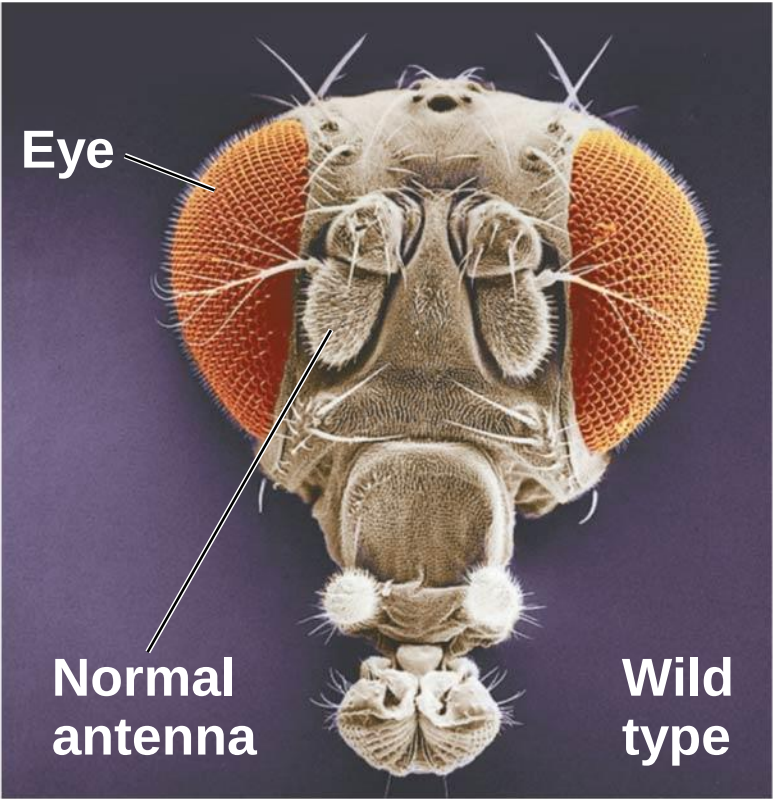


(b) Development from egg to larva.

Genetic Analysis of Early Development: **Scientific Inquiry**

- Edward B. Lewis, Christiane Nüsslein-Volhard, and Eric Wieschaus won a Nobel Prize in 1995 for decoding pattern formation in *Drosophila*
- Lewis discovered **homeotic genes**, which control pattern formation in the late embryo, larva, and adult stages

Figure 18.20



- Nüsslein-Volhard and Wieschaus studied segment formation
- They created mutants, conducted breeding experiments, and looked for corresponding genes
- Many of the identified mutations were **embryonic lethals**, causing death during embryogenesis
- They found about 120 genes essential for normal segmentation

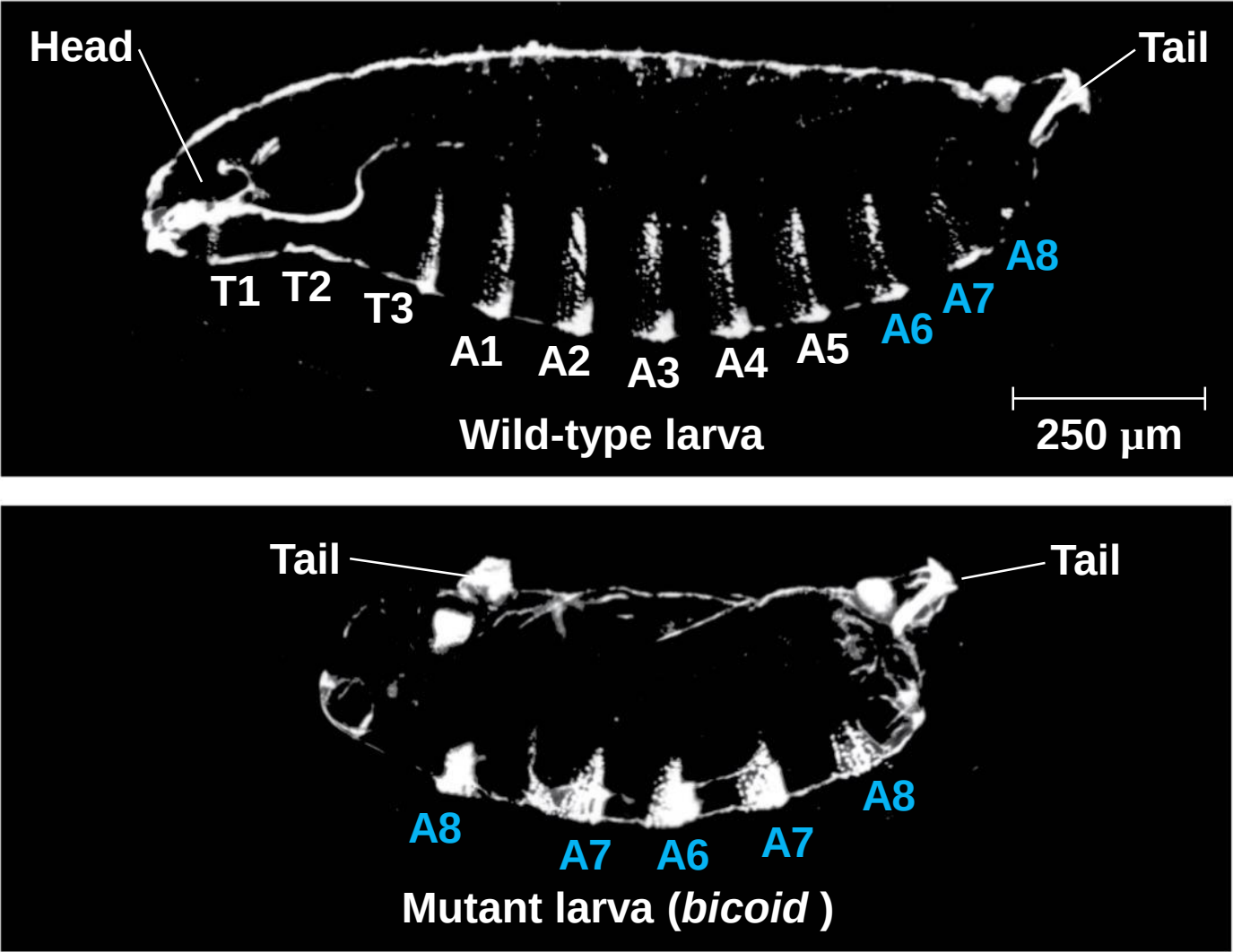
Axis Establishment

- **Maternal effect genes** encode cytoplasmic determinants that initially establish the axes of the body of *Drosophila*
- These maternal effect genes are also called egg-polarity genes because they control orientation of the egg and consequently the fly

Bicoid: A Morphogen That Determines Head Structures

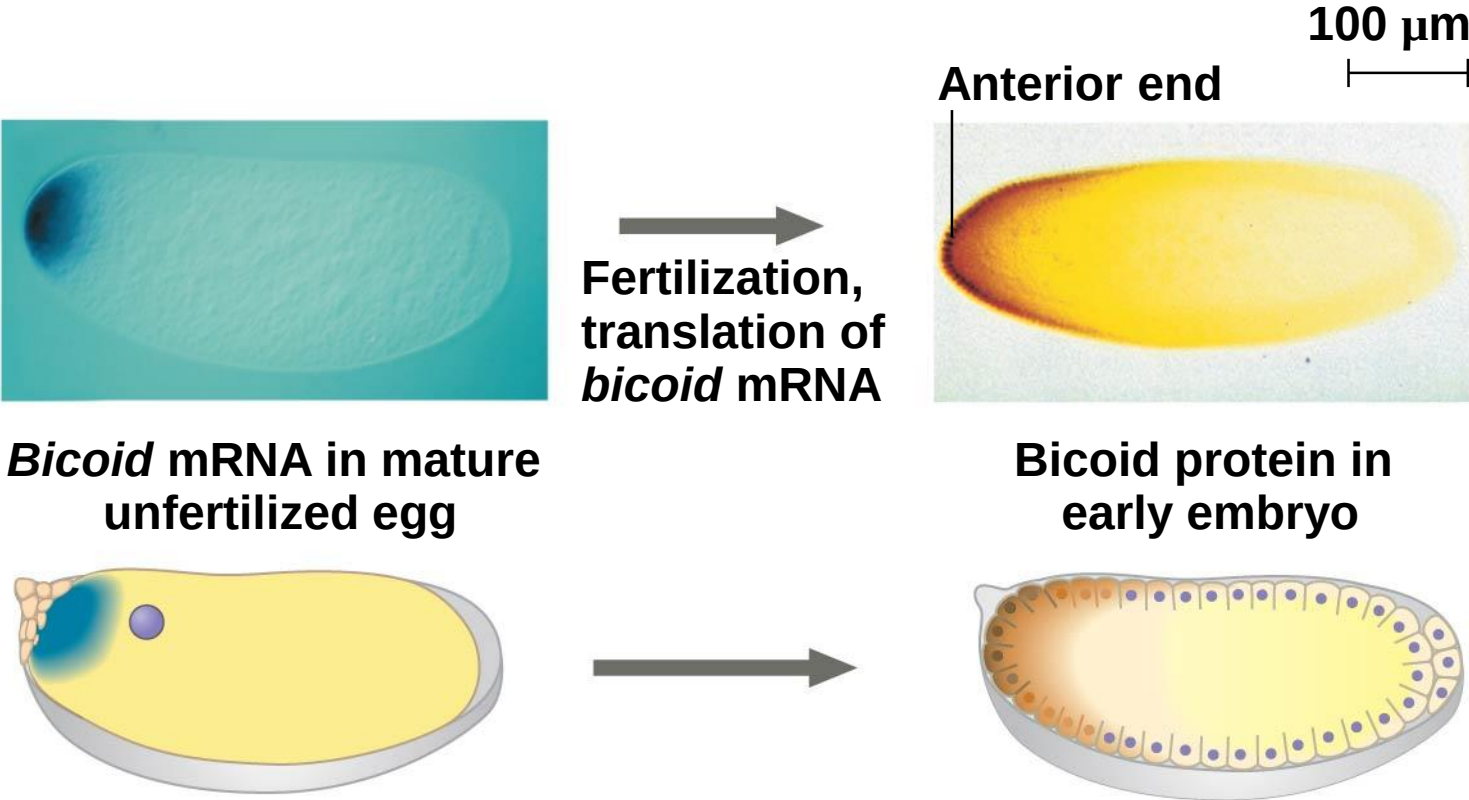
- One maternal effect gene, the *bicoid* gene, affects the front half of the body
- An embryo whose mother has no functional *bicoid* gene lacks the front half of its body and has duplicate posterior structures at both ends

Figure 18.21

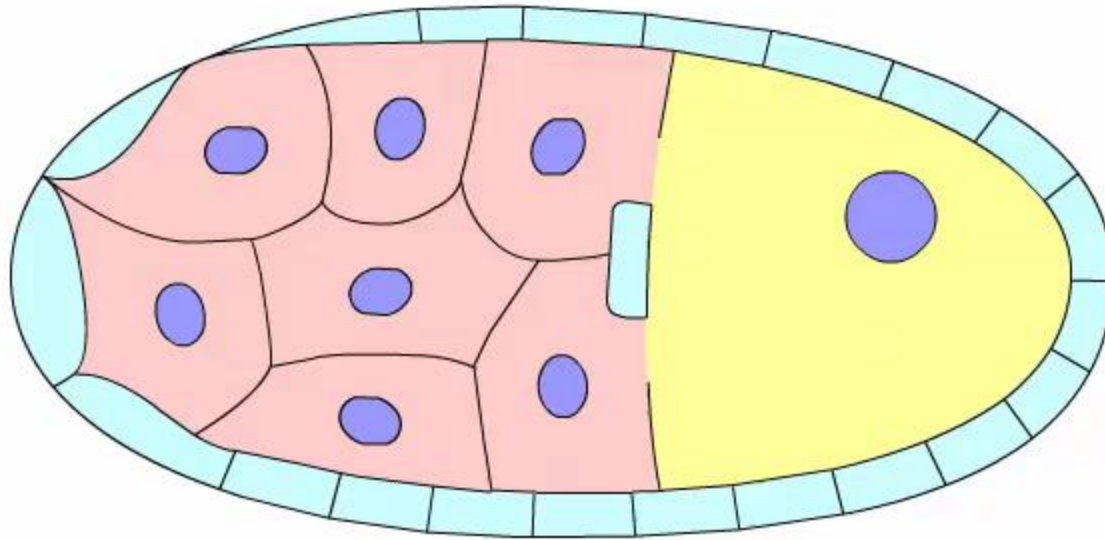


- This phenotype suggests that the product of the mother's *bicoid* gene is essential for setting up the anterior end of the embryo
- This hypothesis is an example of the morphogen gradient hypothesis, in which gradients of substances called **morphogens** establish an embryo's axes and other features of its form
- Experiments showed that bicoid protein is distributed in an anterior to posterior gradient in the early embryo

Figure 18.22



Animation: Development of Head-Tail Axis in Fruit Flies



- The *bicoid* research was groundbreaking for three reasons:
 - It identified a specific protein required for some early steps in pattern formation
 - It increased understanding of the mother's role in embryo development
 - It demonstrated that a gradient of molecules can determine polarity and position in the embryo

Evolutionary Developmental Biology

(“Evo-Devo”)

- The fly with legs emerging from its head in Figure 18.20 is the result of a single mutation in one gene
- Some scientists considered whether these types of mutations could contribute to evolution by generating novel body shapes
- This line of inquiry gave rise to the field of evolutionary developmental biology, “evo-devo”

Concept 18.5: Cancer results from genetic changes that affect cell cycle control

- The gene regulation systems that go wrong during cancer are the very same systems involved in embryonic development

Types of Genes Associated with Cancer

- Cancer can be caused by mutations to genes that normally regulate cell growth and division
- Mutations in these genes can be caused by spontaneous mutation or environmental influences such as chemicals, radiation, and some viruses

- **Oncogenes** are cancer-causing genes in some types of viruses
- **Proto-oncogenes** are the corresponding normal cellular genes that are responsible for normal cell growth and division
- Conversion of a proto-oncogene to an oncogene can lead to abnormal stimulation of the cell cycle

- Proto-oncogenes can be converted to oncogenes by
 - movement of DNA within the genome
 - amplification of a proto-oncogene
 - point mutations in the proto-oncogene or its control elements

Figure 18.23

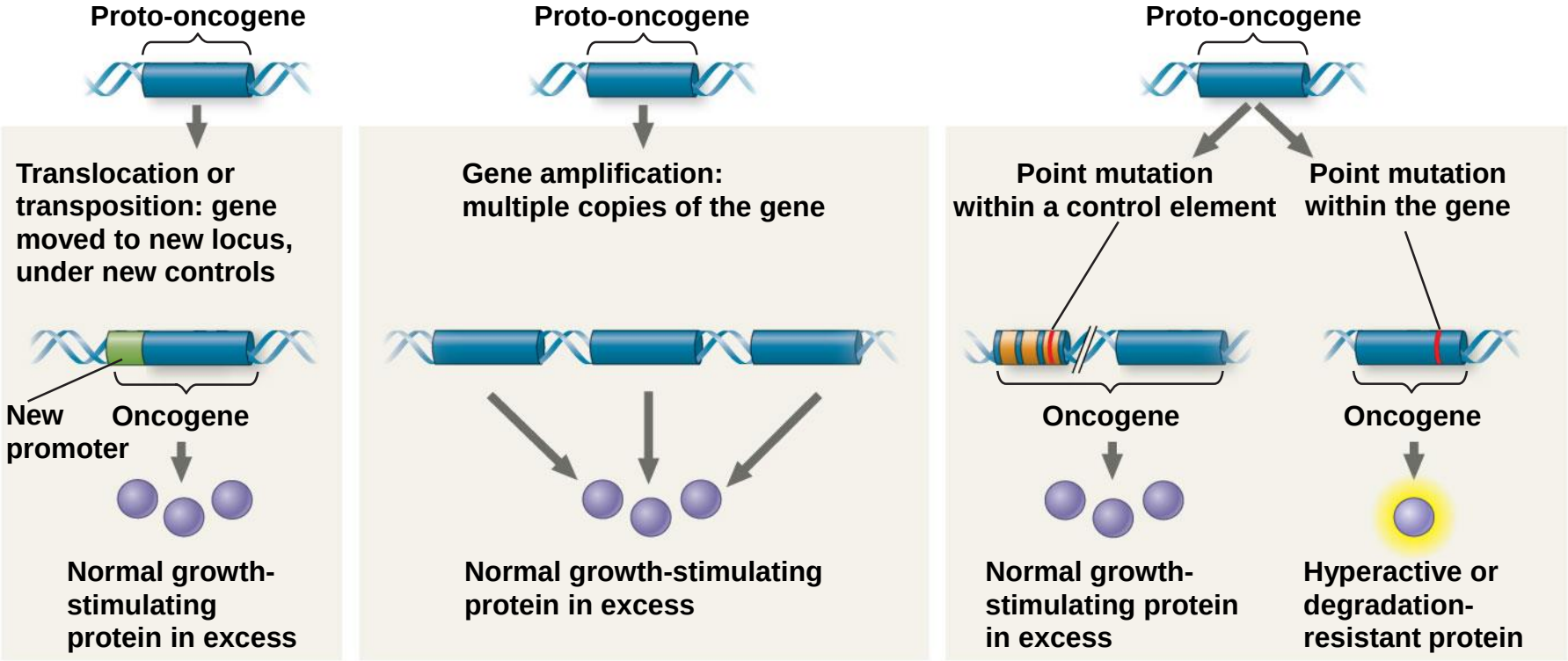


Figure 18.23a

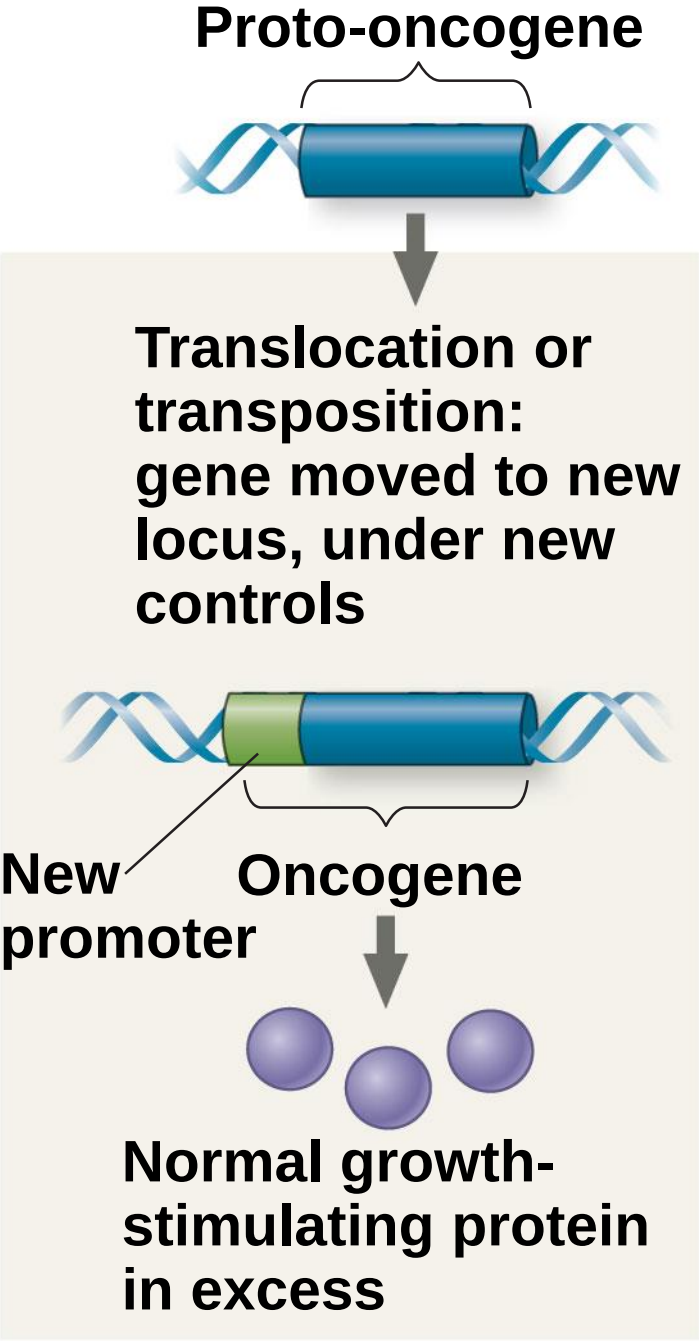


Figure 18.23b

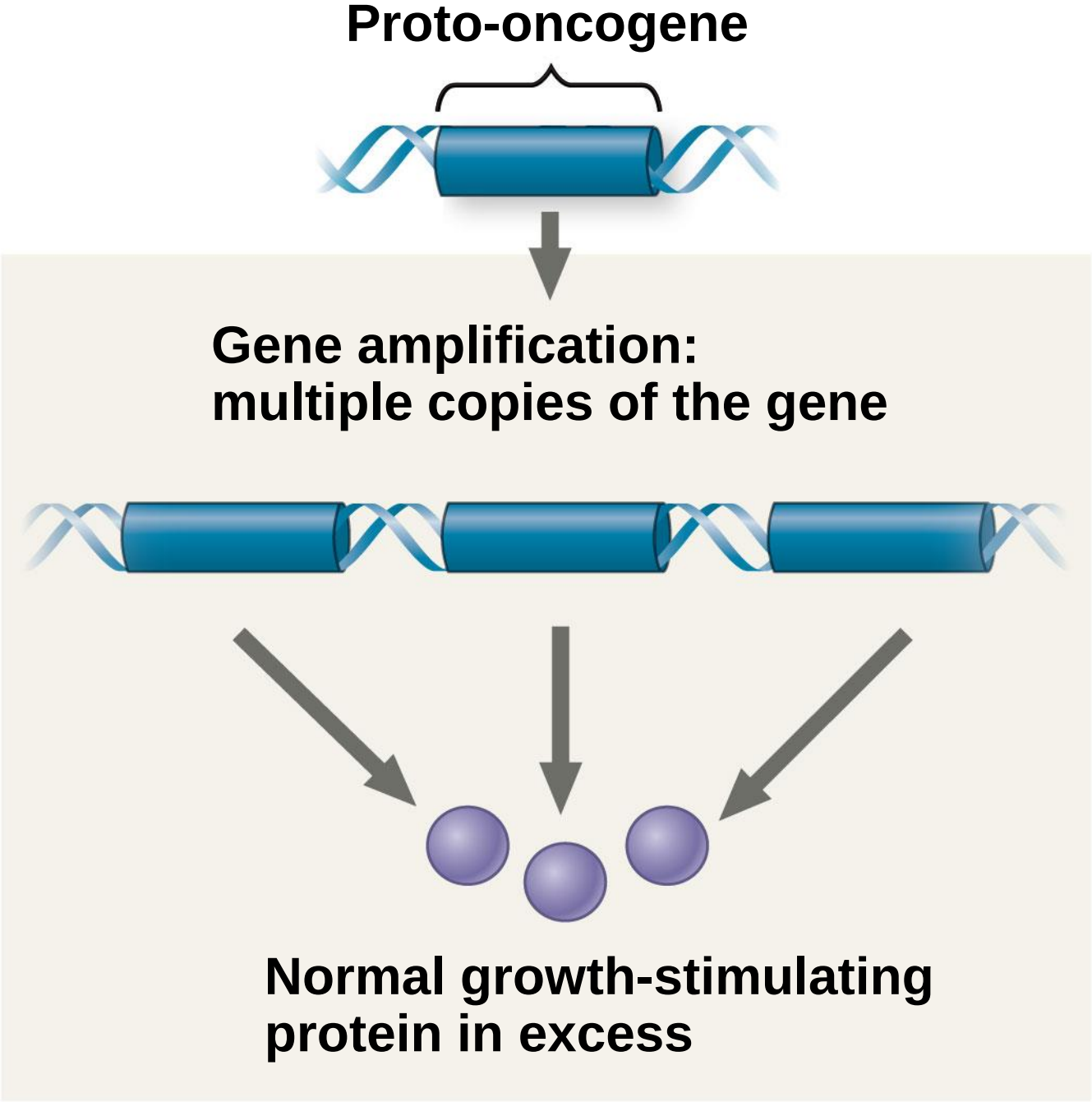
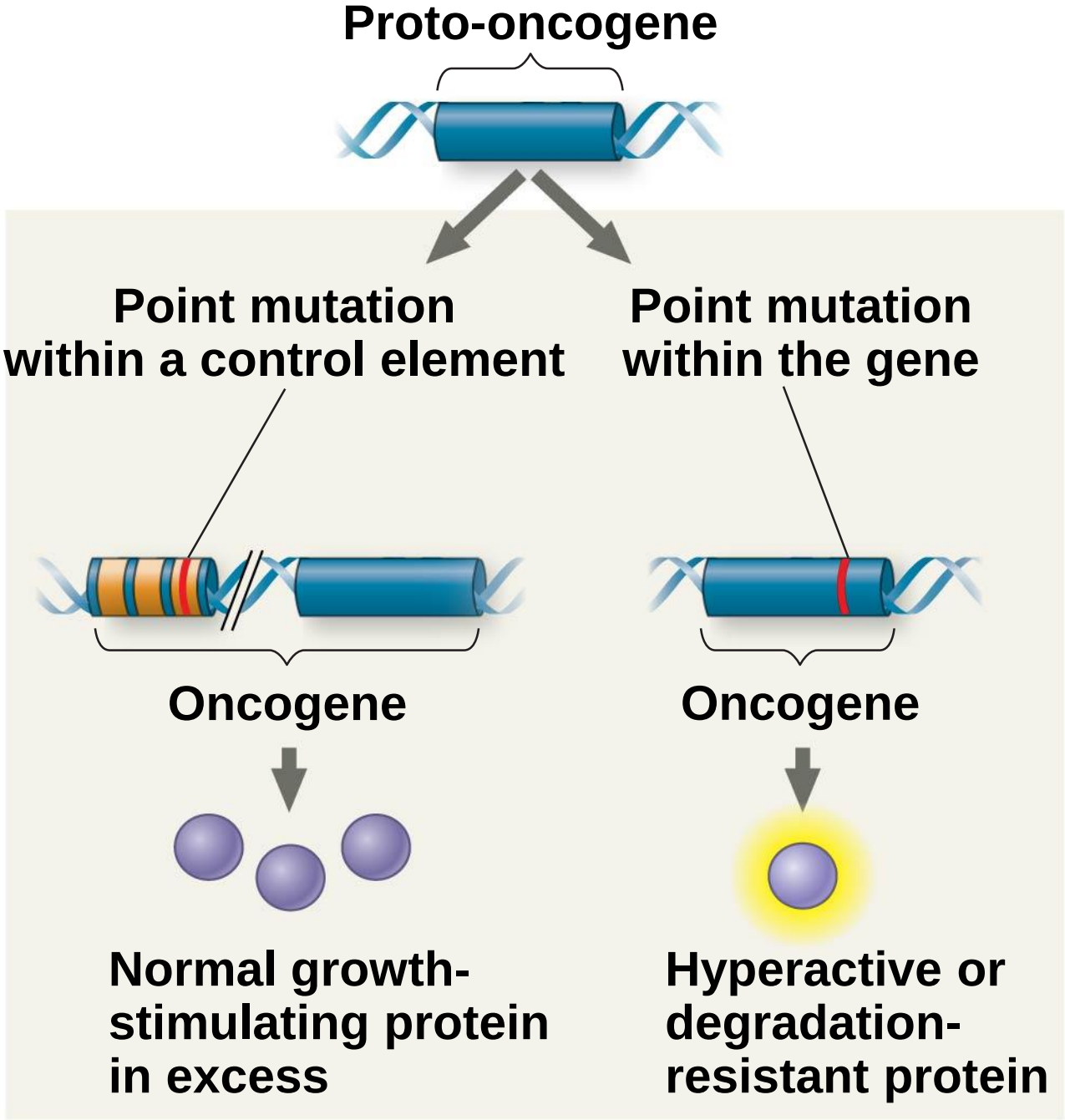


Figure 18.23c

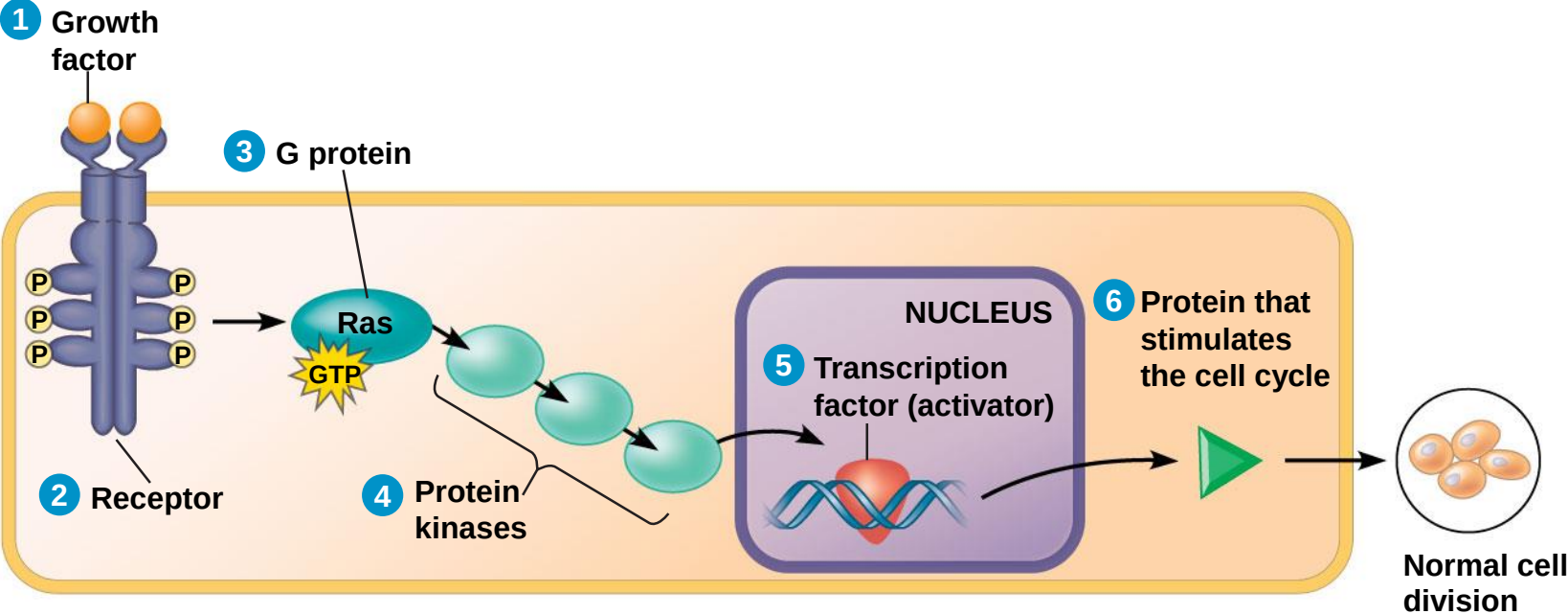


- **Tumor-suppressor genes** normally inhibit cell division
- Mutations that decrease protein products of tumor-suppressor genes may contribute to cancer onset
- Tumor-suppressor proteins
 - repair damaged DNA
 - control cell adhesion
 - act in cell-signaling pathways that inhibit the cell cycle

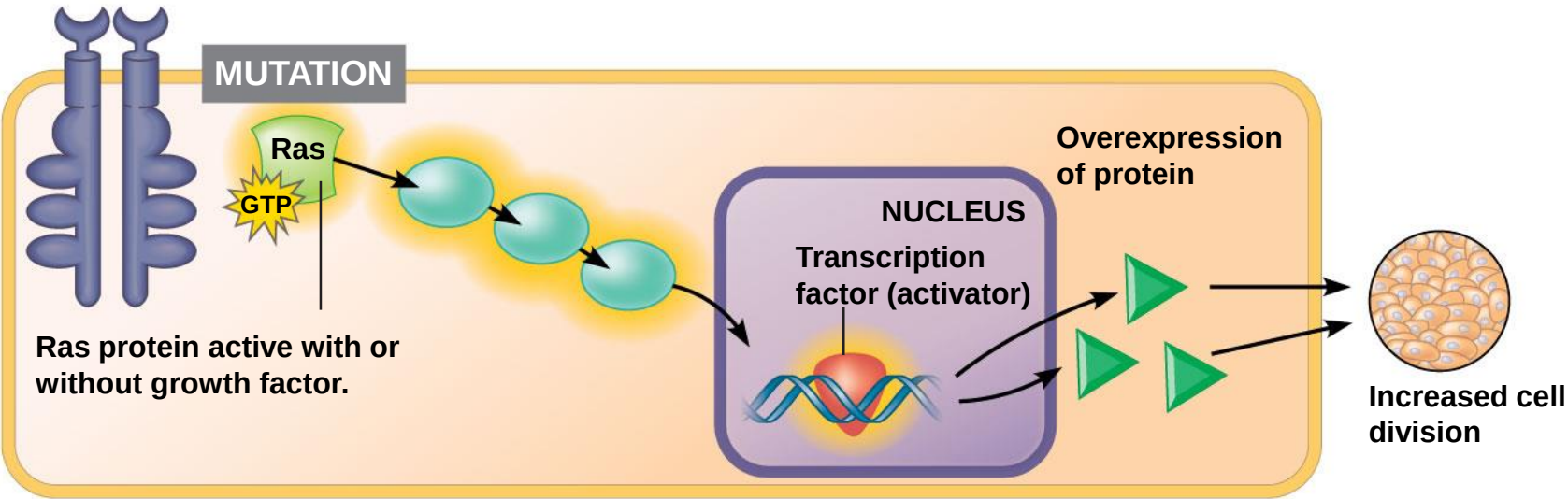
Interference with Normal Cell-Signaling Pathways

- Mutations in the *ras* proto-oncogene and *p53* tumor-suppressor gene are common in human cancers
- Mutations in the ***ras* gene** can lead to production of a hyperactive Ras protein and increased cell division
- The Ras protein is a G protein that relays a signal from a growth factor receptor on the cell surface
- The response to the resulting cascade stimulates cell division

Figure 18.24

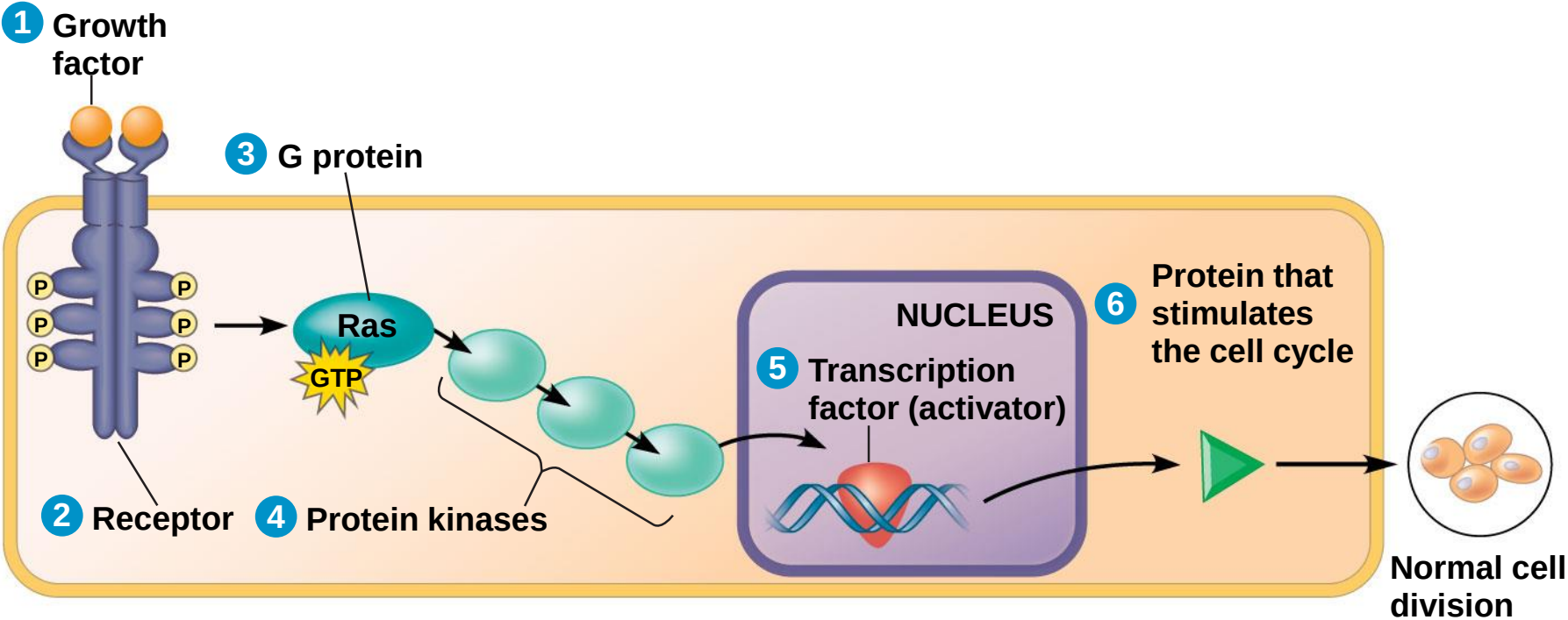


(a) Normal cell cycle–stimulating pathway.



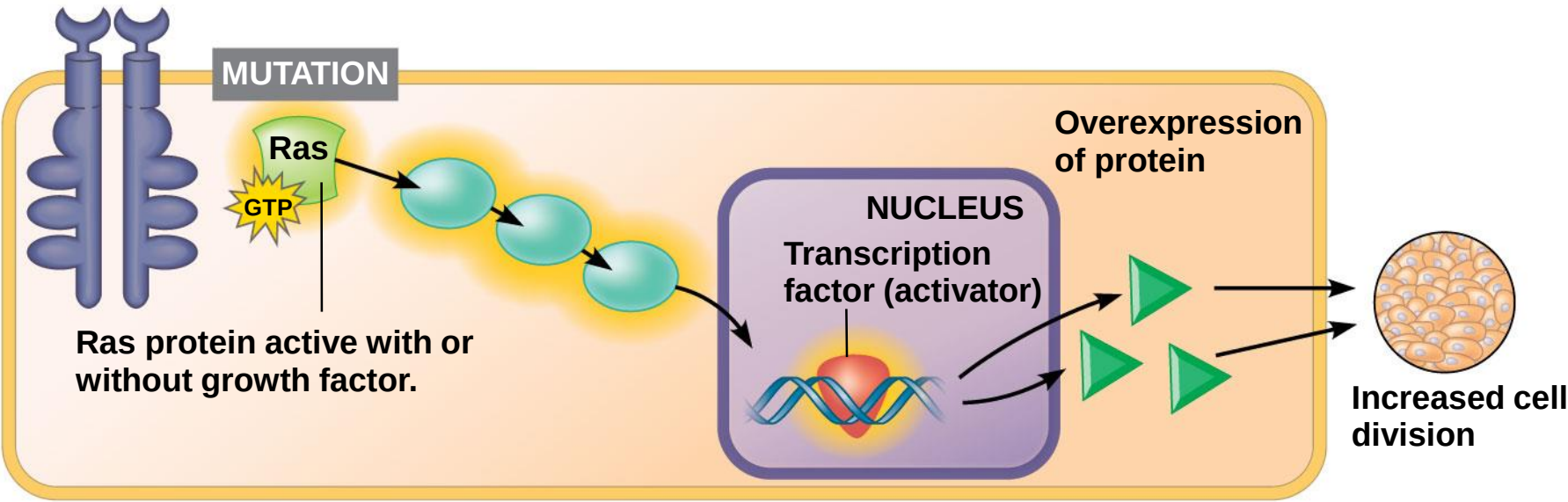
(b) Mutant cell cycle–stimulating pathway.

Figure 18.24a



(a) Normal cell cycle–stimulating pathway.

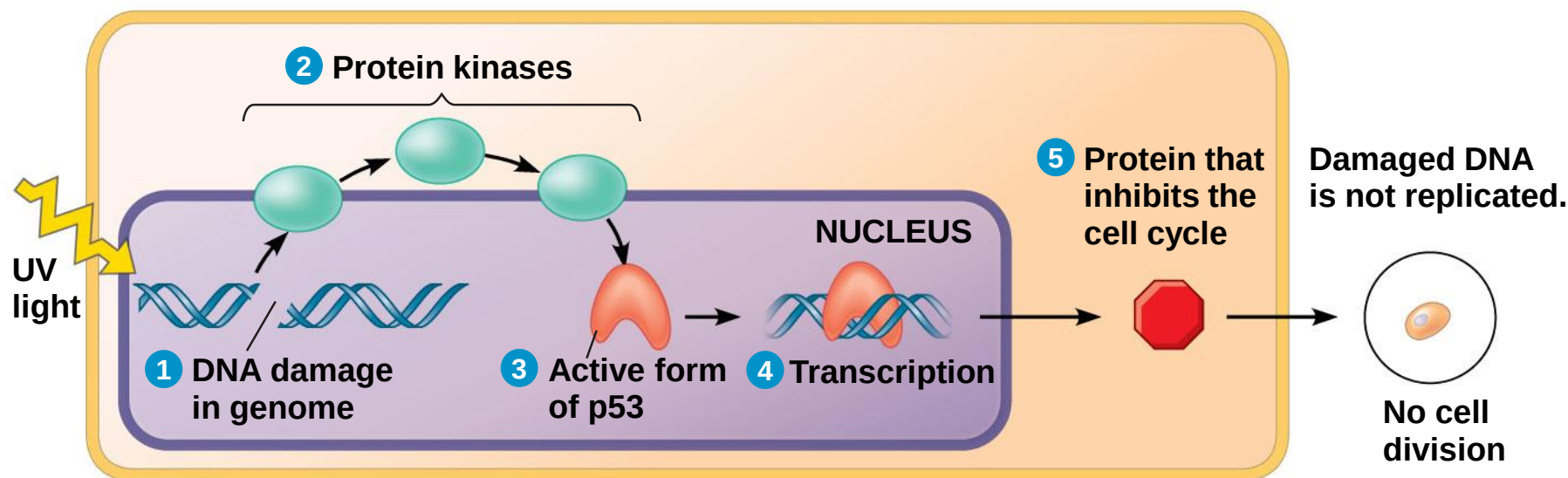
Figure 18.24b



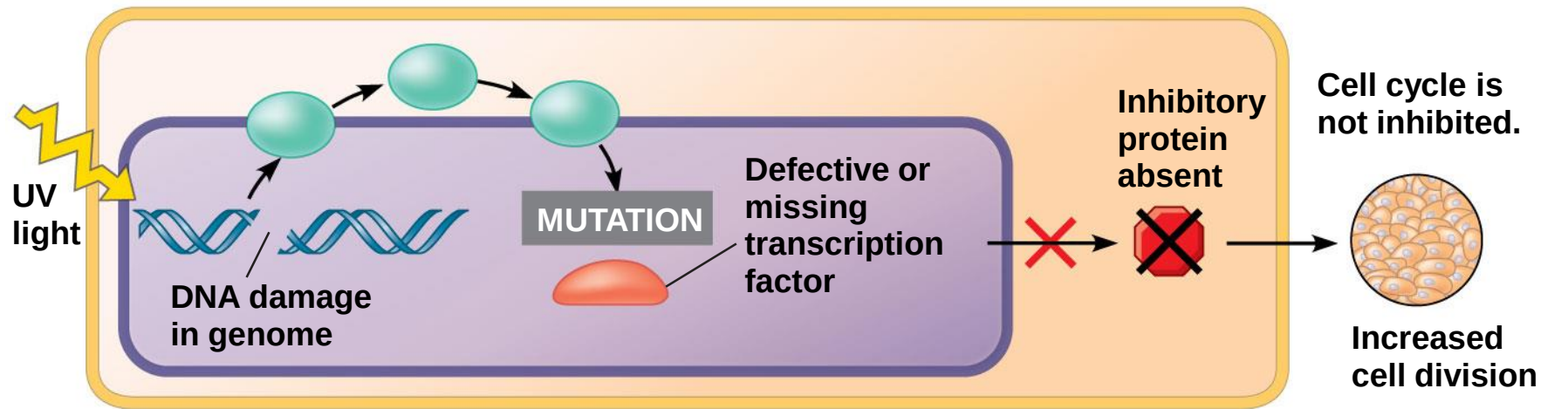
(b) Mutant cell cycle–stimulating pathway.

- Mutations in the **p53 gene** prevent suppression of the cell cycle
- Suppression of the cell cycle can be important in the case of damage to a cell's DNA; normal p53 prevents a cell from passing on mutations
- It also activates expression of miRNAs that inhibit the cell cycle, and can turn on genes directly involved in DNA repair
- If DNA is irreparable, p53 activates cell “suicide” genes

Figure 18.25

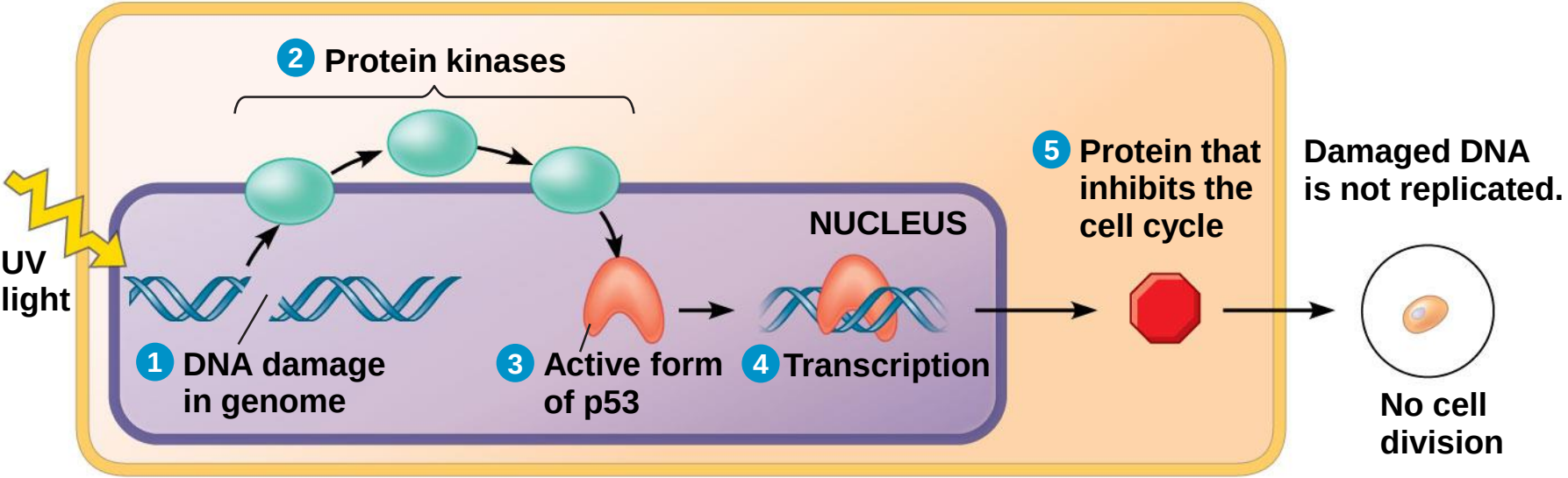


(a) Normal cell cycle-inhibiting pathway



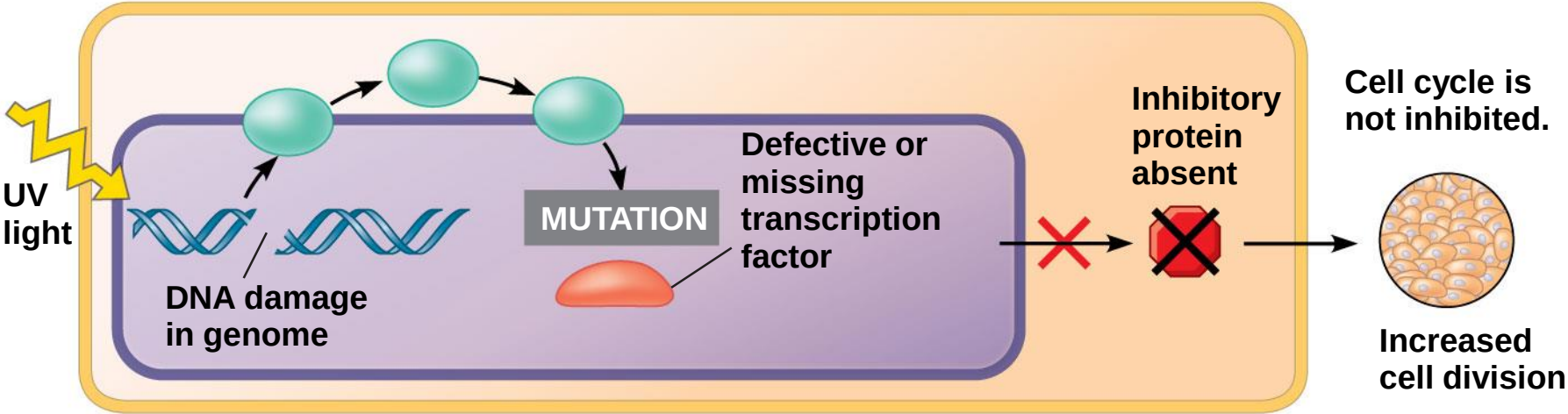
(b) Mutant cell cycle-inhibiting pathway

Figure 18.25a



(a) Normal cell cycle-inhibiting pathway

Figure 18.25b



(b) Mutant cell cycle-inhibiting pathway

The Multistep Model of Cancer Development

- Multiple mutations are generally needed for full-fledged cancer; thus the incidence increases with age
- At the DNA level, a cancerous cell is usually characterized by at least one active oncogene and the mutation of several tumor-suppressor genes

Figure 18.26

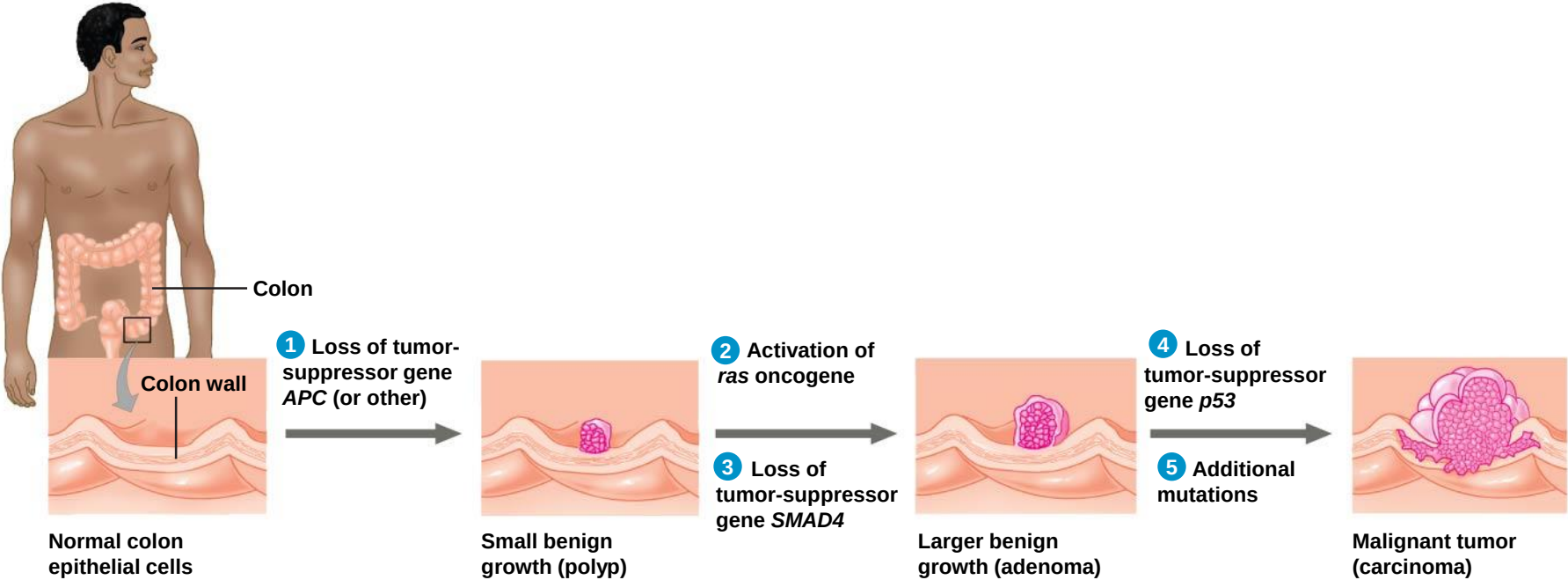
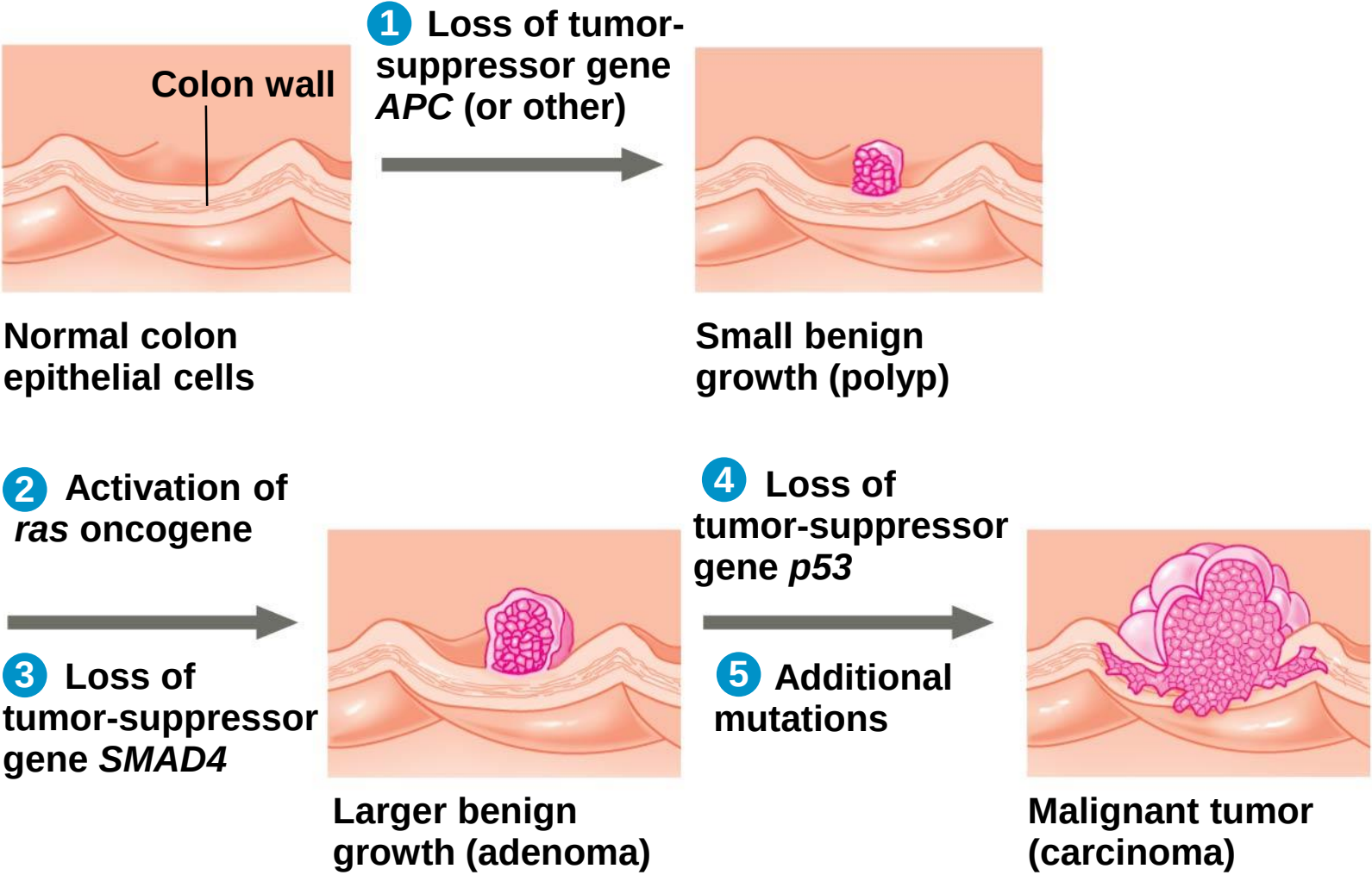


Figure 18.26a



- Routine screening for some cancers, such as colorectal cancer, is recommended
- Suspicious polyps may be removed before cancer progresses
- Breast cancer is a heterogeneous disease that is the second most common form of cancer in women in the United States; it also occurs in some men
- A genomics approach to profiling breast tumors has identified four major types of breast cancer

Inherited Predisposition and Environmental Factors Contributing to Cancer

- Individuals can inherit oncogenes or mutant alleles of tumor-suppressor genes
- Inherited mutations in the tumor-suppressor gene *adenomatous polyposis coli* are common in individuals with colorectal cancer
- Mutations in the *BRCA1* or *BRCA2* gene are found in at least half of inherited breast cancers, and tests using DNA sequencing can detect these mutations

The Role of Viruses in Cancer

- A number of tumor viruses can also cause cancer in humans and animals
- Viruses can interfere with normal gene regulation in several ways if they integrate into the DNA of a cell
- Viruses are powerful biological agents